# How Forms Become Reconstructible

## Supporting Information

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Version 2.1 - 16 October 2026
Companion methods and figures; hosted with the paper on GitHub

## Guide to the supporting information

Open the main paper v2.1. Download both PDFs together. The guide below supplies method identifiers and page numbers; the repository also supplies a complete methods page index.

This document supplies methods, model correspondence and supporting figures for the five-question main paper. All **C.1–C.45** identifiers are retained; **C.46–C.51** describe the new selection. The v2.1 main/SI PDFs and versioned review package are prepared locally for commit and push; new versioned web links require that publication step. No Zenodo deposit or DOI is claimed.

The **97-note evidence ledger**, extended experimental comparisons and development history remain separate repository documents. Evidence codes identify stable notes, rather than a new sample or replication. References to D.1–D.10 denote the repository's *Extended evidence* document, not missing sections of this PDF. The full original v1.0 and both reviewed v1.1 PDFs remain preserved in the working archives.

| Main question | Principal methods retained here | Supporting detail |
|---|---|---|
| History across replacement | C.3 (p. 4), C.7–C.10 (pp. 10–18), C.16 (p. 26) | Spatial capture, exact Q score, budgets, reference acquisition and editable cells. |
| Embodied relationships | C.1–C.2 (pp. 2–3), C.8–C.9 (pp. 14–16), C.41 (p. 57), C.45–C.46 (pp. 60–61), C.51 (p. 67) | Reciprocal repair, writer placement, Hopfield readers, fresh body-source confirmation and serial capture limits. |
| Functional record structure | C.17 (p. 27), C.23 (p. 36), C.25–C.26 (pp. 42–43), C.31–C.32 (pp. 47–48), C.39 (p. 55), C.48–C.50 (pp. 64–66) | Two-rule function, numerical conditions, sign–magnitude compatibility, partial recovery and pruning reversal. |
| Forecasts and repair | C.12 (p. 20), C.16 (p. 26), C.27 (p. 44), C.33–C.34 (pp. 49–50), C.38 (p. 54), C.43 (p. 59), C.47 (p. 63) | Recipient-informed choices, frozen repair algorithm, serial cohorts, failed timing policies and repeated serial futures. |
| Predictive theory | C.12 (p. 20), C.18–C.22 (pp. 29–34), C.24 (p. 40), C.35–C.37 (pp. 52–53), C.40 (p. 56), C.42–C.44 (pp. 58–60), C.47 (p. 63) | Response transfer, turnover, forecast limits, conditional-capacity estimates, cellular readiness and exact-state rescue. |

Methods preserve native scores, cohort definitions and uncertainty coverage. Independent blocks are distinguished from repeated reads, related cellular panels and solver resolutions. Post hoc analyses retain that designation. Missing raw archives or external models remain explicit limitations; source identities do not substitute for scientific replay.

The five supporting figures preserve measured supporting endpoints and two conceptual comparisons; all sixteen original main figures, including the continuous cellular history, remain in the main paper. Main figures are 1–16; supporting figures are S1–S5. Figure references in the methods have been updated to this arrangement. Earlier source quotations, protocol labels and chronological draft references retain their historical meaning.

Current evidence ledger · Current extended evidence · Figure correspondence · Repository availability. The v1.1 package is retained separately as a historical checkpoint.

## C. Methods and protocol specifications

Methods retained from the earlier evidence selections, followed by the dated 16 October 2026 addition in C.46–C.51. Cohorts, native endpoints and earlier contracts remain distinct. The new intake checks retained protocols, interpretations, implementation sources and completion receipts without simulation, fitting, new intervals or original audit reruns. Full raw archives for the newly selected runs were unavailable in this checkout; the selection manifest distinguishes manuscript checks from the lanes' reported verification. Unfinished outcomes are excluded.

### C.1. Tissue model and experimentally distinct records

The central tissue cases use a 16 × 16 periodic voltage sheet, with a scalar state V at each site. These are dimensionless simulated fields, not calibrated membrane potentials. The native update combines a double-well drift, a site-specific threshold field theta, four-neighbor coupling, and mean-centered Gaussian noise. With spatial mean denoted by an overbar, neighbor mean N(V), clipping to [-2, 2] denoted by C, and mean removal P(z) = z - mean(z), its two successive body updates are

$$\widetilde{V} = C\{V + \Delta t\, P[V - V^3 + a + \theta + g(N(V) - V)] + P[\sigma\xi]\}, \qquad V' = C\{\widetilde{V} + \Delta t\, P[c(R - \widetilde{V})]\}.$$

The reference comparison uses the incoming R. When writing is enabled, R then updates from the resulting body: R′ = R + rho(V′ - R). The native settings used here have time step 0.2, coupling 1, noise amplitude 0.3, reference rate rho = 0.002, and active comparison gain c = 0.2. A separate fast moving average is part of the engine state and is controlled at assay births. Mean removal precedes clipping, so it should not be described as an unconditional conservation theorem. Tissue-indexed threshold draws and declared heterogeneity provide the experimental blocks; these blocks do not represent different target families.

The repair experiments additionally write theta using

$$\theta' = \Pi_{0.3}\{(1 - 0.001)\theta + 0.001 \tanh(R')\},$$

where the supplied projection subtracts the spatial mean and rescales its standard deviation to 0.3, with a numerical floor of 10^-9. That normalization is part of the engineered writer. D099

instead holds native theta fixed and unprojected. L17 holds incoming R while writing a separate Q from the body. A variable called a record in one case is therefore a supplied parameter in another.

The native update and projection were checked in the retained voltage configuration, sheet update, reference update, and threshold projection. Exact run configurations and seed manifests remain part of the underlying campaign records; this supplement is not a standalone replacement for the full executable archive.

## C.2. Reciprocal repair: D073 and D074

**Objects and samples.** D073 uses sixteen new tissues 30024–30039; D074 uses sixteen further tissues 30040–30055. Each tissue supplies four fixed balanced half-plane messages: horizontal halves, vertical halves, and their sign reversals. Four parent replicates per message produce sixteen records per tissue. Sixteen readout starts per record produce 4096 readouts per arm; inference remains at the sixteen-tissue level.

**Preparation and damage.** Parents undergo 120 settling updates and 1000 imprinting updates with target amplitude 1.2. At fission, half the voltage body survives and fresh values fill the rest. The theta record is damaged by a norm-matched mixture at correlation 0.3. Comparisons within a parent receive the same damaged theta. All main repair arms begin with working R exactly zero; an intact-R backup is a separately labelled control. Recovery lasts 1000 updates at the native rates.

**D073 intervention.** Fixed-reader writing uses the original damaged theta to influence body development while a new theta is written. Live-loop writing additionally permits changes in theta to feed back into that body. The remaining controls include writing off, reference writing off, altered readers, scrambled surviving bodies, and the intact-reference backup. They locate routes of dependence without deleting all surviving information.

**D074 intervention.** Kept, scrambled, and donor-ranked conditions preserve the recipient fragment's exact voltage values and the fresh-cell fill. Scrambling changes their positions; donor ranking rearranges those values according to another message's body rank order. It is not a literal donor-body transplant and does not remove nonspatial recipient information. Matched damaged theta and zero initial R remain present. The donor endpoint measures donor-message recognition, not merely loss of the original message.

**Readout boundary and observer.** Only the final 256-entry float64 theta packet enters fresh, centered Gaussian bodies; V, its moving average, R, and fixed-reader copies are excluded. Theta is held throughout 1000 readout updates; R is zero and its writing/comparison are disabled. Recognition uses the fixed four-template decoder, with equal splitting of maximum-score ties. Wrong-packet and reader-off controls test content dependence. This is one record-repair episode followed by readout, not a renewing lineage.

**Inference.** Average readouts within records and records within tissues, then weight tissues equally. D073's three primary contrasts are live versus damaged, fixed versus damaged, and live versus fixed. D074's are useful repair, kept versus scrambled own-message recognition, and donor-ranked versus scrambled donor-message recognition. Each uses 10,000 paired whole-tissue bootstrap draws with percentile bounds at 0.05/6 and 1 - 0.05/6: nominal 98.33% individual intervals under the three-endpoint allocation. Qualification additionally requires at least twelve positive tissue differences. The donor criterion remains failed at eight positive tissues despite its positive interval. No new p-values or uncertainty intervals were calculated for this revision.

Sources: D073 protocol, D073 report, D074 protocol, and D074 report. Manuscript evidence notes: E09–E10.

### C.3. Acquired-reference test: D099

**Design.** Eight main tissue blocks 45000–45007 each supply two paired lives and both old/new target orderings. Eight separate qualification blocks test constructed known-old, known-new, and erased references before main evaluation. The targets are left/right and top/bottom halves; reversal swaps their roles without rotating theta or noise. This tests two patterns, not arbitrary shapes or orientation invariance.

**Acquisition and assay.** After 120 formation updates, six equally aged exposure histories cross old-only, brief-new (980 old/20 new), and sustained-new (1000 new) with enabled versus held R writing. V is imposed before each native step; R reads the resulting post-step V rather than receiving the template directly. Only terminal R enters the subsequent matched-state assay. Public direction fixes the common initial V and moving average within each ordering; native theta is unchanged. This is a within-life record-isolation diagnostic, not a hereditary generation or a multigeneration transfer claim.

Readout lasts 200 unforced updates before branching into toward-old, toward-new, and sham pulses. Pulses add $\pm 0.8$ on the 128 of 256 sites where targets differ. They do not directly change R or theta. Every branch then receives 80 unforced updates; reader gain is either 0.2 or zero. All readouts update R at 0.002, even when writing was held during exposure.

**Endpoints.** New-pattern sign fidelity is the fraction of disagreement-mask sites with the new target's sign. The duration-writing contrast is (sustained enabled - brief enabled) - (sustained held - brief held). Average equally over the two pulse directions and lives, then tissues; the prespecified primary is the smaller of the two direction means. Ten thousand whole-tissue bootstrap draws preserve all pairings and recompute this minimum for the saved 95% interval. Direction-specific and absolute-recovery intervals are pointwise secondary summaries. The large relative contrast and the separate absolute-recovery benchmark answer different questions. All 40 qualification gates passed, but the combined main criterion failed at 22/23 components because reversed acquired-new recovery averaged 79.15%, below the fixed 80% sample-mean criterion.

Sources: protocol and report. Manuscript evidence note: E08. Supplementary Figure S1 retains the original summary's intervals and tissue means.

### C.4. Repeated record replacement: L17

**Objects and samples.** Eight retained G0 tissue blocks 10000–10007 carry four translated stripe messages, each with eight paired noise replicates. The messages are distinct from the repair study's half-planes. This is outcome-informed work on the same parent panel, not fresh-tissue confirmation.

**Transmission and renewal operation.** Each generation starts with a saved message-independent naive V and moving average. A 2048-byte incoming R is held fixed while a separate Q starts at zero and updates after each body step: $Q' = Q + 0.002(V' - Q)$, for 1000 updates. Only Q crosses the next boundary and becomes the next incoming reference. No direct R-to-Q term, target label,

ancestral fragment, gain amplification, normalization, or pristine backup enters this lineage. Rules and generation-specific noise remain supplied conditions.

**Controls and endpoint.** Nine lineage conditions include reset, wrong content, reader-off, coupled writing, native-reference carry, and protected serial copying of the original acquired record. The G12 body reads Q11. A separate terminal fresh-body assay reads Q12 with held writing; it is not a thirteenth renewing descendant. The four-message observer is fixed, and reset/reader-off controls give 25% recognition. The twelve plotted generations are equal-tissue means from the saved summaries; faint lines show the eight observed tissue courses, not confidence bands.

The three predefined terminal contrasts use 10,000 paired tissue-bootstrap draws and the same three-endpoint allocation as above. Each additionally requires at least six of eight positive tissue differences. All three show five positives and therefore fail, despite positive intervals. The saved protected-copy G12 result is 100% in every tissue. The contrast identifies loss in this reconstruction procedure without showing zero information, indefinite impossibility, or a causal explanation of differences from CA architectures.

Sources: protocol and report. Manuscript evidence note: E12.

### C.5. Campaign identities and relationship to the companion v5

The companion v5 is the current cited version; its version note declares endpoints, analyses, and results unchanged from v4. References to v4 campaign identities below retain their historical meaning.

Sample size alone does not identify a campaign. A completed study in one implementation must not be substituted for an unrun proposal or a different cohort in another branch.

| Campaign | Cohort and status | Relationship and retained evidence |
|---|---|---|
| Reference acquisition, D099 | Eight fresh main tissues; eight separate qualification tissues | Later confirmation study; protocol, report, native code, and summary inspected. |
| Reciprocal repair, D073 | Sixteen fresh parent/tissue blocks; four fixed messages | Fresh follow-up; no D071/D072 pooling. Report, protocol, and saved audit summary inspected. |
| Body contribution, D074 | Sixteen further tissue blocks; same fixed message family | Separate cohort; spatial benefit qualifies, donor redirection fails. |
| Tissue replacement, L17 | Eight reused tissue blocks, four stripe messages, twelve generations | Same-panel mechanism study; saved summary plotted, raw audit not repeated. |
| Acquired carrier, D116 | Eight blocks, four lives, two supplied roles | Later instructed acquisition; only theta enters one fresh-body assay. |
| Tissue renewal, D117 | Eight blocks, four lives, two roles; three rewrites and terminal read | Later engineered read-erase-rewrite qualification; distinct from L17. |

| Campaign | Cohort and status | Relationship and retained evidence |
|---|---|---|
| Guide renewal, D132 | Eight blocks 73000-73007; one fixed guide | Later L20 chain; known material reused by D135, not independent confirmation of it. |
| Serial alignment, D135 | Eight blocks 75000-75007 mapped to D132 founders | Matched within-model intervention; report tables plotted. |
| Fresh alignment, D140 | Eight new blocks 78000-78007 | New-founder replication of D135; no guide refit or pooled cohort estimate. |
| Renewal forecasting, D141 | Sixteen untouched blocks 90000–90015; development reuses D135/D140 | All 256 forecasts sealed before outcomes; primary challenge advantage unresolved. |
| Repeated futures, D142 | D141 record–host cohort; four new futures per case | Variance characterization, no new founders; original D141 outcomes excluded from its primary estimates. |
| Crossed sources, D143 | Same cohort; four address by four other-noise histories | 1024 reused outcomes, 1536 exact aliases, 1536 novel ordinary combinations; no independent new cohort. |
| Compatible feedback, D144 | Sixteen fresh blocks 91000–91015 | One rewrite, two noise replicas; positive primary B comparison after exploratory D143 lead. |
| Serial compatible feedback, D145 | Sixteen fresh blocks 92000–92015 | Three rewrites; positive primary and secondary copy comparison, separate from D144. |
| Extra-time allocation, D146 | Sixteen fresh blocks 93000–93015 | Equal additional read/write budgets; primary allocation unresolved. |
| Contact recovery, P10R | Deterministic schedule panel, historical founders | Corrected integration method after failed P10; one common graph. |
| Editable lineage, P11 | Two founder histories, shared preparations, paired resolutions | Fresh bounded-clock preparation; three M then three E handovers. |
| Lineage robustness, P12 | Separate two-seed factor screens; three held-out seeds paired across two origins and resolutions | Newly simulated editable histories from shared audited P11 stock; fixed strong/strong selection, 3/3 control-qualified per origin. |
| Cellular return, P13 | Actual P12 final E trios, three shared challenge histories per origin | Completed instructed return to M and three further handovers; reused ancestry, ordinary origins also pass. |

| Campaign | Cohort and status | Relationship and retained evidence |
|---|---|---|
| Tissue context and isolation, P14/P15 | Preselected P13 sources and matched saved tissue states | P14 sham already succeeds; P15 separates surrounding-tissue instruction from internal communication. |
| Turnover limit, P16 | One P15-derived trio per historical origin plus ordinary-E stock, paired resolutions | Fixed rate/batch screen, two full turnovers, selected onward tests; shared ancestry, not independent population samples. |
| CA motif-lineage Stage 4 | 128 fresh founder pairs, 64 futures per history, sixteen generations | Source branch here (E21); report, results, design, and exclusion audit retained. |
| CA compact-carrier direct replication | Separate 128 founder pairs, same nominal futures/horizon | Already in v4; outcome-known reimplementation with separate estimates. |
| Wagner corrected carrier campaign | 240 rulebooks in the v4 campaign | Already in v4; separate Wagner memory implementation. |
| Wagner lineage-carrier bounded follow-up | 128 fresh gre-ideas rulebooks, 16 futures per comparison, 1,916,928 records | E22: selected full latch; retained log and audit, raw tree absent. |
| Wagner full-confirmation proposal in gre-ideas | Planned 240 Wagner/96 Boolean rulebooks; reported unrun | Unrun within gre-ideas; distinct from the completed v4 campaign. |
| Wagner renewal-frontier follow-up | Separate copying-damage and selection study | Separate log/protocol (E22): +0.130 G8 renewal advantage; selection extension fails. |
| Unprescribed development, E001 | Eight paired blocks per background, 32 founders each | Development; heterogeneous complete sequence passes, homogeneous formation fails. |
| Fresh unprescribed confirmation, E004 | 32 new blocks 1000–1031, 128 founders | Same procedure/codebase; no pilot pooling, all founders retained. |
| Formation diagnostic, E005 | All 64 E001 founder histories | Post-hoc saved-field analysis; no new trajectories or changed gates. |
| Compact renewal development, E006 | 64 known E004 founders, fresh recipients | Outcome-informed capacity development, not fresh formation confirmation. |
| Compact confirmation, E007 | 32 new blocks 2000–2031, 128 founders | Frozen 128-byte budget, paired full-field reference. |
| Exact access maps, E002 | Sixteen new rulebooks, four founder starts each | Exact finite census; exploratory rulebook-level uncertainty. |
| Response transfer, E003 | 16 development blocks 3000–3015; 32 test blocks 4000–4031 | Frozen models/predictions; fails direct-transfer comparator. |

| Campaign | Cohort and status | Relationship and retained evidence |
|---|---|---|
| Rewired transfer, E008 | 16 development blocks 5000–5015; 32 test blocks 6000–6031 | Outcome-informed new question; all four comparisons pass. |
| Repair selection, E009 | 64 new test blocks 7000–7063 | E008 model unchanged; no added benefit over source-response choice. |
| Weak-record repair, E010 | 64 new blocks 8000–8063 | Primary comparison fails; all three foreign founders in observer. |
| Crossed record strength, E011 | 64 new blocks 9000–9063 | Five of seven criteria pass; complete criterion fails. |
| Graph-informed repair, E012 | E009 records reused for fitting; 64 new test blocks 10000–10063 | First six-gate positive evaluation; no wrong-graph comparator. |
| Frozen repair confirmation, E013 | 128 new blocks 11000–11127 | Frozen E012 models; all seven gates pass, including wrong-graph control. |
| Longer renewal, E014 | 64 new blocks 12000–12063 | Sixteen rewrites, 128-/2048-byte records; specificity persists while measured reconstruction fidelity declines. |
| Functional development, E015–E019 | E015: 16 blocks; each E016–E019: 32 separate new blocks | Failed feasibility/renewal, followed by positive timing and capture comparisons; no pooling. |
| Functional serial capture, E020 | 32 new blocks 18000–18031, 128 founders, 512 nested lineages | Three intact renewals; primary and full qualification pass. |
| Damaged renewal, E021/E022 | Separate 32-block cohorts 19000–19031 and 20000–20031 | E021 serial failure; E022 one-step primary rescue, full qualification fails. |
| L20 component and forecast follow-ups, D147–D159 | Reused D146 or earlier forecasting cases | D149 causal readout improvement; subsequent prediction work remains development. |
| Finite FEP calculations and inference comparisons | Declared finite laws/policies; eight selected grid fixtures and saved beliefs | Exact or bounded theoretical results; deterministic queries are not biological samples. |

The CA numbers illustrate why this bridge matters. In Stage 4, ordinary G8/G16 fixed-form crossover is 0.95447/0.94751 and moderate-stress crossover is 0.84583/0.75830. The separate direct replication reported in v4 gives 0.9700/0.9596 and 0.8405/0.7316. Both use a 64-bit Walsh payload and 656 shared codebook bits, but these are separate estimates, not revisions of one number or independent laboratories. No-rewrite attenuates decoded coefficients by 50% before re-encoding; it does not delete half of the bits and is not a perfect-copy comparator.

The gre-ideas log's 240 tested mutated offspring belong to a selection/robustness follow-up. That count is neither a replacement for 128 independent rulebooks nor evidence that its original planned 240-rulebook confirmation ran. The separate renewal-frontier result remains report-attributed until its complete run identity and original outputs are recovered for publication.

Sources: companion v5, results unchanged from v4, CA Stage 4 report, Stage 4 structured results, CA direct-replication folder, Wagner v4 campaign folder, gre-ideas chronology, and bounded gate correction.

### C.6. Figure data and verification scope

Figure 4 presents E014's own-record, matched foreign-record and protected-copy qualified agreement at both prespecified budgets and all four post-rewrite checkpoints, with the paired own-minus-foreign differences below. The renderer reads the native summary from the existing v0.12 portable archive and verifies its completion/audit bindings. All 32 plotted means and original 95% intervals are transcribed; stored block means are checked against the summary's block arrays without rebootstrap. The extract, CSV and asset manifest retain this selection. The common upper axes span 60–80% of the 0–100% native score; lower axes report agreement-point differences. Protected-copy probes are exactly constant because checkpoint births/noise are paired across horizons. The separate completed saved-field decomposition appears in Supplementary Figure S5; it is not inferred from these aggregate scores. C.16 specifies both analyses and their provenance.

Figure 7 and Supplementary Figures S1/S2 are rendered with Matplotlib from a portable JSON extract, endpoint CSV, and generation CSV. The extract preserves source paths, SHA-256 hashes, cohort identities, tissue-level values, reported intervals, and fixed verdicts. It uses the existing D073/D074 saved-array audit summaries, D099's main summary, and L17's scientific summary. Original arrays remain at their recorded archive locations.

The extraction script reads those sources, verifies selected published endpoints and arithmetic relationships, and checks that the source files do not change during extraction. The plotting script uses only the portable extract. It performs no simulation, rebootstrap, hypothesis test, target selection, or adjustment of a failed criterion. Original summary hashes support traceability, not a claim that this revision repeated the original audits.

The longer strict-eight and observer accounts are retained in the detailed background supplement. The review response distinguishes completed editorial and presentation changes from experiments and public-release work that remain open.

For v0.8, the report-table extract and CSV retain D135/D140's rounded published estimates and pointwise 95% intervals. The update extraction script parses the named report rows and hashes their inputs; the renderer uses only this portable extract. The plot, now Figure 8, contains no synthetic observations or newly estimated uncertainty. The former Figure 7 (v0.8) drew the P11 protocol and reported stages, with report/protocol hashes; that schematic remains a preserved asset. Neither script accesses live jobs or starts experiments.

For v0.10, a separate renderer combines the same estimates (Figure 3 in v0.15/v0.16, now Figure 8) with a schematic of the existing D135/D140 mechanism. It reads the unchanged portable extract, without importing scientific modules or recalculating statistics. The implementation-inspection record identifies the guide, translation, writer, and transmission sources; the asset manifest pins the renderer, inputs, and new figure files. The original v1 plot and its data remain unchanged.

For v0.9, the four-question and collective-capacity schematics were revised using saved prompts and asset hashes. The v0.17 overview remains Figure 1 with two wording changes; the chapter schematics introduced in that version are now Figures 3, 6 and 15. The image record retains exact built-in-tool prompts and visual checks; the active manifest pins all six conceptual images. Original versions remain preserved. Figures 9 and 5 present unchanged saved D140 fields and P11 classifier trajectories; C.9 documents their selection and extraction. Figure 11 retains E041 report estimates and original intervals, with the E046 same-cohort history count, using the v0.14 extract, CSV and renderer; C.23 specifies its endpoint and cohort. Figure 7 presents reciprocal repair, while Supplementary Figures S1/S2 present reference acquisition and replacement loss, and S3/S4 retain the continuity and collective-capacity schematics. Those earlier measured assets, retained histories and supplementary images are unchanged. Figure 11 is a report transcription, not a new saved-array audit; the chapter schematics add no observations.

Figure 12 displays retained E051/E052/E053 estimates using the v0.20 renderer, selected native values, CSV and asset manifest. Its 28 displayed statistics match archived native summaries exactly. Panels A–C retain the previous numerical, fallback and policy comparisons. Panel D adds E053's reference/combined success and paired differences for the independent-noise primary and matched-noise secondary; the dashed line identifies the five-point noninferiority margin. Each study has its own 32 blocks and 512 lineages per condition. Original pointwise block intervals and qualification verdicts are unchanged, and no cross-cohort effect is estimated. C.25/C.26/C.31 specify methods. The earlier three-panel asset remains preserved.

Figure 13 presents TT008's original post hoc reliability, subgroup contribution and ranking summaries. Its v0.20 renderer checks exact equality of every summary with the prior extract, then adds count tables beneath the two calibration plots. Counts come from retained reliability rows, sum to 4,096 for each all-record model and 672 for each current-failure model, and include zeros for empty bins. The underlying estimates, bins, forecasts and outcomes are unchanged. The asset manifest binds the revised display; C.27 retains the 64-block inference unit, supplied task knowledge, operator cost and distinction between original and post hoc analyses.

Introduced for v0.19, Figure 2 presents the three main simulators using the tissue equations in C.1, functional updates and capture in C.17/C.23, and cellular state/transport in C.7. Current Figures 10 and 16 illustrate active-form tests and the discussion of embodiment. All three are explanatory diagrams rather than measured outputs. The current infographic manifest identifies the nine selected conceptual assets; original prompts and earlier versions remain preserved. The simulator-overview record documents its placement and the resulting figure-number changes.

### C.7. Cellular-state acquisition, replacement, and commitment

The cellular extensions use the reported native BITSEY revision `b2cb370aaecbc8299f259bc4251d5396c81f19a6`, with an epithelial-mesenchymal regulatory circuit coupled to ionic dynamics and junction transport. The source model is attributed to Grodstein, Mathews, and Levin (2026); the lineage assays are our program's extensions, not a reproduction of that paper's tables. Retained wrappers call the native miRNA regulatory, ionic and junction laws. The run manifest identifies 36 native Python files at the pinned revision; the integration adapter calls native `sim_slopes` while changing the clock as described below. Fixed source implementation, numerical procedure and correspondence to publication tables therefore have separate provenance.

**Primary, fixed and derived state.** The continuing cell state has 17 stored entries, not 17 independently evolving variables. The zero-based layout retained in the P11 output and component-intervention code is:

| Entries | Stored quantity | Role in these panels |
|---|---|---|
| 0–3 | Na, K, Cl, P | Primary ionic concentrations used to compute membrane voltage. |
| 4 | Snail | Supplied regulatory input, fixed at 72. |
| 5–6 | Total miR200; total Zeb mRNA | Primary RNA concentrations; their difference defines E/M/X. |
| 7 | ZEB protein | Primary regulatory concentration. |
| 8 | Free miR200 | Algebraically refreshed; the configured non-iterative approximation sets it equal to total miR200 at refresh. |
| 9–15 | `bound0` through `bound6` | Seven algebraically calculated RNA-binding entries, derived from the RNA totals. |
| 16 | `ST` | Additional source-model entry, held at zero in these panels. |

Voltage is calculated from ionic concentrations and membrane constants, not inherited as an independently assigned eighteenth entry. Classification uses the difference between total miR200 and total Zeb mRNA: E above 2, M below -2, X otherwise. This coarse classification is not a complete cell phenotype. Whole-state handoffs retain all 17 live entries. Component interventions instead replace specified primary values and apply one native algebraic refresh to entries 8–15, leaving the other entries unchanged; C.24 records its matched refresh controls. These artificial substitutions are not calibrated molecular delivery procedures.

**Retained parameterization and added operations.** The selected source option uses non-iterative RNA binding, Snail 72, effective miRNA valence $-1$ and the medium-size miRNA junction gate. P11 and the preparation/timing branch use the declared `depol` setting, expression power 60 and miRNA diffusion parameter $K_d = 1.25 \times 10^{-14}$; these are selected operating conditions, not a claim to reproduce every source-paper configuration. Native state-dependent voltage/expression gates remain active. The wrappers supply layouts, cell identities, cues, contact masks, fresh-cell placements, component substitutions and transport pauses. Retained cells continue evolving between encounters. Removal/replacement is prescribed; it does not implement cell division or endogenous scheduling. Intracellular dynamics retain state and supplied junction mechanisms transmit influence.

**Established source connection and unresolved publication mapping.** The original bounded pilot reports reproducing small-network isolation comparisons and saved neighbour-switching thresholds using the pinned implementation. Its grouped threshold run timed out; a separate recorded follow-up supplied the remaining threshold. These are historical source-output comparisons, not independent implementation or full source-paper replication. The retained mapping note identifies three publication-correspondence gaps: Table 1's stated three-dimensional geometry is not the pilot's lower-dimensional geometry; Table 2's dimensional description does not map unambiguously to the saved size sweep; and Table 3's algorithm-column order and several parameter-row labels appear inconsistent with the matching saved numerical triples. The exact publication

run/revision or an undocumented transformation has not been resolved. These observations are not an author-confirmed erratum. A table-by-table configuration/output manifest is needed before claiming exact publication replication.

The v0.18 source record binds the mapping note, pilot report, P11 manifest and selected frozen wrappers, and the later component/timing documentation in a portable package. This revision checks the retained state layout and wrapper/provenance correspondence; it does not rerun native dynamics, repeat the original biological-equation audit or resolve the publication discrepancies. The native checkout and original source-paper PDF were unavailable locally during this documentation review; the table discrepancies remain attributed to the preserved mapping note. The following campaign descriptions distinguish the later numerical changes from the retained biological laws.

For P4, P8 and P9, retained E/M requires its live endpoint label and at least 95% target-label occupancy among late-quarter samples in both final 10,000-unit isolated holds. All donors are removed before those recipient holds; junction transport and neighbor reading are disabled during isolation. The prepared bath and model parameters remain supplied. Native names such as depol and hypol describe source configurations, not calibrated voltage clamps. Expression power scales a state-dependent factor rather than imposing constant conductance.

**P8 writing and transfer.** Four three-cell E backgrounds include ordinary fresh cells, isolated age-matched cells, and actual P5 round-six carriers from two ancestries. Six M cues contact the three targets on a 3x3 sheet for 20,000 units, under depol, power 60 and miRNA Kd 1.25e-14. After cue removal and two isolated holds, the three actual targets contact one fresh E recipient on a 2x2 sheet, followed by target removal and two recipient holds. Writing- and probe-stage RNA/boundary blocks are distinct interventions. The 61-case panel contains historical replays, age preparation, and 28 new conditions with finer counterparts; reported numerical agreement is not independent cohort replication. Historical P5 ancestry remains a fixed input.

**P9 complete replacement.** Founders are the actual three retained-M P8 targets before the one-recipient probe, with ordinary M and E-sham comparators. The same teachers contact one pupil at a time on a 2x2 sheet for three successive 20,000-unit intervals. Teacher live states pass directly between encounters; previously exposed waiting pupils continue in isolation. All teachers are removed, and all three pupils undergo two 10,000-unit holds before becoming the next teachers. The branch stops after its first incomplete retained-M round or six successful rounds, without selecting pupils or supplementing teachers. Simultaneous controls use a 2x3 sheet and 20,000 or 60,000 contact units; total round duration is 80,000 throughout. Geometry and pupil histories therefore differ alongside scheduling. The 34 cases comprise 16 new branches, their finer counterparts and two exact historical replays. Finer runs refine new phases, not the entire P8/P5 ancestry.

**Commitment and component intervention.** Selected receiver checkpoints have six/eight-contact prefix ages of 110/150 units. Both are M-labelled. The factorial chooses each of three blocks from either checkpoint: ions Na/K/Cl/P, the paired RNA totals, and ZEB protein. Derived RNA species are recomputed without changing the chosen primary values. Eight combinations at two numerical settings yield 16 branches. Actual held pairs enter two fresh-pair relays; only unchallenged recipients continue, while shadow and 20%-opposite-regulator perturbation forks test local recovery. Commitment requires retention plus target acquisition, retention and recovery in both re-

placements. These are synthetic component interventions on reused histories, with unequal source ages, not independent contact-dose trials or physically validated molecular substitutions.

**H4b environmental comparison.** Four fixed new PCG64 streams cross four environmental profiles, two founder polarities and four independently enabled parent/cue policies, giving 128 main cases. Eight finer cases and eight historical controls bring the count to 144. Each trajectory spans twelve epochs; primary target matching averages epochs 2-12. Paired random draws govern world changes, cue availability and cue error. Neither hidden truth nor the score is supplied to the cellular trajectory rule. Source ancestry uses the earlier stabilized-M and externally pooled rewritten-E pairs, not P8/P9 descendants. Costs are assigned parent/cue contact penalties. The prespecified binary reference uses only visible history; its checkpoint agreement is an explanatory comparison, not another native simulation or a biological fitness measure.

**P10 failure and P10R correction.** The original adaptive Euler loop could overshoot phase endpoints near equilibrium. P10 retained six invalid single-sample time axes and ten large overshoots despite agreement on tested categorical labels. P10R clips steps to observation and phase boundaries: 100-unit sampling includes zero and the exact endpoint; maximum steps are 100 coarse and 50 fine, with finer base steps/increments and complete proposed/accepted step traces. This changes integration and observation, not the biological equations. P10 remains failed. Older time axes need separate validation; the defect is not evidence that every prior biological conclusion is false.

P10R compares schedules on a six-cell, eight-edge common graph with a common pupil clock. All cells evolve while only contact masks change. Sequential and 50%-overlapping schedules qualify in the screen and continue actual first-round pupils through six full replacements; RNA/cut controls branch at round two. Other qualifying schedules are not automatically extended. Equal contact duration is not equal flux. Historical founder and reference stocks remain inputs, so numerical refinement does not rebuild their full ancestry.

**P11 preparation and continuous lineage.** At each resolution, new stocks come from the native 13x13 reference preparation with a fixed 4x4 M seed and a 1000-unit preparation, entirely under the bounded clock. Fixed E cell 168 and M cell 1 of the new endpoint initialize external stocks; no historical final state seeds the scientific chain. Six temporary cues write three targets on the P8 3x3 layout for 20,000 units, followed by cue removal and two 10,000-unit isolated holds. The time-matched ordinary-M comparison ages without acquisition.

Each handover uses the common graph: sequential teacher-pupil contacts at 0-20,000, 20,000-40,000 and 40,000-60,000 within an 80,000-unit window. Waiting cells keep evolving. Every teacher is removed, and all three pupils receive two further 10,000-unit isolated holds. Each must have the target endpoint label and at least 95% target occupancy among the 26 late samples in each hold. No successful-subset selection or teacher supplementation is allowed. Three M handovers precede E-cue rewriting of the actual current carriers; three E handovers into fresh M pupils follow. Logical lineage time is 680,000 units plus the separate stock preparation, with three handovers on each side, not six on each side.

Stage-specific RNA/boundary blocks, same-state shams, no-cue controls, and ordinary-founder calibrations distinguish acquisition, retention and transfer. All five gates reportedly pass in 74 deterministic cases and 37 coarse/fine pairs. Shared preparations, counterfactual prefixes, and numerical pairs do not estimate independent population success. All 17 live state entries continue

across stages. Fresh cells, cue sources, schedules, and resources are external; no cell division is simulated. Numerical agreement does not prove convergence.

Campaign-wide outcomes remain attributed to retained reports. The original inclusion review and source snapshot are preserved; the v0.8 snapshot records the new report/protocol selection separately. C.9 describes the subsequent v0.10 check and extraction of one retained P11 chain, without rerunning the full campaign audit. P12 is included separately in C.14 and E37, without changing the P10R/P11 campaign identities.

Sources: P4 protocol, P8 protocol, P9 protocol, component protocol, H4b protocol, and source-paper mapping.

Updated methods: P10 failure, P10R protocol, P11 protocol, and P11 results.

## C.8. L20 acquired records and serial writer alignment

**Targets and acquisition.** D116/D117 and the later guide series use two balanced supplied recipes: a horizontal half-plane and vertical four-column bands, disagreeing at 128 of 256 sites. They are not D099's two half-plane orientations or L17's four stripe translations. Instructed founders settle for 120 updates and undergo 1000 writing updates with the role pattern imposed at amplitude 1.2. Only theta crosses the acquisition boundary. The replacement of native theta and mean-centered dynamics cancel some nominal parameter differences; eight blocks must not be presented as eight distinct effective physics families.

**D117 reconstruction.** Each descendant begins with a fresh message-independent V, moving average, and R. It reads the incoming 2048-byte theta for 300 steps with reference feedback and theta writing disabled, then loses those bytes. During 1000 writing steps, theta begins at zero and updates from the descendant's working R using the projection in C.1. Only the resulting theta is inherited. Three such operations precede a fourth fresh, non-writing terminal read. The first write satisfies $\mathrm{Pi}(0.001\tanh(\mathrm{R})) = \mathrm{Pi}(\tanh(\mathrm{R}))$ outside the numerical floor, so normalized feedback begins at full strength. This mechanism cannot be described as gradual amplification of theta amplitude. Exact-copy, disabled-reading/writing, wrong-content, shuffle, and externally assisted G1-rescue conditions qualify the finite route; algebraic aliases remain dependent controls.

**Guide fitting and transformation.** D131 fits one public guide from 16 separate development records, eight per supplied family, already in a supplied common coordinate frame. Each template is the arithmetic mean of its family's eight fields. The two 256-entry float64 templates occupy 4096 bytes. They remain fixed and shared outside the lineage payload. Let $p_g$ be the incoming 16 by 16 field, $G_f$ a template, and $T_s$ the periodic translation defined by $(T_s x)_{ij} = x_{i-s_1,j-s_2}$, with indices modulo 16. The selector computes normalized correlations across both families and all 256 translations:

$$c_{f,s} = \mathrm{corr}(p_g, T_s G_f), \qquad \tilde{p}_g = T_{-\hat{s}} p_g.$$

The selected family has the larger maximum correlation. If the two family maxima differ by at most $10^{-12}$, the selector abstains and leaves the packet unchanged. Otherwise it chooses the lexicographically first shift within $10^{-12}$ of the winning family's maximum and applies its inverse to the original field. A constant field also produces an unchanged packet. Centering and standardization are used for scoring only: the output is a rigid permutation of the incoming values, without

reprojection, template substitution, or blending. The selector receives packet and guide bytes, not the target label, historical displacement, source body, or outcome. The inferred family and shift are not inherited.

**Reading and removal of packet access.** D135 maps blocks 75000-75007 to D132 source founders 73000-73007; D140 generates eight new founder blocks 78000-78007. Each has four lives and both roles. The aligned and translated chains start from the same translated F0 packet, with the same guide, dynamics, phase lengths, and matched per-phase random streams. Each descendant starts V as Gaussian noise with scale 0.2 and initializes its moving average and R to V, independently of the preceding body. It reads the selected field for 300 updates with theta writing and reference-to-body comparison disabled. Before writing, packet access is removed and the writer's theta is initialized to zero. The complete read-end (V, moving average, R) continues into writing. Thus packet erasure does not erase its effects in the developing body/reference; the guide can influence those states during reading but is not an input to the writer.

**Writer-alignment intervention.** During 1000 writing updates, both arms use rho 0.002, comparison gain 0.2, noise 0.3, theta rate 0.001, and the projection defined in C.1. Only the placement of applied theta differs:

$$\theta_t^{\text{applied}} = \begin{cases} \theta_t & \text{aligned,} \\ T_{\delta_g}\theta_t & \text{translated,} \end{cases} \qquad \theta_0 = 0,$$

$$\theta_{t+1} = \Pi_{0.3}\{0.999\theta_t + 0.001\tanh(R_{t+1})\}.$$

The body and reference first update using the applied field; the new reference then updates the writer's theta. For each block, life, and generation, $\delta_g$ is sampled uniformly from the 255 nonidentity translations in $\{0,\dots,15\}^2$, excluding $(0,0)$, and held fixed throughout that writing phase. This is a periodic displacement of feedback relative to the body, distinct from the guide's correlation-selected inverse translation before reading. Translating a field preserves its values and norm; the two arms' evolving fields need not remain identical. Random streams are experimentally matched, not transmitted as lineage state.

**New packet and body reset.** Each ordinary outgoing packet is $p_{g+1} = \Pi_{0.3}[\tanh(R_{1000})]$, serialized as 256 little-endian float64 values (2048 bytes). It is derived from the writer's final reference, not its final latent theta; D117 instead transmits latent theta. No previous V, working R, moving average, writer theta, selector history, family label, or random-generator state enters the next fresh body. Both arms retain the same public guide, while each arm's newly derived packet becomes its own next input. Later starting bodies and records are allowed to diverge through transmission; they are not forced back to saved common states. Three successive rewrites in G1-G3 precede the fresh, non-writing G4 terminal assay.

**Observer and score scale.** The implementation classifies $V_i > 0$ as positive and $V_i \le 0$ as nonpositive. Let $u_i$ and $a_i$ be the corresponding binary sign-agreement indicators for the inherited and alternative targets. The qualified and sign-only preferences are

$$P_q(V) = \frac{100}{256}\sum_{i=1}^{256} \mathbf{1}\{|V_i| > 0.2\}(u_i - a_i),$$

$$P_s(V) = \frac{100}{256}\sum_{i=1}^{256} (u_i - a_i).$$

The amplitude threshold is strict: sites with $|V_i| = 0.2$ receive no qualified credit. Neither score renormalizes over selected sites. At the 128 positions where the two targets agree, $u_i = a_i$, so those terms cancel. At each of the other 128 positions a qualified term is -1, 0, or +1 before multiplication by 100/256. Consequently both scores range from -50 to +50 points. This is an algebraic bound from the targets and scoring, not a new analysis of trajectories. An aligned-minus-translated difference between two scores can range from -100 to +100 points. Neither a preference score nor its treatment contrast is ordinary own-pattern accuracy. The prespecified qualified primary measures relative target expression while withholding credit from sites below the amplitude criterion.

**Retained sign-only results.** The D135 report gives terminal pooled aligned-minus-translated sign-only preference as +4.63867 [1.86768, 7.44629] points; D140 reports +3.11 [1.92, 4.39]. The main text reports the D140 contrast as +3.11 [1.92, 4.39]. These are existing secondary results, calculated on the same terminal bodies, with pointwise unadjusted 95% whole-block bootstrap intervals. The scoring code changes only the amplitude mask between observers. Positive sign-only contrasts show that the direction is not confined to threshold-qualified scoring. They do not establish equivalence between the two effects, partition the qualified effect into causal components, or rule out amplitude-dependent changes in the dynamics. No new bootstrap or sign/amplitude interaction test was performed. Report and implementation hashes are retained in the preference inspection record.

**Controls and inference.** Protected copying retains the imperfect F0 bytes exactly. Disabled writing emits zeros; it is not the copy comparator. Disabled reading applies at G1-G3, with terminal reading enabled. Wrong content begins from the opposite role; lesion/rescue share a shuffle, and rescue alone receives an extra intact 2048-byte backup at G1 step 101. Nothing carries that backup onward. The primary is terminal aligned-minus-translated qualified preference, averaging roles within lives and lives within each of eight blocks. Ten thousand paired whole-block percentile bootstrap draws give the reported 95% interval; qualification also requires positive role means and the prespecified renewal controls. Role-specific, copying, stage, and sign-only comparisons are secondary, pointwise, and unadjusted. No 80% fidelity threshold or equivalence margin was used.

Sources: D116 report, D117 protocol, D131 guide, D135 protocol, and D140 protocol. The v0.10 clarification also checks guide fitting/selection, translation, reading/writing, D135 transmission, and D140 transmission. Evidence entries E27-E30 and their verdicts are unchanged.

## C.9. Retained reconstruction histories

**Selection and scope.** The editorial selection record was saved on 17 September 2026, before inspection of the selected raw fields and before the P11 archive became available locally. Aggregate campaign results were already known. This is an editorial selection rule, not an experimental preregistration. Figure 9 takes the smallest registered D140 block identifier, 78000, and first life, 0, retaining both supplied roles and both aligned/translated arms. Figure 5 takes P11's registered

written-founder/open-route chain with M-open acquisition and E-open rewriting at the finer numerical resolution. Neither selection searches for a visually favorable history. The figures are descriptive illustrations of previously reported campaigns; they supply no additional independent replicates, effect estimates, or uncertainty intervals.

**D140 fields and transmission.** The completed archive is `l20_fresh_founders_v1_attempt1`. The extractor checks the selected block JSON/NPZ against the original output hash manifest, and the manifest against the completion record. It extracts the shared F0 packet, read-end V at 300 updates, writer-end V and R at 1000 updates, outgoing packet, and terminal V after 300 fresh reading updates. Inherited fields are decoded directly from 2048 bytes as 256 little-endian float64 values. It verifies identical F0 bytes and first read-end bodies between arms within each role, the saved input/output packet hashes, and exact equality between each outgoing packet and the next incoming packet, including the terminal input. Only the plotted selection is inspected; the campaign bootstrap is not repeated.

The portable arrays and metadata retain the selected states, byte packets, indices, source paths and hashes. The renderer displays all 44 selected tiles in stored coordinates using nearest-neighbor pixel display. It performs no additional translation, shape registration, smoothing, or per-tile normalization. One symmetric color scale spans all displayed V fields, and a separate symmetric scale spans the initial/outgoing record fields; no values are clipped. This preserves amplitude differences within each quantity without treating record and body values as the same variable. Guide routing remains part of the simulated read operation, not a display transformation.

**P11 trajectories and identities.** The retained archive is `paper-editable-lineage-20260915T125104Z-teji03a9`. The selected cases are `write_M__M_open__fine`, `before__written__open__fine`, `rewrite_E__written__E_open__fine`, and `after__written__E_open__open__fine`. Their 42 phase arrays cover logical lineage time 0–680,000, excluding the separate 1000-unit stock preparation. The extractor verifies four result-file hashes against the saved summary, all 42 phase-file hashes against those results, stage-audit hashes, and the relevant frozen protocol/implementation hashes. It checks saved times at 100-unit intervals, a maximum integration step of 50, first/last snapshots against saved live endpoints, and final labels against the recorded classifier. Across the selected phases, 177 same-cell handoffs preserve all 17 live state entries exactly. Three explicit cross-case checks confirm founder-to-relay, actual R3-to-rewrite, and rewritten-R3-to-onward-teacher continuity. These are extraction/integrity checks, not a rerun of the complete 27,610-check campaign audit.

The portable P11 arrays retain all saved 17-entry states in the selected phases, including cues and teachers; the metadata records cell IDs, phases, time offsets and hashes. The renderer plots all saved samples of total miR200 minus total Zeb mRNA for the three targets/recipients. No cells are averaged or omitted from those groups. Separate panels mark replacement identities; color/style labels 00–02 repeat within each group and do not identify one persistent cell across the full lineage. The R3 recipients alone continue into the rewriting panel as the same cells. Plotting lines join samples within a cell without smoothing or estimating unsaved crossing times. A common linear vertical range covers all plotted values without clipping. E is above $+2$, M below $-2$, and X includes both thresholds.

Each relay round has three successive 20,000-unit recipient-contact intervals, a 20,000-unit off interval, teacher removal, and two 10,000-unit isolated holds. Writing/rewriting instead has a 20,000-

unit cue interval followed by cue removal and the same two holds. Darker shading marks the last quarter of each hold, where the original retention criterion requires at least 95% target labels and the target label at the endpoint, in both holds. The plot does not newly qualify other histories or controls. Native model units are retained, and E/M labels remain distinct from biological diagnoses.

The two figure asset manifests pin the portable inputs, metadata, renderers, PNGs and SVGs: D140 manifest and P11 manifest. Re-rendering needs only the portable extracts; re-extraction needs the corresponding retained archives. No scientific model is imported or executed by either workflow.

### C.10. Unprescribed formation and compact renewal: E001/E004–E007

**Dynamics and formation.** The paper-local implementation uses the dimensionless body/reference equations of C.1 with a = 0, step 0.2, four-neighbor coupling one, noise amplitude 0.3, reference rate 0.002, and projected theta writing at rate 0.001. The moving average updates as EMA′ = 0.9 EMA + 0.1 U before reference feedback. No target image, label, reward, guide, or template enters formation/capture. Birth V is centered Gaussian with standard-deviation parameter 0.2; EMA and R initially equal V. Formation lasts 1000 updates with feedback gain zero and theta held. Theta is zero in the homogeneous pilot or a block-shared Gaussian field projected to standard deviation 0.3 in the heterogeneous condition. Four founders share this field but have independent birth/noise streams.

The offline prototype F averages V over updates 901–1000. Individual formation requires both signs on at least 10% of sites, at least 80% of sites with $|F| > 0.2$, cardinal-neighbor sign agreement at least 0.65, and sign agreement at least 0.80 between the two half-window averages. The cohort gate requires at least 75% of founders to pass and mean within-block pairwise sign disagreement above 0.10. All founders, including failures and similar endpoints, remain in later scoring.

**Capture and information boundary.** Remove the formation background at update 1000. Matched V/EMA/R histories branch into writing-enabled and writing-held capture, both starting theta at zero. Capture lasts 1000 updates with reference feedback 0.2; only V→R→theta supplies record content. The founder prototype is retained solely for observation. Descendant births use independent centered Gaussian states with EMA = R = V. Only a little-endian row-major float64 theta packet enters ordinary descendants; founder V, EMA, R, background, prototype, labels, and RNG state do not cross. Shared code, coordinates, and schedule remain supplied.

Each of three descendants reads held theta for 300 steps with reference feedback zero, then erases theta and writes for 1000 steps with feedback 0.2. A fourth fresh body reads the third output for 300 nonwriting steps. Two recipient lineages per founder use independent futures, paired across arms and capacities. Controls start from an own record, the next founder's record within the same block, an outcome-independent shuffled own record, zero/held-capture record, or the unlearned background. Further controls disable inherited reading during each nonterminal read, disable writing after erasure, preserve the exact original packet, or flip 25% of body signs at read step 150 in every generation including the terminal probe. Read-off's terminal probe uses a normal reader. Held-capture and erased packets are exact aliases and are not independent corroboration.

**Scores and inference.** Qualified Q is the percentage of all 256 sites with matching prototype sign and $|V| > 0.2$, without alignment. Primary own-minus-foreign compares Q against the same own prototype across differently initialized packet lineages. Sign-only agreement and within-body own-

minus-mean-other-founder preference are secondary. Average the two recipient futures and then four founders within each block; use 10,000 whole-block percentile bootstrap draws. Formation, G1 copy-minus-erased readability, positive own-minus-foreign lower bounds at G1/G2/G3/terminal, and terminal dependencies on reading/writing form the complete sequence criterion. This is not a criterion of perfect reconstruction or equality to copying.

E001 uses eight blocks per background and exploratory pointwise 95% intervals. E004 uses 32 fresh heterogeneous blocks 1000–1031 with unchanged scoring and 95% intervals, without pilot pooling. The reused scorer's legacy "pilot" label in its detailed report does not redefine the cohort. E005 uses saved formation quarters: a translation fitted on Q1/Q3 is evaluated on Q2/Q4, alongside foreign-pattern alignment and an optimistic full half-window search. Those diagnostic outcomes do not change formation gates and do not uniquely separate deformation, amplitude fluctuation, and motion.

**Compact encoding.** E006 uses 64 known E004 founders with fresh recipients. Fixed spatial blocks of side 1, 2, or 4 yield respectively 256, 64, or 16 float64 means: 2048, 512, or 128 bytes. Decoding repeats each value over its original block with no post-decoding normalization. Encode capture and every subsequent output at the same budget; protected copying retains identical original bytes at that budget. E006's two compressed primary tests use 97.5% intervals, with 95% full-field and secondary intervals. E007 preselects 128 bytes for 32 fresh blocks 2000–2031, paired with 2048 bytes; its primary intervals are 95%, with 97.5% reported as sensitivity information. The complete criterion must pass at both budgets. Compression changes amplitude and resolution together; it does not isolate abstract information capacity.

Sources: E001 protocol, E004 protocol, E005 protocol, E006 protocol, and E007 protocol. Evidence notes E31–E32 give completed results and report links; frozen copies are in the v0.11 source-record archive.

### C.11. Exact finite accessibility: E002

Sixteen independently sampled Gaussian ten-gene matrices are normalized to unit row L2 norm. A sweep updates genes sequentially 0–9, setting positive/negative local fields to $\pm 1$ and retaining the bit at exact zero. All 1024 states are enumerated, with complete successor tables, recurrent cycles, basins, and transient lengths; no finite trajectory censor substitutes for cycle detection.

Four unselected founders per rulebook run for 64 native sweeps. The supplied writer averages gene activity over sweeps 49–64. A public outcome-independent address permutation selects two or five entries, whose values form the record. A gain of 0.25 or 0.5 applies an additive field at those addresses during each update. The fixed-record maps compare own, next-founder foreign, address-shuffled, and neutral records; erased/read-off conditions are exact neutral aliases. No full-expression overwrite latch or founder attractor label is given to the writer.

Native-orbit access is the fraction of uniformly weighted initial states entering the same recurrent orbit by sixteen sweeps. The whole original orbit must persist in the conditional map; a destroyed orbit has zero access even if an old state is visited transiently. Exact cycle canonicalization distinguishes basin redistribution from new/lost destinations. The study retains 784 maps, with related conditions dependent within each rulebook. Exploratory 10,000-draw 95% intervals resample the sixteen whole rulebooks. These statistics do not turn the exact finite census into a held-out forecasting experiment or a claim about unsampled network families.

Sources: protocol, report, and evidence note E33.

### C.12. Response transfer and externally supplied repair: E003/E008–E012

**Embodiments and prediction boundary.** Source A uses the 16 × 16 cardinal-neighbor tissue and heterogeneous founder procedure in C.10. E003's B uses all four cardinal and four diagonal neighbors with weight 1/8. E008's B starts from the 256-node/512-edge native graph and replaces exactly 128 native edges through 64 degree-preserving double-edge swaps. Each new link has toroidal Manhattan length at least four; connectedness, degree four, and total coupling one remain fixed. Each block has a new graph shared by its four founders. E009–E012 use that same rewired-body family. Founders form and capture in A; none of these assays establishes native B formation.

Each fresh body reads a held 2048-byte packet for 300 steps, without rewriting. Eight recipient futures per founder/body have independent A/B births and noise, with matched streams across interventions within a body. Development interventions are isolated body-sign damage at step 150, foreign-record mixture, or record deletion; the fresh test crosses damage 0.25/0.50 with mixture/deletion severity 0.25/0.75, giving eight joint conditions. Masks are fixed without outcomes; packet mixtures/deletions are not renormalized. Intact/foreign/erased references are excluded from primary forecast loss.

The target is the pair of changes in qualified own and cyclic-foreign agreement relative to matched intact B. A's actual test responses are permitted inputs. B's intact and perturbed outcomes are inaccessible until all forecasts are frozen, then serve evaluation only. The five forecasters are zero, direct A-response transfer, intervention-only ridge, appearance ridge, and appearance-plus-response ridge. Interventions contribute severities, squares, and pairwise products; appearance adds fixed founder/intact-A summaries and descriptor interactions; response adds A's changes in own agreement, foreign agreement, resolved fraction, and neighbor agreement. Regressions use development RMS scaling, no intercept, fixed penalty 0.01, and clipping to [-1, 1] in fractional score units. There is no tuning on test outcomes.

Each transfer study uses sixteen development blocks and 32 fresh test blocks. Primary block MSE averages four founders, eight interventions, and two coordinates. Ten thousand whole-block percentile draws give 95% baseline-minus-response-map MSE intervals; all four lower bounds must exceed zero. Displayed RMSE is converted to score points, while MSE contrasts are in squared points. E008 adds a separate two-point practical tolerance for direct transfer. Its learned predictor does not receive graph features. E003's failed direct-transfer comparator remains a failed full qualification.

**Repair menu and observers.** E009 uses the frozen E008 predictor on a new 64-block cohort. Erase 192 of 256 packet entries; each candidate restores 64 exact values from an external clean backup, leaving 128 entries zero. The shuffled eight-option menu contains the strongest-value subset and seven distinct uniform subsets. All options have a 512-byte restored-value budget; this does not account for supplied backup/probe/observer costs. Native probes and prototypes inform selection, but fresh B outcomes cannot. E009's utility is recovery of own-minus-one-foreign Q relative to unrepaired; the map must improve over source-based, strongest, and uniform choices under its declared conjunction.

E010 uses 64 new blocks and compares the weakest, strongest, and random 64 erased entries. Its primary specificity is own Q minus mean Q for all three other founders. E011 uses 64 further

blocks and holds each chosen subset fixed while applying either original values or original signs at magnitude 0.15/0.40. Equal magnitudes do not mean equal signed sums, complete-field RMS, or physical energy. Each study averages eight recipient futures and four founders within blocks and uses 10,000 whole-block 95% intervals. E010's four gates and E011's seven gates require all stated contrasts to pass; a positive component or secondary foreign reference does not change a failed conjunction.

**E012 graph-informed choice.** All 64 archived E009 blocks are explicitly development data. Their raw endpoints were rescored against all three foreign founders by that experiment, producing 2048 candidate rows; E010/E011 contribute no fit rows. The native model has seventeen features from source readouts, applied-record geometry/amplitude, prototype relationships, and native coupling. Six extra graph features measure the difference between target and native neighbor operators on the record and source terminal V, its projection onto own/foreign prototype signs, and changes in smooth agreement after one algebraic coupling increment. The smooth surrogate is sigmoid$((sV - 0.2)/0.2)$, with prototype sign s. These features do not simulate a target future.

Center features and target within each eight-option founder menu. Scale by training RMS, replacing values below 10^-12 by one, and fit ridge without intercept:

$$\beta = (Z^\top Z + n\,0.01I)^{-1} Z^\top y.$$

The target y is centered B specificity minus centered A specificity; add the predicted correction to A specificity. Both models use the same development rows and fixed penalty, without clipping, refitting, feature selection, or test-data tuning. The graph model adds predictors as well as target-related information; E012 alone does not isolate correct graph identity from that added capacity.

On 64 fresh blocks 10000–10063, freeze all source observations, graphs, menus, features, and policy choices before B readout. Policies are graph model, native model, greatest source specificity, strongest entries, and exact mean utility of uniform menu selection. Ties select the lowest shuffled menu index. The primary six-gate conjunction requires graph choice to beat all four alternatives and to have positive specificity and own-agreement recovery. All 95% whole-block lower bounds must exceed zero; these are individual contrast intervals, not simultaneous bands. Noisy hindsight-best gaps are optimistic diagnostics, not deployable policies or optimality proofs.

**Later confirmation.** E013 uses 128 fresh blocks, both E012 models unchanged, and one independent wrong graph supplied to the identical graph model while holding the actual simulated target fixed. Its six-gate replication and additional correct-versus-wrong graph comparison pass in the v0.12 selection (C.16; E42). The completed repair studies use one fresh reading with externally supplied repairs; they are not a continuation of the multigeneration lineage in C.10.

Sources: E003, E008, E009, E010, E011, E012, and E013 prospective protocol. Evidence notes E34–E36 link the reports.

## C.13. v0.11 source selection and checking scope

The v0.11 source record names nineteen completed run phases from E001–E012 and thirteen selected report records, including E012's development report. Technical benchmarks are excluded. Every selected REPORT/summary hash matches its COMPLETE record; every COMPLETE hash matches the associated recorded passing audit. These are editorial consistency checks, not a rerun

of those audits, the simulations, or their statistical analyses. The audit term "independent" denotes separate checking within the same human-directed program, not an independent laboratory.

The portable source-record archive retains frozen input code/protocol copies, reports, summaries, completion and audit records, and selected prediction/model freezes. The source record hashes every included file. Raw trajectory and endpoint arrays remain in the original experiment archives and are not included in this portable text/metadata package. E013 contributed only its protocol to that historical selection. The v0.12 additions have a separate source record and package; they do not rewrite the v0.11 selection.

### C.14. Cellular P12: perturbations, selection, and editable-lineage controls

**Model and numerical boundary.** P12 retains BITSEY revision b2cb370aaecbc8299f259bc4251d5396c81f19a6, the native biological equations, six-cell/eight-edge relay graph, six-cue/three-target writer, power 60, and Kd 1.25e−14. It uses the P10R bounded explicit-Euler method described in C.7, with maximum steps 100/50; the finer configuration also halves base steps and adaptive increments. Samples every 100 logical units include the initial and exact final state. Every cell continues evolving while waiting. Whole 17-component states and identities are retained, rather than reconstructed from E/M labels. Coarse/fine agreement is a numerical check, not proof of convergence or validated biological time calibration.

**Perturbations.** Timing changes spacing only. Three nonnegative gaps occur before recipient 0 and between the successive contacts, in fixed order 0, 1, 2. Draws use a 100-unit grid from 0–2000 (mild) or 0–5000 (strong). Each pupil receives 20,000 open-contact units. The final idle interval is 20,000 minus the sum of the gaps, preserving the 80,000-unit window. Two subsequent isolated 10,000-unit holds make each handover 100,000 units. Contact order, duration, overlap pattern, and total window are not randomized.

Incoming-cell variation multiplies only ZEB protein (state index 7) by cell-specific factors using integer percentage changes from −10 to +10 (mild) or −25 to +25 (strong). The other sixteen components are unchanged. Initial fresh founder targets and fresh recipient trios vary; existing carriers and the six writing cues do not. ZEB protein is not one of the two RNA totals used in the E/M/X classifier. Starting labels are checked without selecting or redrawing cells. This is initial within-state variation, not genetic diversity or continuing molecular noise.

**Draws, screen, and untouched comparison.** The frozen draw file derives each word from the first sixteen hexadecimal digits of SHA256 of `P12_lineage_robustness_v1|SEED|DOMAIN|INDEX`; integer mapping places words on the declared uniform grids. Domains distinguish six timing rounds and founder/recipient protein draws. Common words pair strengths, directions, origins, routes, and numerical resolutions. Screen seeds are 101/102; combined-test seeds 201/202/203 are held out from the screens.

Four zero-stress baseline cases replay P11 handovers with exact full-state agreement. Separate timing and protein screens each use 32 cases: both levels and labels, two open-route seeds and two resolutions, plus matched RNA and boundary-cut controls at seed 101. A level qualifies only when both open seeds in both directions and resolutions succeed and all corresponding controls remain opposite-state. For each factor select strong if qualified, otherwise mild, otherwise none. Failure of both levels for either factor precludes combined tests; the protein screen still runs if timing fails. The protocol permits no extra draws, retuning, or retries. Both factors qualify strong.

The twelve combined cases pair three seeds, written/ordinary-M origins, and two resolutions. They use shared audited P11 stock but newly simulate all dynamic lineage phases: M writing or time-matched ordinary-M aging, three M handovers, temporary E writing of the actual current carriers, and three E handovers into initially M cells. Each complete chain spans 680,000 new logical units. It does not start from a historical dynamic lineage endpoint. A failed chain stops at its first incomplete block and remains in the denominator.

**Qualification and controls.** E/M/X classification uses total miR200 minus total Zeb mRNA: E above +2, M below −2, X otherwise. Every cue or teacher is removed at the corresponding boundary, and all three pupils are retained. Each of the two isolated 10,000-unit holds requires the desired endpoint and at least 95% of its 26 samples at or after 7500 in that state. Each recipient must still be opposite-state immediately before its first scheduled signal, excluding spontaneous precontact switching as teaching.

Seventy-eight controls branch from exact available live prefixes: RNA and boundary-cut controls for the first M and first E handovers, M-sham and no-cue controls for E rewriting, plus a no-cue initial-founder control for each written trial. Missing prefixes would be explicit structural skips and could not qualify a chain. All required controls are available and negative. Raw and control-qualified complete-chain counts are each 3/3 for each origin. The same three challenge words pair the histories; shared preparation and numerical duplication preclude treating six histories or twelve numerical chains as independent biological replicates. No acquired-history advantage is observed.

**Provenance and verification.** Run paper-lineage-robustness-20260916T094348Z-quf8m72y completed on 17 September 2026. The saved final audit reports 158 cases, 1746 integrations, 110,444 checks, seven passing integrity flags, and 79 agreeing finer comparisons; prior software verification reported 989 passing tests. These counts describe implementation checks, not additional scientific samples. The preceding editorial review verified all 158 case-result and 1746 raw-phase hashes, 99 frozen files, 99 current local execution/protocol/configuration files, and five stage audits. Incorporation after the repository sync reconfirmed selected report/protocol/summary/audit hashes. No simulation, numerical rescoring, or full mathematical audit was rerun. The source package retains reports, frozen files, summaries, and recorded audits; raw arrays, full P11 ancestry, and external dependencies remain in their original archives. The audit is separately implemented checking within this human-directed program. Source-paper table/geometry correspondence remains unresolved.

**Subsequent continuation.** Cellular P13 applies temporary M cues to the actual P12 final E trios, then completes three more full M handovers with new challenge draws. It reuses the full states and identities of the existing descendants. The return label is supplied; success does not establish hidden ancestral recall, exact recovery of ancestral state, or an independent replication of the prior six handovers. C.18 and E47 record the completed v0.12 addition. It is distinct from paper-local E013's predictor/wiring comparison.

Sources: P12 protocol, P12 report, P12 source record, portable results, earlier static integrity check, and prospective cellular P13 protocol.

## C.15. L20 extension: forecasting, geometry, and reconstruction time

**Shared apparatus and endpoints.** D141–D146 retain the tissue dynamics and the fixed D131 guide described in C.1/C.8. The 2048-byte little-endian float64 record is lineage state; the 4096-byte guide and host parameters are shared apparatus, unchanged across the interventions. Each or-

dinary replacement routes the incoming record with the guide, initializes fresh V/EMA/R, reads for 300 updates, removes direct incoming-packet access, resets writer theta to zero, and writes for 1000 updates. The guide has no direct writer input; it can affect writing through the developed body/reference. The outgoing record is the centered, standard-deviation-0.3 projection of tanh(final R), not final theta. No parent body, working reference, guide, RNG state, or optimizer state crosses as inherited state. Host parameters remain fixed along a case's lineage; fresh embodiment does not mean resampling its rulebook.

Qualified preference is P = (qualified own-sign matches − qualified alternative-sign matches)/256, with strict $|V| > 0.2$ and positive sign defined by $V > 0$. Reports multiply P by 100 for points. Since the two targets disagree at 128 sites, P lies in $[-0.5, +0.5]$. Shape correlations use continuous centered fields, normalized by their population standard deviations, maximizing over all 256 toroidal translations of the ideal own-family target; zero-variance fields receive zero correlation. Packet, reference R, and active V scores describe different fields. Original coordinate-based preference and translation-tolerant record shape are not interchangeable endpoints.

**D141 prediction.** Development uses the sixteen unique D135/D140 founder blocks, all four lives, both families, and both writer conditions. The held-out blocks are 90000–90015. Each diagnostic policy receives four starting features and six features from each of two complete read–rewrite–fresh-read cycles, plus four family-by-writer intercepts. Features concern own/alternative template correlations, guide margins, fresh preference, and V RMS; raw fields, identities, seeds, and future states are excluded. The passive policy gets intact U0/U1; the challenge policy gets the same U0 plus C1, which uniformly permutes values among 64 randomly chosen sites before routing. Permutations may leave some selected sites unchanged; no normalization or favorable perturbation search is added. Diagnostic copies never enter the actual lineage.

Both policies fit ridge predictors, choosing alpha from {0.001, 0.01, 0.1, 1, 10, 100} by leave-one-development-block-out MSE, with training-only centering/scaling and larger-alpha tie preference. Final predictors are frozen; predictions are clipped to $[-1, 1]$, a wider bound than the actual two-target score. A four-feature starting-only predictor is secondary. All 256 holdout forecasts precede all actual held-out outcomes. Actual lineages perform three rewrites and a fresh fourth read. The primary difference is passive squared error minus challenge squared error, equally averaged within each of sixteen blocks; its interval spans zero. No post-holdout refit or new challenge enters this comparison.

**D142/D143 variation.** D142 uses four new futures for each D141 record–host–writer case: 1024 outcomes, without including original D141 outcomes in its primary variance estimates. For each family/writer, W is mean within-case sample variance, B is variance of case means minus W/4, and repeatability is B/(B+W). Estimates are not clipped; between-case variation includes both record and host differences. D143 crosses those four address histories with four other-noise histories, retaining all 4096 labelled cells. Its primary characterization uses finite-level-corrected address, noise, and interaction components. The original 1024 diagonal trajectories are reused and 1536 aligned off-diagonal cells are exact address aliases; only 1536 ordinary off-diagonal combinations are new. Conditional variances at fixed address/noise each include interaction and cannot be added as independent contributions. Post-hoc geometry and stage reviews do not change either protocol's primary status.

**D144/D145 addressing.** For target T and displacement d = (dy, dx), define q(d) = mean[T × roll(T,d)]. On the 16 by 16 sheet, A has q = 1 − min(dy,16−dy)/4; B has q = 1 − min(dx mod 8,8−(dx mod 8))/2. D144 retains every nonzero compatibility class: nine levels for A and five for B, plus zero shift separately. The q = +1 nonzero pools have fifteen A and thirty-one B members. The experiment draws uniformly with replacement within lexicographically ordered classes, not uniformly across all 255 shifts. This externally supplied rule uses target family; the numerical writer receives only the selected shift. Zero shift has q = +1 too and is excluded from the nonzero pool. Compatibility with an ideal template need not imply compatibility with every imperfect acquired field.

D144 uses fresh blocks 91000–91015, four lives, both families, and two noise replicas, after one rewrite. D145 uses blocks 92000–92015 and three rewrites; it retains zero, q = +1, q = −1, protected-copy, and matched reading/writing-disabled conditions. Intermediate classes are not retested. D145 redraws shifts each generation from a complete schedule sealed before any founder generation. Separate address streams and common body/noise streams match interventions without letting address sampling change other randomness. The first ordinary reading states match; later inputs legitimately diverge as each arm transmits its own output.

At every D145 boundary, a separate fresh 300-step reader assays the outgoing packet. Its states and random streams do not enter subsequent lineage development. Protected copying passes original F0 unchanged; writing disabled passes zeros; lineage reading disabled does not disable the separate terminal assay. D144's sole primary is B compatible-minus-zero preference after one rewrite; D145's is that contrast after replacement 3. Copy and shape comparisons are predeclared secondary endpoints, with pointwise unadjusted intervals. D145's saved-state review retains every case and distinguishes score decomposition from causal explanation; it supplies no new dynamics or prospective test subgroup.

**D146 timing.** Blocks 93000–93015 supply the same sixteen-block/four-life/two-family/two-replica structure. Baseline is 300 reading plus 1000 writing updates; the equal-extra-time arms use 600 + 1000 and 300 + 1300. Only phase duration changes. Assigned compatible shifts and read/write noise prefixes are matched; additional reading does not advance the separate writer-noise stream. All external assays remain 300 updates. Each timing retains copying and disabled-reading/writing controls. Timing-labelled exact-copy aliases do not add independent evidence. The sole primary is terminal B extra-read minus extra-write preference after three replacements; comparisons with baseline and copy are secondary. The unresolved primary remains visible alongside those positive secondary contrasts.

**Uncertainty and provenance.** All six studies retain sixteen founder blocks. All use 10,000 whole-block bootstrap draws, with seeds 14120260917, 14220260917, 14320260917, 14420260918, 14520260918, and 14620260918 respectively. Lives, roles, conditions, and stochastic replicates stay grouped within blocks. D142/D143 intervals and secondary D144–D146 intervals are pointwise, unadjusted. All observed cases remain; no shape or preference filter selects founders. Fresh cohorts use the same supplied pattern families and guide, not new family/guide generalization. Internal prospective freezes and separate checking belong to one human-directed program.

The source selection identifies six completed runs, protocols/reports, and retained summaries/verification records. The static review checks all 285 indexed output hashes and each summary/manifest/completion link; incorporation rechecks selected metadata. The portable extract

preserves native-unit estimates, intervals, and D145 table inputs without recomputing uncertainty. The source package omits raw arrays and the complete executable dependency closure. Reported original replay and scientific-audit checks remain attributed to their records, not rerun here. E38–E41 link each protocol and report.

## C.16. Fresh repair confirmation and longer unprescribed renewal

**E013.** Retain C.12's E012 ridge coefficients, feature transforms, target encoding, eight-option repair menus, seventeen native features and six graph features. Blocks 11000–11127 are new, with four founders per block and eight recipient futures per body. Each packet has 192 entries erased and each repair restores 64 values from a supplied clean backup. The correct graph and an independently generated structurally matched wrong graph are passed separately to the identical frozen graph predictor; the target simulator always uses its actual graph. Native, source-based, strongest-entry and uniform choices remain. Predictions and selected options are sealed before target readout. The original six positive-lower-bound criteria and the additional correct-versus-wrong-graph criterion all pass. Specificity is qualified own-founder agreement minus mean agreement to the other three founders sharing that block's bias. Recovery subtracts the unrepaired specificity. Inference uses 10,000 whole-block bootstrap resamples, with pointwise 95% intervals; no test-set refit, selection, or pooling with E012 occurs.

**E014.** Retain C.10's E007 heterogeneous formation, generic capture, reader/writer, and encodings at factors one (2048 bytes) and four (128 bytes). Blocks 12000–12063 contain four founders each; all 256 founders remain, including formation failures. The packet is block-averaged at every boundary and decoded by repetition without extra normalization. Each current body reads, loses the incoming packet, and writes its successor under the frozen protocol. Fresh checkpoint readers follow rewrites 1, 3, 8 and 16 without affecting the chain. Own, foreign, shuffled, erased, held-acquisition, reading-disabled, writing-disabled, protected-copy, background and challenged conditions remain separate. The primary is the 128-byte after-16 own-minus-foreign-record agreement contrast. Formation, acquisition and route controls are retained in the full gate. Protected copying preserves the original packet at the same budget. Checkpoints are paired along the same history, not independent generations. Native summary intervals and all failed founders are retained; no equivalent-fidelity or indefinite-stability test is substituted.

**Completed post hoc decomposition.** The scorer uses the binary sign class $b(x) = \mathbf{1}\{x > 0\}$, including nonpositive values in the other class. Let $S$ be all-site sign agreement and $L$ the fraction of all sites that match the prototype's sign but have $|V| \leq 0.2$, both expressed in percent. Then $Q = S - L$ exactly. The fraction above threshold, regardless of sign, is a separate marginal; Q is the intersection of sign matching and amplitude qualification, not generally their product. No amplitude filter is applied to the founder prototype.

The analysis counts four exhaustive categories at every saved activity field: matching/above-threshold, matching/subthreshold, mismatching/above-threshold and mismatching/subthreshold. It retains all 256 coordinates, 64 blocks, four founders and two recipient replicates, ten arms, both budgets, sixteen lineage reads and four fresh checkpoint reads. Activity is state index zero of `generation_read` or `checkpoint_read`, not the working reference or its moving average. G1 precedes the first rewrite and G16 precedes the sixteenth; these sixteen lineage reads

are kept separate from fresh assays after completed rewrites 1, 3, 8 and 16. Every arm uses the same own-founder prototype. Checkpoint assays remain outside the lineage.

Founders and recipients are averaged within blocks before using E014's original 10,000 whole-block bootstrap index sets. New sign/amplitude and paired endpoint intervals are post hoc, descriptive and pointwise 95%; they do not revise the original success criteria. A separate endpoint cross-tabulation pairs coordinates in the fresh after-three and after-sixteen bodies. It accounts for Q gains and losses involving sign-only differences, amplitude-only differences or both, but is not a trajectory of one cell or a causal decomposition. The portable report retains that accounting and all controls; main-text section 3.1 reports the marginal sign and amplitude changes; SI Figure S5 shows the joint-count decomposition. The separate eight-block-per-setting E001 pilot is reaggregated from its own summary, not pooled with E014.

The analysis handoff records 128 original lineage archives checked against completion hashes on the producing host. Its retained audit independently recounts those fields. They remain excluded from git by the NPZ rule and were not available in this manuscript checkout. Two compressed count tables preserve 204,800 site-category rows and 163,840 endpoint-category rows; they are sufficient for the reported statistics, not a backup of the continuous fields. During incorporation, derivation and a separate statistical reaggregation were run from those tables and the nine supporting tracked dependencies in an isolated directory with no NPZ files. All 3,180 means and intervals, the pilot context and checkpoint CSV reproduced exactly, including the 480 selected original qualified-score summaries. No dynamics or raw-field audit was rerun here. The source package, inspection/reproduction record and figure manifest preserve the handoff, source hashes, four supplied figures and reproducible scripts. Supplementary Figure S5 uses its sign/amplitude figure unchanged.

The first E014 detach request stopped before generating scientific blocks; `confirmation_v2` used the same frozen workflow and cohort. The E013 prediction array is unavailable at the inspected local paths, although its recorded hash, completion and passing audit remain. Selected text/metadata and original native summary fields appear in the v0.12 package. These provenance checks do not rerun either experiment.

Sources: E013 protocol, E013 report, E014 protocol, E014 report. E42/E43 identify the claims.

### C.17. Local functional acquisition, capture timing, and damaged renewal

**Dynamics and task.** This branch changes apparatus from the theta/reference tissue model. On a 16 × 16 torus, each site has signed activity v and four reciprocal nearest-neighbor edges. The 512 edge weights are serialized as little-endian float64 values (4096 bytes). With dt = 0.2, gain k = 2, and independent standard-normal update noise z, the ordinary step is

$$v_i' = \mathrm{clip}_{[-2,2]}\left(v_i + 0.2\left[v_i - v_i^3 + k\left(\frac{\sum_j w_{ij}v_j}{4} - v_i\right)\right] + 0.02z_i\right).$$

There is no whole-field normalization. Staged reading omits the positive linear amplification term v_i for the first 256 of 2048 updates; cubic drift and coupling remain. A 2 × 2 input patch begins at zero-indexed row 7, column 1; output patches begin at (3,10) and (11,10). For cue $c \in \{-1, +1\}$ and taught orientation $s \in \{-1, +1\}$, the required output polarities are $(A, B) = (sc, -sc)$: Rule I has $s = +1$ and Rule II has $s = -1$. These specify signs; the success criterion below addition-

ally requires output-mean magnitudes greater than 0.25. The tested task distinction is the choice between these two mappings at the fixed ports. Teaching clamps all three ports for 512 updates per cue under a unit-positive scaffold. The consistent-acquisition procedure pairs sign-reversed initial states and noise across opposite cues; local correlations $\tanh(3v_i)\tanh(3v_j)$, averaged over the final 128 updates per cue, form the acquired weights. The writer receives activities, not a desired full-body image.

E015 tests gains one and two on 16 development blocks; neither qualifies. E016's 32 fresh blocks compare ordinary, consistent and engineered acquisition at gain two; its primary improvement passes but full reliability remains unqualified. E017's 32 fresh blocks compare baseline, equally long extended, and staged readers. E018 attempts three replacements with delayed writing on another 32 blocks and fails. E019's 32 fresh blocks compare immediate, delayed and time-averaged capture at one replacement, with immediate capture qualifying. Outcome-informed hypotheses are frozen before each subsequent cohort; earlier outcomes are not pooled into the new test.

**E020 boundary.** Blocks 18000–18031 have four founders and four lineages per founder. Independently born renewal activity reads the predecessor packet in two matched cue episodes. Only the input is clamped; output labels, rule labels and desired targets are absent. After reading, effective incoming weights are erased. The writer averages read-end local correlations across the two cues before any scaffold update and encodes a new packet. A supplied 128-update holding window uses a unit-positive scaffold, independent writing noise and continued input clamping; the already written packet is protected and receives no later feedback. Discard the body and pass only that packet. Repeat three times with new body/noise streams, retaining matched prefixes across intervention arms. The writer never receives incoming packets, scores, templates or offline observation buffers.

Protected-copy chains preserve exact founder packet bytes; reading-disabled chains zero incoming effective weights; writing-disabled chains calculate the snapshot for evidence but transmit zeros. Each lineage propagates its own outcome without reinitialization to the founder record. Independent terminal readers at horizons 0, 1 and 3 never enter the chain. Every assay retains own, opposite-rule foreign, shuffled, erased, untrained, communication-disabled and input-isolated arms.

**Scoring and gates.** A cue succeeds when both output-patch activity means, averaged over the final 32 updates, have their required polarities strictly beyond $\pm 0.25$. Both-cue success requires the two fresh cue assays to succeed for the matched lineage. The primary after-three intact both-cue 95% lower bound must exceed 0.75. Full renewal additionally requires the after-one joint criterion; the original 24 individual-cue gates at both horizons (six rule-by-damage reliability lower bounds $>0.75$ and eighteen own-minus-control lower bounds $>0.10$); and twelve reading/writing-dependence lower bounds $>0.10$. Ten thousand percentile resamples preserve whole blocks. Other intervals are pointwise; nested founders, lineages, cues, controls and injuries are not independent units. All-success bootstrap intervals describe the observed sample, not a universal probability-one claim. Copy and serial-change contrasts are secondary, with no equivalence or noninferiority design.

**Injury and E021/E022.** Terminal injury is either 24 scattered non-port cell removals or a non-port $3 \times 3$ lesion, without rejecting disconnected cases. Dead activities and incident edges are removed; surviving dynamics are otherwise unchanged. E020 renews in intact bodies. E021 uses blocks 19000–19031 and redraws a 24-cell mask at every renewal birth. The masked-copy comparator applies the same edge erasures without rewriting surviving values. E022 uses blocks 20000–

20031 and one damaged replacement, changing only the renewal reader denominator: four, surviving physical degree, or sum of absolute surviving weights. A zero denominator gives zero incoming drive; the −k v term, writer and terminal reader remain fixed. Its primary strength-minus-baseline intact both-cue lower bound must exceed 0.10. Full compensation separately requires candidate intact reliability, all original 24 gates, intact-reference qualification and six reading/writing-dependence gates. The primary passes while two scattered-injury reliability gates fail. Full verdicts and all cases are retained.

Sources: E015 protocol, E016, E017, E018, E019, E020, E021, E022. E44–E46 link the completed reports; the v0.12 extract copies their native summaries without recomputing uncertainty.

### C.18. Cellular return, tissue dependence, and turnover schedules

**Shared model and ancestry.** P13–P16 retain the corrected native BITSEY equations, complete 17-component states and cell identities, bounded integration clocks, E/M/X classifier, and two isolated-hold criterion described in C.7/C.14. They are imposed contact/replacement experiments, not simulated cell division. Coarse/fine numerical comparisons share scientific conditions and are not biological replicates.

P13 continues each P12 final E trio through temporary six-cell M instruction, cue removal, retention and three further complete replacements. Three challenge histories per origin yield M→E→M over nine handovers. The return instruction is supplied, and ordinary-origin histories also pass. P14 uses one preselected current source per origin plus fresh M in a two-dimensional E-dominated tissue, comparing sham, internal/boundary/both expression changes, whole-sheet RNA blockade and patch-boundary cutting. Its registered composite concerns withdrawal persistence, replacement and final isolation. Delayed correction after restoring communication can pass that endpoint; it cannot be described as correction while blocked. No expression intervention improves the composite over sham.

P15 branches matched pre-replacement states into surrounding-tissue contact, boundary-cut/internal-contact-preserved, and boundary-cut plus RNA blockade, with ordinary-E substitution controls. Actual incoming replacements are followed through isolation and onward transmission to fresh M recipients. Success with the boundary cut removes continued surrounding-tissue instruction as necessary under this interval; failure with RNA blockade identifies an internal transport dependence. The protocol does not establish the source of the original patch correction.

**P16 screen.** Actual P15 onward-recipient trios from the written and ordinary histories enter a native nonperiodic 1 × 3 chain at logical time 1,180,000; each resolution retains its own full state and identity. Ordinary-E stock at time zero is a third benchmark, unmatched for prior age. The native source commit is b2cb370aaecbc8299f259bc4251d5396c81f19a6. Depolarization gating, expression power 60 and miRNA Kd = 1.25 × 10^-14 are supplied. No surrounding tissue or external E cue remains.

Cross tau = 1, 10, 100 and 20,000 with four schedules, each repeated twice: (i) replace position 0, wait tau, then 1, wait tau, then 2, wait tau; (ii) replace 0 and 2, wait 2 tau, then 1, wait tau; (iii) replace 0 and 1, wait 2 tau, then 2, wait tau; (iv) replace all three and wait 3 tau. Every schedule introduces six M cells, replaces each position twice and lasts 6 tau. Surviving cells evolve continuously. Batch size, surviving position and exposure differ together.

Incoming M stocks alter only ZEB protein by fixed integer percentages in [−25,+25], using the declared SHA256 namespace, seed 501 and cycle/site index. The same perturbations pair schedules, rates, origins and resolutions. No source search, retuning or exclusion by outcome occurs. Two final isolated 10,000-unit holds require E at both endpoints and at least 95% E occupancy in each last quarter, for every remaining cell. Intermediate continuous correction is recorded separately from final retention.

For each non-wipe schedule, the fixed rule selects the smallest tested tau where both historical founders pass at both resolutions. If none passes, tau = 20,000 is retained with reason `no_shared_retention`; it is not promoted to a success. At each selected interval, all three origins and both resolutions receive RNA-blocked controls. Every selected open and blocked final trio, including failures, supplies open/RNA-blocked/boundary-cut onward assays. Donors contact three fresh M cells for 20,000 units and are removed before the recipients' two holds. Matched wipe failure and ordinary-E capacity controls are required for the strongest conjunction. No-turnover controls use the maximum 120,000-unit exposure, not each shorter schedule's age.

For a phase of duration H, samples occur every min(100,H/20), and steps are capped at min(100,H/100); fine runs halve that cap and the native step/increment limits. Numerical agreement at the two resolutions is a check, not a convergence proof. P16 retains 234 cases, 1056 integrations and 117 numerical pairs, including failed schedules and control forks. These are deterministic model panels with shared ancestry, not population sample counts.

**Availability.** P13–P16 outcome claims retain report attribution. P14–P16 raw archives were unavailable at the report-relative locations and previously synced `/mnt/bioIce1/runs/runs` during review. The reported audits are not rerun or independently rederived here. Source reports and protocols are packaged; they do not replace the raw histories needed for a fresh mathematical audit or additional measured trajectory figures.

Sources: P13 protocol, P14, P15, P16. E47–E49 link the reports.

### C.19. L20 stopping, component edits, and forecast development

**D147–D149.** Retain C.15's D146 cohort (sixteen blocks 93000–93015, four lives, two roles and two noise replicas). D147 compares immediate fresh readouts of packets after 1000 versus 1300 writing steps along the same extended ancestry; it does not stop the entire lineage early at every generation. D148 crosses the two packets with their two guide-selected placements, retaining values and noise while changing only translation. Primary B/G3 stopping and B/G2 placement effects remain unresolved.

D149 edits the late packet after its natural late placement, without rerouting. Decompose the centered field into orthogonal row, column and double-centered residual components. Rows are own for A and columns for B; the other axis competes. Write the packet as mean + own + c + r, with RMS norms C and R. Set H = hypot(C,R), d = 0.5 min(C,R), and theta = 2 asin(d/(2H)). Rotate the coefficient pair toward the residual: (C cos(theta) − R sin(theta), R cos(theta) + C sin(theta)). This preserves own axis, mean and norm while jointly reducing the competing component and increasing the existing residual component. Two shams preserve c and rotate r toward opposite signs of a fixed projected Walsh residual, with beta = 2 asin(d/(2R)), matching edit distance d. Degenerate records make every edit an identity and remain included.

The primary is B/G2 attenuation minus the mean of both shams; untouched comparisons and other groups are secondary. Intervals use 10,000 whole-block percentile draws, seed 14920260919. Qualified preference retains strict $|V| > 0.2$, all 256 sites in the denominator and the own-minus-alternative definition in C.8. Fresh assays receive only edited records and public machinery, never the original body. The 12,288 labels include 6,912 new readouts and 5,376 exact aliases; none adds new founders. Later readouts do not propagate into a lineage. Completion, summary and verification hashes match the quoted source values.

**D150–D159.** Known cases and earlier held-out material are now development data. Nested founder-block cross-validation, matched regularization, one-copy baselines, mean versus separate readings, four saved future outcomes and founder-deletion panels diagnose forecast failures. These procedures must not be read as an untouched confirmation cohort. The future-average training and omission panels retain all settings rather than selecting a favorable future or founder. D159 performs arithmetic on saved fitted predictions only: each prediction is offset + starting terms + reading level + reading half-difference + clipping correction. For candidate prediction c, reference r and outcome y, each signed squared-loss contribution is (reference component − candidate component)$(r+c-2y)$. Components sum to reference MSE minus candidate MSE, depend on both models and the centering convention, and can cancel. They do not establish the effect of removing a component or quantify independently useful information. The underlying forecast primary remains unresolved.

Sources: E50/E51 and the dated cross-track review; Appendix D gives the individual D150–D159 reports. The portable source record distinguishes new causal assays from later saved-data development.

## C.20. Finite FEP models, preference choices, and bounded decisions

**Record and reader.** Let X be a fair hidden bit, R = X XOR N a record with content-error probability eta, and Y = R XOR E the output of a separate reader. Maintenance changes reader reliability rho without changing R. A controller sees only Y and selects one of four binary report maps; paired with maintain/skip this gives eight deterministic policies. It does not receive X, R, realized faults, or an answer-bearing policy index. Retained information I(X;R) and accessible information I(X;Y) can differ; with eta = 1/2, X is independent of all available signals. The outcome is expected correct reporting minus maintenance cost. Bayesian utility, state risk and its exact decomposition, and explicitly defined information-target extensions are compared with the same channels, costs and access.

Repeated-read identification assumes independent reader errors; a consistent reversal is not identified by agreement alone. The evidence extension crosses unknown intact/erased content with intact/random reading and supplies a noisy independent reference to X. Its reliability and independence are assumptions, not learned facts. The controller may answer directly from that reference; preservation of the original record is not imposed as the only available action.

**Transfer and timing.** In the three-record continuation model, the first two independent targets provide optional calibration and only the third is rewarded. The reader's hidden episode class either preserves orientation (prior 0.75) or redraws it; a separate content class is intact with prior 0.5. Reference accuracy is 0.9. Each calibration pair costs 0.025 and the terminal independent reference costs 0.30. The selected policy uses the shared orientation when the two calibrations agree

and buys the terminal reference when they disagree. Accuracy 0.802 minus 0.050 calibration and 0.114 expected reference cost gives utility 0.638; the best at-most-one-calibration policy gives 0.625. A control commits maintenance/reference purchase before observing the second calibration while allowing the final report to use it. All permitted contingent policies and ties are retained. Subsequent reviews test changed transfer assumptions; matching past calibration does not certify the future law.

**Preferences and continued operation.** Hidden-preference tests retain the physical state and preferred joint over target, report and cost-bearing actions while changing the preferred conditional over hidden details. The conditional-KL term can then change rankings. A consistent relabeling changes neither task performance nor a consistently transformed objective; adding a preference for literal reader agreement does. The seven-policy witness set is explicit and is not a global search over all possible policies. A separate two-energy-unit/two-task model compares economy and precision reading, stochastic recharge, hard versus priced continuation, and reference-policy choices. Energy use changes future task feasibility; risk tolerance, preference strength and the reference controller remain specified. Matched Bayesian choices include the same constraints and costs. These are exact finite calculations, not sample-based biological confirmations.

**Structured inference.** Eight chosen fixtures cross $4 \times 4$ and $8 \times 8$ grids, baseline failure probabilities 10^-6 and 0.01, and all-failure or mixed evidence. Priors and model are known. Independent mean field and disjoint $2 \times 2$ joint-block approximations minimize the same variational free energy under fixed damping and forward/reverse coordinate orders. The primary budget is 4096 logical work units per edge; 256 and 1024 are descriptive checkpoints. Whole updates and initialization are charged. Each directed neighboring pair supplies a decision query. Means average queries within fixture/order and then equally weight the sixteen fixture/order cases. Exact inference supplies a reference answer; productizing completed joint beliefs at unchanged marginals tests the specified correlation mechanism without rerunning inference. The overall gate requires both decision improvement and worsened risk-prediction error after productization. The latter fails; larger conditional fixtures remain unexecuted.

**Certified cap release.** Later derivations use these saved approximate beliefs, known model and risk bounds. With action probability r, true model risk R_pi(r) = A + G r, and the declared objective

$$J_\pi(r) = D_{\mathrm{KL}}[\mathrm{Bern}(r) \| \mathrm{Bern}(1/2)] - 5 - 0.6r + 2R_\pi(r),$$

regret is J_pi(r) minus its unrestricted feasible optimum under risk $\leq 0.1$. Existing interval bounds define a certified safe ceiling C and an upper bound G_high on the risk slope. If s is the previous certified choice, the new rule is $\max(s, \min(C, \mathrm{sigmoid}(0.6 - 2G_high)))$. A separate physical-slope control uses $0.1(1-h)$ instead of G_high. Policy construction uses only the declared fourteen available fields and precedes loading the exact reference for evaluation; this is a checked code boundary, not process-level file isolation. The action cap from the old approximation is not retained as a new ceiling.

Both rules improve or retain objective value under the supplied model and meet its risk limit. The audit retains 13,056 query decisions across beliefs, schedules and three budgets; these are deterministic evaluations, not independent trials. Regret reductions are ratios of mean objective losses at the original primary checkpoint. The identical actions of the two rules prevent attributing extra action benefit to the tighter slope guarantee. No new inference, budget selection, biological safety

claim or rescue of the original failed mechanism gate is implied. Exact inference remains faster at the tested sizes.

**Selection and verification.** E52–E54 identify reports, prospective specifications and later assumption reviews. The v0.12 package retains the selected text/metadata and finite results, including the structured-inference verdict and safe-cap manifest. Hash checks verify file identity; no checker, inference routine, fit or model trajectory was executed for manuscript incorporation. Arithmetic checks reported by the FEP studies are their own verification counts, not independent observations. The separately selected sensor studies below do not alter these earlier verdicts.

## C.21. Prospective reconstruction prediction and diagnostic feedback location

**D162 cohort and boundary.** The separately registered pilot uses 64 new founders 100000–100063, each with four lives, two supplied target families and two writer conditions: 1,024 cases. Four stochastic futures per case yield 4,096 terminal outcomes; founder is the statistical block. The existing 2048-byte record and frozen D131 guide are retained. Each lineage performs three fresh read–erase–rewrite cycles followed by a fresh G4 read. Reading lasts 300 steps and writing 1000; only the outgoing record crosses. Full controls run under the first future, while the remaining futures supply primary ordinary/aligned paths. Diagnostic copies and their records never enter an actual lineage.

**Challenge and observation budget.** Passive receives independent intact diagnostic U0 and intact U1; the primary receives the same U0 and residual-minus in place of U1. Each complete diagnostic is read300–write1000–fresh-read300. All second-copy alternatives share U1's birth, noise and assigned feedback shift, separately from actual futures. Every model also receives seven starting measurements and public family/writer labels; each permitted diagnostic contributes six endpoint features. No observer receives a third copy, true future outcome, seed or founder identifier.

The D151 residual edit is applied after the original incoming routing and is not rerouted on entry. Decompose the centered packet into own-axis field o, competing-axis field c and double-centered residual r. Let C and R be the RMS norms of c and r, d = 0.5 min(C,R), and beta = 2 asin(d/(2R)). Residual-minus replaces r by cos(beta)r − R sin(beta)q, where q is a fixed-seed unit-RMS residual direction orthogonal to r. Residual-plus uses the opposite sign. The same fixed two Walsh candidates and projection rule as D149/D151 construct q; own and competing fields, packet mean and norm remain unchanged. Near-zero min(C,R) is an identity case and is retained. The outgoing newly written packet is routed normally. This supplied geometry is not a learned target vocabulary or the original D141 shuffle.

**Training and freeze.** All 17 observers at penalties 0.001, 0.01 and 0.1 are fitted on the same sixteen development founders' four-future means, using training-only centering/scaling and ridge regression with public group intercepts. The primary ties corresponding raw slopes of the two readings under the original penalty; this differs from separately standardizing their average. Primary alpha 0.01 and separately selected starting-only/one-copy settings 0.001/0.01 precede pilot founders. Models precede founder generation, and all 52,224 forecasts were sealed on 21 September at 12:14:30.557744 UTC, before actual outcomes began at 12:14:31.751734 UTC. None clipped within the allowed $[-1,1]$ forecast range.

**Endpoint and inference.** At each of 256 sites, own and alternative sign agreement contribute only when $|V| > 0.2$, with all sites retained in the denominator. Their difference is raw preference

in [−0.5,+0.5], or [−50,+50] points. Prediction MSE averages individual-future squared errors equally over futures, lives, groups and then founders. The primary gain is passive MSE minus residual-minus MSE. A 20,000-draw whole-founder percentile bootstrap, seed 16120260920, gives the saved 95% interval. These intervals condition on the one frozen model set; they do not estimate variation over independently retrained predictors. Cheap-baseline and apparatus intervals are pointwise secondary comparisons. The within-case four-future variance is common to both MSEs and cancels from their paired difference; it is not a demonstrated irreducible-noise bound.

**D169 controlled mechanism.** All sixteen development and 64 pilot founders are retained separately. Y[c,a] uses copy c's saved read-end V, moving average and R, starts writer theta at zero, and applies the feedback shift assigned to copy a. The 2 × 2 intervention holds body-specific writer noise and fresh-reader streams fixed; only displacement changes. Original paths reproduce their retained outputs. Aligned writers ignore displacement, so unchanged full paths are implementation controls. The matched address contrast is A = ((Y01−Y00)+(Y11−Y10))/2; the background contrast is B = ((Y10−Y00)+(Y11−Y01))/2. Original and swapped copy gaps equal A+B and B−A. B bundles saved body and noise; it does not isolate one source. Outcomes include best-translated own-template shape correlation of the outgoing field and qualified preference of the diagnostic fresh reader, not actual three-replacement outcomes. Pointwise intervals use 20,000 whole-founder resamples, with seeds 16920260922/16920260923. The development focus was chosen after earlier results but before these counterfactuals.

**D170/D171 prediction boundary.** Saved-metadata checks confirm equal shifts and streams for passive/challenged second copies. D171 adds the same two ideal-template compatibility summaries q0/q1 to all seven observers, with tied reading slopes for two-copy models and all three penalties. Leave-one-founder-out development fitting produces 336 new models and 5,376 forecasts, scored on all four saved futures. No pilot deployment or new dynamics occurs. Score-only omissions do not refit models or supply confidence intervals; independently selected cheap-baseline settings are development-selection context. The additive extension also increases fitting capacity. Its failure does not test all setting-dependent interactions or exclude useful alternative representations.

Sources: E55–E57, the linked frozen protocols and reports, and the D160–D171 bridge in Appendix D. Selected native summaries and completion/audit records are in the v0.13 source package. The present check verifies selected file identities and saved-audit bindings; it does not rerun tissue dynamics, fits or bootstraps.

## C.22. Learned sensor interpretation, costed acquisition, and deployment feedback

**Model and supplied resources.** A fair hidden bit W produces warnings Y_i = W XOR E_i. Error pairs are stationary IID with law Q over 00/01/10/11. Calibration supplies known-positive inputs, revealing error pairs; the same symmetric error law transfers to deployment. The independent model M0 has separate Beta(1,1) error-rate priors. The joint model M1 has a four-cell Dirichlet(1/2,1/2,1/2,1/2) prior; structural prior weights are 0.9 and 0.1. The candidate family, reference access and calibration transfer are supplied, and paired dependence does not identify a physical common cause. These studies neither reuse C.20's specific task nor inherit its risk guarantees.

Within each model, minimized VFE equals negative log evidence. Exact posterior model weights minimize the structural VFE, giving conventional Bayesian inference without an approximation

gap. The primary deploys M1 only when its Bayes factor exceeds 9; otherwise it uses M0. Given each warning pair, it reports the bit with lowest expected error or buys a perfect reference, taking the reference at ties. Wrong reports cost 1, references 0.1, calibration pairs k = 0.001, and deployment lasts H = 100 tasks. Full cost is expected errors plus reference expenditure plus calibration expenditure, divided by H. A learned warning-conditioned table stays fixed during deployment except in the expressly defined feedback studies.

**Fixed and adaptive acquisition.** Factor revision compares four learned rules after 64 calibration pairs: independent, evidence-selected, joint and posterior-average. Both model families retain the same marginal predictive error rates at each history. Adaptive calibration instead buys one pair at a time or stops, with cap 64 and no free initial observations. For count history c, stop cost S(c) is H times expected task loss under that controller's declared deployment table. Remaining loss satisfies V(c) = min{S(c), k + sum_e p(e|c)V(c+e)} until the cap, with exact ties stopping. The primary planner retains the full structural posterior while respecting the fixed MAP deployment rule. Averaging has its own schedule. No truth parameter enters planning and no separate information bonus is added. Original two-fixture efficiency, later nine-law efficiency and later 125-law population benefit are different comparisons, retaining their different verdicts.

**Prospective 125-law panel.** Marginal error rates a,b independently range over {0.05,0.15,0.25,0.35,0.45}; dependence parameter t ranges over {−1,−0.5,0,0.5,1}. With l = max(0,a+b−1) and u = min(a,b), q11 = (1+t)ab−tl for t ≤ 0, and q11 = (1−t)ab+tu for t ≥ 0. Then Q = (1−a−b+q11, b−q11, a−q11, q11). This parameter interpolates joint-error bounds, not Pearson correlation. All 125 laws have equal weight, including swapped sensor orders and 36 laws offering no advantage over always referencing. The ten known controls are separate. Exact rational integration over stopped calibration histories and future tasks evaluates unchanged policies; software assertions are not independent experiments.

The evaluator knowing Q receives the same warning and pays for references; it does not receive a free target label. Its population floor is 0.068664. Available improvement is 0.1 minus that floor, and captured improvement is (0.1 minus agent mean cost) divided by available improvement. The ratio is neither accuracy nor the mean of per-environment percentages. The prospective criteria are mean cost below 0.1 and cost below 0.1 in every law with positive available improvement. The first passes and the second fails. Completing the budget or changing deployment afterward does not revise those original verdicts.

**Joint deployment and attribution.** After inspecting the new population, the comparison crosses both saved stopping schedules with MAP, posterior-average and joint deployment. The primary remains the MAP acquisition schedule. Joint prediction uses (c_e+0.5)/(n+2), with the same saved counts and within-model prior; acquisition is not reoptimized. Calibration is exactly matched within a schedule. All physical-law means, including harms and older controls, remain visible. Posterior averaging is verified no worse under its own full-mixture belief at every saved history; the equally weighted physical grid is a different population. A preceding substitution analysis uses true within-model parameters only in evaluation, retaining model choices/weights, and separately completes calibration to cap 64. Its signed terms are declared-order accounting, not disjoint universal causes of failure.

**Second deployment feedback.** The known nine-law panel crosses a = {0.05,0.15,0.25}, b = 2a and independent/midpoint/nested error laws, equally weighted. Matched arms share calibration,

a first-task reference-acquisition rule and its retained evidence, and the second-task action. They then retain either no additional observation, the second warning alone, or that warning plus its true target only if a reference was already purchased. Unchecked answers never become labels. The resulting table stays fixed for tasks 3–100; the first acquisition policy is not reoptimized for this extension. All second-task costs match, so differences concern the final 98 tasks. Three criteria compare full retention with frozen, warning-only and original no-update behavior; all fail for both MAP and averaged controllers. Own-prior expectation and fixed-panel costs are kept separate.

**Provenance.** E58–E60 link the named specifications and reports. Generalization's original integration stopped at a wrong old-record field lookup before exporting or inspecting new outcomes. The preserved revision changes that lookup and qualification/provenance, leaving population, policies, costs and criteria fixed; successful execution is attempt two. Selected manifests, summaries and frozen-source identities are retained in the v0.13 package. No model, optimizer or scientific checker was executed for this incorporation, and no off-grid successor result is included.

### C.23. Functional reconstruction under repeated damage: E023–E047

**Shared task, inherited state and reader.** Retain C.17's 16 × 16 periodic network, 512 reciprocal nearest-neighbour weights, 4096-byte float64 record, two supplied conditional rules, three public ports, time step 0.2, gain 2, birth standard deviation 0.05 and update-noise scale 0.02. Acquisition teaches the two output polarities under opposite input cues. Every renewal uses a fresh activity field and a new mask removing 24 non-port cells. Birth, noise and damage streams are paired across policies within a lineage and refreshed at each boundary. Only the current packet passes onward; there is no restart from the founder packet. The 32 independent blocks, each containing four founders and four descendant lineages per founder, are the resampling units.

E023 varies the fresh receiving reader while keeping record production fixed; E024/E025 apply compensation to both renewal and terminal reading. For surviving edges, the compensated incoming drive is

$$u_i = \begin{cases} \dfrac{\sum_j w_{ij} v_j}{\sum_j |w_{ij}|}, & \sum_j |w_{ij}| > 0, \\ 0, & \text{otherwise.} \end{cases}$$

This replaces the fixed denominator four in C.17; the negative gain-times-activity term, cubic drift and clipping remain. Denominators are recalculated from the effective record after injury or erasure. In E038–E041, the positive linear amplification is suppressed for the first 1024 of 2048 reading steps in renewal and terminal assays. Only the input port is clamped during renewal. The two cue episodes share matched birth/noise structure; desired output values, task labels and observer scores do not enter the renewal writer.

**Capture and copying routes.** Each capture receives the final activity fields from both supplied cue trials, with matching cell indices. An edge update uses its two endpoints' activities from both episodes. Spatial locality therefore coexists with two-trial access; the cue schedule and coordinate correspondence remain supplied even when rebuilding noise becomes independent (C.25–C.26). Let $x_i^+ = \tanh(3v_i^+)$ and $x_i^- = \tanh(3v_i^-)$ denote the read-end activities transformed under cues $+1$ and $-1$, respectively, with $d_i = (x_i^+ - x_i^-)/2$ and $c_{ij} = (x_i^+ x_j^+ + x_i^- x_j^-)/2$. Ordinary snapshot capture

writes $c_{ij}$, while pure cue-contrast capture writes $d_i d_j$. The fallback capture introduced in E038 and frozen in E041 writes

$$\widehat{w}_{ij} = \begin{cases} d_i d_j, & d_i d_j \neq 0, \\ c_{ij}, & d_i d_j = 0. \end{cases}$$

These are products of the final developed activities, averaged across cues only for ordinary coactivity; renewal capture does not average a later holding trajectory. There is no epsilon, weight floor, rescaling, output label, donor template or use of past damage. Every exactly zero cue product is eligible for fallback, not only a connection identified as returning. Physically unavailable edges are zero. The **full-rewrite** candidate sets every available outgoing edge to its newly captured value. A distinct **preserving** policy keeps physically available incoming weights with $|w| > 0.5$ and writes the other available edges. Thus preservation includes a direct historical copying route that full rewriting does not. The reading input is erased before writing; the current developed state can retain its effects. Records are captured before an unchanged 128-step scaffold holding phase and protected during that phase. Only the resulting packet enters the next fresh body.

Protected copying carries the exact acquired record without a rebuilding body. Matched-erasure copying cumulatively zeros the same incident edges lost under each renewal mask, without rewriting remaining values. These have different information-loss contracts. The full-rewrite reading-blocked arm zeros the effective incoming record before normalization and development, then writes from its own body without a preserved-strong route. Its writing-blocked arm reads and computes normally but transmits zeros. Every arm continues its own history. Older preserving-policy ablations are retained for comparison but cannot satisfy E041's full-rewrite dependence requirements.

**Analytic structure of pure-contrast capture.** The following is a deduction from the writer and reader equations, not a new measured mechanism result. Let $a_{ij} = a_{ji} \in \{0, 1\}$ describe available edges after the relevant injury, and suppose the entire effective record is $w_{ij} = a_{ij} d_i d_j$ from one capture. On vertices incident to nonzero edges, define $s_i = \operatorname{sign}(d_i) \in \{-1, +1\}$ and $y_i = s_i v_i$. Here $s_i$ labels individual cell orientations; $s$ in C.17 labels the taught rule. Then

$$\operatorname{sign}(w_{ij}) = s_i s_j, \qquad \prod_{(i,j)\in C} \operatorname{sign}(w_{ij}) = \prod_{i \in C} s_i^2 = 1$$

for every surviving simple cycle $C$. Multiplying both coordinates of each coupling by their vertex signs makes it nonnegative: $s_i w_{ij} s_j = a_{ij} |d_i| |d_j|$. This is the standard structural-balance relation for signed networks; its use here identifies structure imposed by this particular writer, rather than a new general graph theorem. The consensus results for other dynamics are not assumed to hold for this noisy, clamped, finite-time model. Altafini, 2013

Taking the sums over surviving nonzero neighbours and writing $Q_i = \sum_j a_{ij} |d_j|$, the positive-strength reader satisfies

$$u_i = \frac{\sum_j a_{ij} d_i d_j v_j}{|d_i| Q_i} = s_i \frac{\sum_j a_{ij} |d_j| y_j}{Q_i},$$
$$s_i u_i = \sum_j P_{ij} y_j, \quad P_{ij} = \frac{a_{ij} |d_j|}{Q_i}, \quad \sum_j P_{ij} = 1.$$

Thus normalized coupling is a positive weighted average in the reoriented coordinates. The receiving cell's $|d_i|$ cancels locally; contrast magnitudes still weight its neighbours and affect other rows. Zero-strength vertices retain the implemented zero-drive rule, with no division or orientation claim. Although capture uses one contrast vector, the sparse masked weight matrix is not generally rank one. Each connected component's signs are determined only up to a common reversal; compatible input clamping can fix that orientation where it reaches the component.

The identity also distinguishes the numerical interventions. With $D_i = |d_i|Q_i > 0$ and a denominator floor $\epsilon$, the same ideal pure-contrast record gives

$$s_i u_i^{(\epsilon)} = \frac{D_i}{\max(D_i, \epsilon)} \sum_j P_{ij} y_j.$$

The extra factor attenuates input where $D_i < \epsilon$, restoring sensitivity to absolute strength. Edge deletion preserves cycle-sign balance but can disconnect the cue from an output. Sign-preserving rounding likewise preserves balance on surviving edges while generally changing the product magnitudes, normalized coefficients and support. Additive corruption can change signs as well. These are algebraic distinctions, not a decomposition of E052's observed losses.

The identities assume the pure-product weights and real arithmetic. Floating-point products and later channels need not preserve exact magnitude factorization; nonzero product signs retain the stated relation unless altered. Ordinary-coactivity fallback and incoming strong-weight preservation can mix incompatible orientations, so neither whole-record factorization nor balance is guaranteed for those policies. Nor does a positive cycle product establish the taught port relation, sufficient connectivity, amplitude, convergence within the budget or successful renewal. The supplied writer guarantees this structure for pure-contrast capture; the particular orientations come from the developed cue-trial activities, which can include noise. A mechanism test must separate those properties rather than use balance alone as a success predictor.

**E041 endpoints and fixed decision.** Blocks 32000–32031 are fresh; earlier experiments motivate the frozen policy but supply no cases to its estimates. Assay the acquired packet and generations 1, 3, 8, 16, 24 and 32 in fresh terminal bodies. The primary reader uses local strength normalization and the primary terminal injury removes 24 scattered non-port cells. Intact and square-nine injury panels, fixed-four reading, and own/foreign/shuffled/erased/untrained/communication-off/source-cut controls remain retained. Terminal readers never write into the lineage. A cue succeeds only when both output-patch means over the last 32 updates have the instructed signs with signed margin strictly greater than 0.25. Both-cue success requires this for each opposite cue in the matched lineage; it is neither site accuracy nor qualified pattern preference from the theta-field experiments.

The single relative primary is full fallback rewriting minus fallback writing with strong preservation at generation 32, normalized/scattered own-record both-cue success. Its 95% whole-block bootstrap lower bound must exceed five percentage points. Full qualification also requires seventeen durability conditions: intact and scattered both-cue lower bounds above 75% at generations 8/16/24/32 (eight); all original 24 context gates at each of those horizons (four); scattered advantages over each matched full-rewrite reading/writing ablation with lower bounds above ten points at generations 16 and 32 (four); and initial acquisition passing the original 24 gates (one). C.17 defines the original gates. Use the registered 10,000 whole-block resamples; intervals are pointwise

and there is no global familywise claim for all alternatives. The 2,881,536 individual terminal-cue scores and 294,912 body-cue scores are repeated observations within 32 blocks, not independent samples. A 100% sample bootstrap endpoint does not establish population certainty.

E039's confirmed preserving component and failed full-rewrite superiority remain distinct; E040's preserving candidate remains below its absolute durability requirements. E041 qualifies its fresh candidate, without retroactively changing either verdict. The figure uses the six post-acquisition checkpoints for five policies in one cohort. Every point and original rounded interval is extracted from the report, with no recalculated bootstrap or interpolation treated as an observation. Its separate E046 annotation describes the full-rewrite policy's rebuilding-body history: 307 of 512 lineages fail at least one of the 32 saved body readouts, so 205/512 (40.04%) pass every readout. The complement and percentage are simple arithmetic from the report; no interval is newly estimated. This tests success at each recorded body assay, not at every instant, and differs from E041's additional fresh-recipient terminal assay. It is not an added qualification criterion or an independent cohort.

The history counts in main-text section 7.1 use the same E046 report: of 428 terminal successes, 236 have an earlier failed body readout; of 84 terminal failures, 71 do. The complementary cells are therefore 192 and 13. Counts sum to the original 205 always-passing and 307 ever-failed histories. The portable extract and extraction script bind the report to its retained completion hash and the existing v0.14 source archive. This is report arithmetic, with no new trajectory, raw-array recount, interval or qualification. Terminal success follows an additional injury in a fresh body; it is not the generation-32 rebuilding-body readout. Neither this cross-tabulation nor the recovery fraction estimates repeatability from a particular frozen record.

**Interventions on saved histories.** E026 adds/removes the connections restored in E025 final packets and recomputes receiving denominators; it is a terminal packet intervention. E028 branches known generation-two histories into weak-edge availability and writing treatments. E037 measures returning-edge restoration in E036, where zero inherited connectivity makes paired cue responses identical. E046 uses all saved E041 body scores and counts the first failure, recovery and censored follow-up; groupings by final outcome are descriptive. The 243/281 recovery fraction includes only first failures with three subsequent bodies observed. It neither certifies strict-eight resemblance nor creates independent recovery trials.

E047 selects each lineage's first failed body at generations 1–31 before its new intervention, retaining 301 eligible cases; 205 never fail and six first fail at generation 32 without an observed successor. At the outgoing failure packet, targeted rollback restores the exact incoming value on physically available edges changing from $|w| > 0.5$ to $|w| \leq 0.5$. Random rollback restores the same number from available old-strong changed coordinates, allowing overlap; it does not match weight mass or sign. Current erasures and new weak repairs remain. Replay only the next body with original full-size random streams and next injury, recalculating denominators. There is no new capture or later serial continuation. Pooled eligible-case ratios use all 32 blocks in 10,000 resamples, with empty-denominator draws handled explicitly. Joint benefit requires lower bounds above +5 points versus unchanged and above zero versus random; the first requirement fails. Original next-body outputs exactly reproduce E041. The executed v2 corrects a signed-zero assertion, without retuning the intervention or selecting a new cohort.

**Provenance.** E61–E64 link native reports. E041 protocol and E047 protocol specify the frozen candidates. The v0.14 package retains protocols, reports, available native summaries and selected completion/audit records. Several scientific summaries and arrays are absent locally; the E041 figure therefore uses its completion-bound report, not fixture or benchmark data. Passing original audits are retained records, not audits rerun during manuscript preparation.

### C.24. Cellular preparation, component substitutions and communication timing

**Model, ancestry and endpoints.** P17–P31 retain C.7/C.14/C.18's corrected BITSEY regulatory/ionic model, full 17-component cell states, cell identities and imposed replacement schedules. E is classified when miR200 minus Zeb mRNA exceeds +2, M below −2, and X otherwise. These are regulatory labels, not diagnoses. Retention requires the matching endpoint and at least 95% matching samples in the final quarter of each of two isolated 10,000-unit holds. The stricter every-replacement-window criterion applies the corresponding occupancy/endpoint requirement during each renewal window and is reported separately. Onward assays expose three fresh M recipients to actual final donors for 20,000 units, remove donors, and assess both isolated holds. Open, RNA-blocked and boundary-cut readouts use the same donor outputs, including failures. Whole-trio wipe, ordinary-E capacity and historical controls remain required where declared; reuse of previously qualified controls is not a new independent replication. P31 is a focused numerical study without new onward assays.

P17 forks the actual first-newcomer trio after intervals 100, 300, 1000, 3000, 10000 or 20000 into isolated retention and continued replacement. The 300-unit final-renewal/onward result narrows P16's earlier tested gap. It does not qualify every intervening window; longer P17 schedules can meet that stricter endpoint but were not all given new onward assays. P18 crosses first and later intervals in {100,300,1000}, preserving complete saved states; all five subsequent placements follow the same sequence. Preparation and continued turnover therefore vary separately, while all original carriers are removed.

**Preparation and component interventions.** P20 uses the pre-existing trio after its first 100-unit interval and its saved state after a further isolated-200 preparation. At a common registered logical clock it substitutes complete before/after states for newcomer column 0 and surviving-neighbour column 2; column 1 is immediately replaced. This controlled state assembly is not an uninterrupted unmanipulated history. P21 substitutes selected primary concentrations from each cell's own prepared state: ZEB protein, the paired miR200/Zeb mRNA totals, and Na/K/Cl/P ions. Eight masks at each of two locations share one unaltered baseline, giving fifteen unique component assemblies per history/resolution. Three historical controls give 72 renewal parents across two origins and two numerical resolutions. Snail and ST remain fixed. One native algebraic RNA-binding refresh recalculates only derived variables while preserving primary concentrations; historical/refreshed controls qualify that operation. Voltage is not independently overwritten.

All parents continue replacements at positions 1,2,0,1,2, 300 units apart, followed by two isolated holds and all three onward readouts. Primary component changes preserve every unselected variable; they do not model injection doses. P22 separately tests the RNA species, incoming-draw variations and selected longer schedules. P25 reciprocally replaces only the first post-fork incoming ZEB concentration between legacy and additional draw sequences, retaining the remaining sequence. Alternative sufficient newcomer components, the neighbour's RNA dependence and

the context-sensitive incoming-protein result apply to these backgrounds, not a unique universal memory carrier. Direct mRNA/protein transport is absent in the native law; free-miR200 transport and ongoing ionic dynamics remain.

**P26 timing grid.** Use the unprepared assembly, written/ordinary histories, legacy/17021 incoming sequences and two numerical resolutions. After the archived first replacement, five further placements yield two total turnover cycles. Only the first post-fork 300-unit challenge receives a one-time RNA-transport interruption. Compare immediate durations 25/50/75/100, duration 100 beginning at 25/50/75/100/150/200, continuously open transport and a full 300-unit block. The block suppresses RNA transport on [start,start+duration); ionic dynamics remain, transport reopens afterward, and all later intervals stay open. No prepared-state substitution or repeated pause is added. Each schedule's eight combinations are related deterministic settings, not eight independent trials. Every parent output, successful or not, undergoes all onward assays.

The common diagnostic samples the tracked first newcomer at 0,15,...,300, distinguishing sampled resistance from sampled loss followed by recovery. No continuous-time resistance is inferred between samples. P26 integrates the challenge in twelve 25-unit phases, saving samples every 1.25 and using maximum steps 0.25/0.125 at its two settings. All 198 registered numerical comparisons pass. Its eight parent combinations per schedule and 396 total cases are not a population success-rate study. The 25-unit result does not inherit longer-cycle durability from a different pause schedule; no exact duration or onset threshold is inferred between tested values.

**Numerical follow-ups and stopping.** P27's finer timing grid stops on intermediate coarse/fine label disagreements before new onward assays. P28 finds that downstream refinement alone cannot remove all saved-source dependence. P29 independently varies preparation-prefix and downstream resolution, qualifies its four targeted schedules, and supplies no new transmission endpoint. P30 retains that repaired preparation treatment but stops at a newly tested start-15/duration-100 schedule: three intermediate labels disagree at relative time 175, even though all final outcomes agree and numeric errors meet their separate tolerance. Intermediate-label agreement was an additional registered gate; it is not waived. None of P30's 624 onward assays runs.

P31 reproduces those disagreements, then varies preparation and downstream resolution independently and jointly through divisors 32 and 64. It retains the native equations, classifier, incoming sequence 17021, two histories, two source bundles and four fixed schedules. Finest normalized errors must remain below 0.01 and contract relative to the preceding refinement; registered outcomes and sampled labels must agree, including the target's 41-label neighborhood. All 28 finest comparisons and ten source comparisons pass. The temporary label is resolved but final retention outcomes are unchanged. These finite refinement checks do not establish exact-solution convergence, an exact pause threshold or onward transmission from the refined parents. P27/P30 retain their failed gates; the subsequently completed P32 panel is specified separately in C.40/E85.

**Attribution and availability.** E65–E68 link the retained reports/protocols. Large P17+ trajectories and associated external files are unavailable at this checkout's `/mnt/workspaces/levin-cancer-gj-exploration` paths. The P19/P24 result links are broken external symlinks; their protocols do not establish an outcome, and later reports' contextual mentions are not a fresh verification of those campaigns. New cellular numbers retain report attribution. The 10 October source selection adds the P31 completed report, protocol and saved integrity review to the earlier launch record; it is not a complete execution archive. Neither simulations nor original mathematical audits were rerun. No

cell division, autonomous turnover timing, unique molecular carrier, acquired-history advantage or treatment efficacy is established by these panels.

### C.25. Rebuilding cue-pairing robustness: E051

**Cohort and information boundary.** E051 tests the original functional apparatus of C.23 on fresh blocks 34000–34031, with four founders and four recipients per block: 512 lineages per condition. Seven rebuilding pairings cross five fixed writing policies, giving 35 counterfactual histories per lineage. Prior scientific cases are neither reused nor pooled. Each history passes its own 4096-byte packet through 32 fresh damaged bodies, with 24 non-port cells absent under the matched injury schedule. Desired outputs are not retaught. Teaching, the post-capture holding phase and terminal assays keep their original paired-cue randomness; only the rebuilding trials before capture change.

**Correlation intervention.** Initial-state correlation and update-noise correlation vary separately. Their pairs are (1,1), (1,0.999), (1,0.99), (1,0.9), (1,0), (0.99,1) and (0,1). The first cue receives the unchanged Gaussian draw $z$; the second receives $\rho z + \sqrt{1-\rho^2}\, e$ with independent Gaussian innovation $e$. At $\rho = 1$, the implementation copies the original draw exactly. Both marginals retain their original distribution and scale. Common draws match policies and perturbations, while new streams apply at later boundaries. Reset mismatch concerns the starting activity of the two trials within a rebuilding body, not persistence of activity across bodies. Joint noise-and-reset mismatch is not tested.

**Fixed policies.** With C.23's transformed activities, let $q_{ij} = d_i d_j$ and let $c_{ij}$ denote ordinary two-cue coactivity. Exact rewriting writes $q_{ij}$ unless it is exactly zero, in which case it writes $c_{ij}$. Tolerance rewriting instead selects $c_{ij}$ when $|q_{ij}| \leq 10^{-4}$. This one dimensionless threshold is frozen before outcomes; no alternative is selected from a sweep. Ordinary writing always uses $c_{ij}$; pure contrast always uses $q_{ij}$. The fifth policy retains physically available incoming weights with magnitude above 0.5 and applies exact rewriting elsewhere. This strong-preservation policy is a mixture of copying and writing, not protected storage. Unavailable edges remain zero. Reading uses 2048 updates with 1024 settling updates, capture precedes the unchanged 128-update holding phase, and local strength normalization remains unchanged.

**Endpoint and decision.** Own-record fresh recipients are assessed at acquisition and after 8, 16, 24 and 32 rebuilds under intact, scattered-24 and square-nine injury, with shared erased-record controls. The primary endpoint is normalized/scattered both-cue success at horizon 32, using C.23's strict output margin above 0.25. The original exact writer has two primary challenges: update-noise correlation 0.999 and reset correlation 0.99. All three paired/challenged success lower 95% bounds must exceed 0.75, and each challenged-minus-paired lower bound must exceed −0.05. These are five conjunctive requirements. A failed noninferiority requirement alone would not establish fragility; a loss exceeding five points is separately identified only if the difference's upper bound falls below −0.05. All five requirements pass.

The registered analysis uses 10,000 shared whole-block bootstrap resamples. Stronger perturbations, other policies, shorter horizons, other injuries and mechanism measurements are secondary with pointwise intervals. The original 17-part E041 durability/control battery is not repeated. Terminal success remains distinct from success at every rebuilding-body readout.

**Mechanism observations.** Available returning edges touch cells absent in the previous body and present now. Counts distinguish exactly zero cue products, ordinary fallback selection, nonzero

and strong outgoing values, and access to the input through nonzero connections. The summaries pool boundaries 2–32 within each block before resampling blocks. Under paired exact writing, ordinary fallback is selected on all available returning edges; under every noise-mismatch regime it is selected on none, while all those edges acquire nonzero values. Preserving policies can bypass capture on old strong edges, so selection counts alone do not describe all transmitted values. Reachability ignores sign and is not itself a functional endpoint. With mismatched noise, a nonzero sampled cue difference can include trial-specific fluctuation; it is not automatically an isolated causal response to the cue.

**Provenance and ongoing scope.** Twenty scientific inputs froze on 8 October before outcomes. The saved independent audit reports replay of teaching, rebuilding, holding, capture, connectivity and assays, checking 1,146,880 rebuilding-body cue observations, 448,512 terminal cue observations and 1,966 summary statistics. These repeated observations are nested within 32 independent blocks. The workflow finished on 9 October at 06:18 UTC; closeout passed at 10:29 UTC. Legacy byte-equality checks apply to excluded fixture/benchmark cases, not to fresh scientific cases. The later complete result supersedes the retained eight-block descriptive look.

The 9 October incorporation checks completion/audit/closeout bindings, all 20 frozen input identities, the 1,966 saved block means and the five decisions without recomputing intervals or replaying dynamics. Its portable package contains 46 text, code and metadata records, including the native summary; the original NPZ arrays are absent locally. The separately completed E052 numerical comparisons enter the 10 October update as E70 and C.26. E051 source selection; native extract; portable records.

## C.26. Precision, stored values and normalization: E052

E052 retains C.23/C.25's exact-zero writer and functional task. Blocks 35000–35031 are fresh; four founders and four descendants per block give 512 lineages. Eleven fixed numerical regimes cross rebuilding-noise correlations 1, 0.999 and 0, for 33 histories per lineage. Initial cue activities remain paired, as do teaching and terminal cue trials. Each condition transmits its own record through 32 damaged rebuilds; no prior cases are pooled.

The primary challenges under correlation 1 are float32 computation and storage, uniform weight-grid spacing $2^{-16}$, independent Gaussian noise of SD $10^{-4}$ on available nonzero weights, and a local denominator floor $10^{-4}$. The reader divides incoming signed drive by $\max(\sum_j |w_{ij}|, \epsilon)$ for positive strength and explicitly uses zero drive at zero strength. Noise is clipped to $[-1,1]$, preserving exact-zero and physically erased slots in the primary corruption arm. Channels apply once after acquisition and once after each capture, not again for diagnostic assays. Teaching and post-capture holding remain float64. Stored-float32-only, stronger grids/noise and alternate floors are separate controls. All regimes retain the canonical 4096-byte container; numerical content, not payload size, changes.

At horizons 0, 8, 16, 24 and 32, each transmitted record receives native-reader and common float64/zero-floor assays under intact, scattered-24 and square-nine injuries. The primary is horizon-32 native-reader scattered both-cue success. Each of four challenges must have a success lower 95% bound above 75% and a challenged-minus-reference lower bound above $-5$ points; the reference lower bound must also exceed 75%. All nine are required. A loss beyond five points is

established only when the difference's upper bound falls below $-5$; failure of noninferiority alone is insufficient.

The original analysis uses 10,000 shared whole-block bootstrap draws. Secondary correlations, regimes, reader comparisons and interactions retain pointwise intervals. The saved audit and close-out pass; incorporation reads their receipts and summaries without replaying trajectories or bootstraps. C.26 adds no combined-stress result. E70

## C.27. Prospective record-based forecasting and injury-transfer development

**TT008 frozen prediction.** A logistic candidate with ridge 0.01 is selected from TT007's exposed development panel and fitted once on all 512 development records. Two primary rivals use current expression/history, and those features plus unsigned access. The candidate additionally receives signed transfer. Secondary frozen rivals include quadratic features and nonlinear expression and full-packet models; their larger development searches do not make them new independent tests.

All primary predictors receive the four generation-14 cue/output-port window means aligned to the taught rule, their minimum, the current pass indicator and three history indicators. The expression/history rival receives no outgoing-record feature; the next rival adds unsigned access, and the candidate adds signed transfer as well. Block, founder, recipient and seed identities are excluded from the design matrix. Task orientation is already used to construct the common expression features, even though it is not a separate model column.

For outgoing weights w, the diagnostic constructs $T_{ij} = w_{ij} / \sum_k |w_{ik}|$, with zero-denominator rows set to zero. A 256-site state starts at zero outside the known input port and one inside it. Each of 64 steps multiplies by T and reclamps that input port to one. The signed feature is $\min(s\bar{u}_A, -s\bar{u}_B)$ at the known output ports, with the taught orientation $s \in \{-1, +1\}$. The unsigned feature repeats the procedure using $|T|$ and returns the smaller unaligned output mean. Ports are supplied to both diagnostics; the signed readout additionally uses the orientation. The candidate therefore uses 64 unsigned and 64 signed sparse operator applications per record; the unsigned rival uses 64 and expression/history none. Matrix assembly, reading the 512 weights and predictor evaluation are additional work. These are offline model-informed features, not extra probing of a physical body or forward replay of its nonlinear continuation. All models retain the same exposed development labels, but representation, training and diagnostic costs differ. No equal-compute or equal-probe superiority claim follows.

Test blocks 8800000–8800063 each contain four histories, four paired-rule founders and four recipients: 4,096 related records in 64 independent blocks. Histories use the original exact-zero writer throughout, one tolerance event at generation 3, spaced events at 3/5/7/9/11/13, or clustered events at 3/9/10/11/12/13; the tolerance is fixed at $10^{-4}$. The forecast boundary follows the generation-14 read and common exact capture. Every branch then uses the same exact writer through generation 32. All six models' predictions are sealed before any post-boundary continuation. The endpoint is strict both-cue success, margin above 0.25, in a fresh generation-32 scattered-injury assay.

Brier error is the mean squared difference between forecast probability and binary outcome. The candidate must reduce equal-block error by more than 0.002 against both primary rivals, with each one-sided 95% Student-t lower limit above that margin. Inference is approximate for independent block differences and conditional on fixed development training. No test-set fitting, resizing or threshold revision occurs. Matched spaced/clustered decisions use the fixed 0.02 output caliper

and remain secondary. The saved audit reports full carried-state replay and independently recomputed targets; prediction chronology is procedural within a shared filesystem. E71

**Forecast presentation and post hoc diagnostics.** Figure 13 uses all 4,096 retained forecasts from the candidate and two registered primary rivals. Pass/fail contributions were specified as descriptive strata in the original protocol; reliability and ROC AUC were added for this review. Reliability groups probabilities in ten equal-width bins $[0, 0.1), \ldots, [0.9, 1]$, plotting each nonempty bin's mean probability against its observed success fraction. The portable CSV supplies every bin count; no points or records are selected by error size. The binned calibration summary is $\sum_b (n_b/N)(\bar{p}_b - \bar{y}_b)^2$. It depends on the bins and is not an exact decomposition of the original, unbinned Brier error. AUC is the fraction of successful/unsuccessful record pairs correctly ranked, counting a tie as one half. Rank-sum and explicit pair-count calculations agree. Subgroup summaries pool records within the declared current-status class; they are descriptive and receive no new confidence intervals. The primary inference remains over the original 64 block differences, conditional on the frozen development fit.

During incorporation, the reproducible renderer checks all 24,576 scored predictions against the sealed prediction CSV, all 4,096 targets against retained terminal CSV rows, the six original mean Brier scores, 128 primary block contrasts and four subgroup contributions. The seal and all 64 future-start receipts preserve the declared chronology; this is a metadata check, not independent custody or a new raw-array replay. Original native/scoring audits remain attributed to their receipts. The source package, inspection record, derived data, bin counts and asset manifest retain the inputs and checks. No simulation, model fit, recalibration or new uncertainty estimate is performed for this presentation.

**P03/TT009 injury comparisons.** Both P03 studies retain the same 64 generation-14 failures from eight exposed blocks. Fresh bodies read incoming, outgoing or deletion-control records under the original injury or fresh uniform 24-cell injuries, with paired repeats and the original strict scoring. Fixed-injury runs retain three timepoints through 4,096 updates; the primary comparison and transfer study use 2,048. Effects equally weight the eight blocks; the original 20,000-draw block-bootstrap intervals are descriptive. The fresh-injury prediction requires a positive mean outgoing-minus-incoming effect, which fails. Direct probing selects using repeats 0–3 and evaluates on 4–15; signed/unsigned diagnostics and direct probing have different costs. TT009 subsequently reproduces all trajectories and scores as an audit of exposed evidence, not another scientific cohort. E72

## C.28. One-time partial rollback and fresh continuation: E048–E050

E048/E049 use eligible E041 first failures; E050 generates new histories in blocks 33000–33031, with 512 lineages and no pooling. The unchanged path alone determines enrollment: first failure at generations 1–29 leaves three successors available. All 294 eligible E050 cases remain, including zero-edit cases; 21 later first failures and 197 never-failing lineages are reported separately.

At the failed body's outgoing boundary, available edges moving from incoming magnitude above 0.5 to outgoing magnitude at most 0.5 receive either no edit, half restoration $w_{new} + 0.5(w_{old} - w_{new})$, or the exact incoming value. Erasures and other coordinates remain unchanged. The intervention has direct access to the old record, unlike ordinary full rewriting. Each branch develops under matched later injuries and randomness, captures its own next packet and continues for three bodies. Rollback is applied only once.

The third body's both-cue success is primary. Ten thousand whole-block bootstrap samples estimate pooled eligible-case ratios. Joint confirmation requires lower bounds above five points for half versus unchanged, above zero for half versus full, and above zero for reduced losses of originally successful cases. All-three success, any recovery and relapse are separate secondary endpoints. E050's original branch exactly reproduces its own unchanged history; the retained audit checks enrollment, all edits and serial records. Source qualification and the finite outcome do not establish an autonomous intervention trigger. E64

## C.29. Finite decision-model timing confirmation and challenge

The two new FEP-track studies retain exact Bayesian/VFE inference and a separately supplied expected task-cost objective over the current plus two future decisions. A wrong report costs 1 and a purchased correct answer costs 0.1; only purchased answers provide deployment labels. Sixty-four calibration pairs cost 0.064 in total. CT3-flat retains calibration and supplied general/A-failed/B-failed change hypotheses, with weak new-rate priors. It differs from calibrated CT3 and from C3's earlier general-change model; the nine fixed policies are retained in both studies.

Each study uses 1,024 fresh paired blocks per fixed base law, 27,648 independent blocks across 27 bases. The confirmation has 513 physical conditions and 525,312 episodes. The timing challenge retains the old conditions and adds off-schedule changes, giving 1,161 conditions and 1,188,864 episodes per policy. Modelled times are 26/51/76; unexpected physical times are 13/38/63/88. Actual change times are not supplied to the learner. Stable cases are simulated once per block; multiple policies and schedule conditions share draws within blocks.

Post-change recovery averages each condition's cost over its remaining tasks and excludes sunk calibration. Complete-episode cost includes calibration; full-core means give stability half the weight. All three confirmation/challenge criteria require negative upper 95% bounds for recovery-minus-0.1, recovery-minus-C3 and full-core-minus-C3. Intervals are nominal stratified Monte Carlo mean $\pm 1.96$ SE, with policy contrasts paired within block; they do not quantify generalization to arbitrary worlds. Reports retain harms, transfer outcomes and zero-headroom cases. Saved-count verification is attributed to the source records, not rerun here. E73

## C.30. Recipient-coupling pilot and source-selection limits

D180 uses four exposed donor blocks (90000/90005/90010/90015), four lives each and both target families, giving 32 F0 records. Coupling multipliers 0.75/1/1.25 cross two writers, two paired future replicas and fixed copying/read-off/write-off controls. Three serial replacements produce 1,152 labelled lineages; these remain nested within four reused blocks. Coupling changes both reader and writer dynamics. A common reference reader assesses final record usefulness, alongside native-reader secondary results.

The focal group is family B with ordinary writing. Record, recipient and interaction components use cross-replica signed estimates; a negative estimate is retained rather than interpreted as negative physical variance. Rank reversals must agree in both futures to meet the fixed consistency rule. All family/writer groups, donor omissions, shape correlations and controls remain. These are descriptive finite-panel comparisons without population confidence bounds or a prospective

prediction test. The formal 10 October closeout supersedes the report's historical pending-audit label. E74

The 10 October source manifest and portable records bind the selected reports, protocols, native summaries and audit metadata for C.26–C.30 and the P31 addition. They preserve that dated selection. The later E053, L20 fresh-futures and change-at-any-task outcomes enter separately below; the broader cellular pause archive remains unavailable. Neither manuscript package is a complete executable/raw archive.

## C.31. Combined numerical challenges: E053

The fresh 32-block cohort has 512 lineages per condition and twenty conditions over 32 damaged renewals. The frozen primary uses independent cue noise during rebuilding; teaching, initial activity and terminal cue trials remain paired. The combined condition applies float32 computation, uniform rounding to a $2^{-16}$ grid, SD $10^{-4}$ noise on nonzero stored weights, and a $10^{-4}$ normalization floor. The packet allocation is unchanged.

**Channel order and zero handling.** At acquisition and after each capture, the channel acts once, giving 33 applications. It first adds one Gaussian draw per physically available reciprocal edge that is nonzero in the pre-channel packet, clipping to $[-1, 1]$. Exact zeros and unavailable edges receive no noise. It then rounds to the grid using nearest, ties-to-even rounding, and finally casts to float32 before exact expansion into the common float64 container. Reading and capture use float32 in that condition; teaching and subsequent holding remain float64. Assays receive the processed record without another channel application. Noise can move a small nonzero value across the grid's zero/nonzero threshold, or change its sign; the composed channel therefore need not be more damaging than rounding alone. The writer still selects its coactivity fallback only when the computed contrast product is exactly zero. A finite writer tolerance is not one of the four challenges.

The primary endpoint is both-cue success in the native-reader fresh scattered-injury assay after renewal 32. Three gates require the reference and combined-condition 95% lower bounds to exceed 75%, and the paired difference lower bound to exceed $-5$ percentage points. Ten thousand paired whole-block bootstrap resamples give the retained intervals. Both absolute gates pass, while the difference lower bound is $-5.078125$ points; the conjunction fails without establishing a loss larger than the allowance. Matched-noise and common-reader comparisons remain secondary. Prior cohorts and the interim eight-block summary are not pooled into the completed estimate. E75

**Original factorial comparisons.** Grid rounding $Q$, stored-weight noise $N$ and normalization floor form a complete $2 \times 2 \times 2$ design in float64; arithmetic precision is crossed only with the reference and all-three-challenge endpoints. At horizon 32 under scattered injury, the protocol specifies the secondary interaction $p_{QN} - p_Q - p_N + p_0$, with the floor absent. Contrasts are formed within each block before the original 10,000 shared bootstrap resamples. D.8.6 transcribes the retained pointwise 95% intervals; no new uncertainty calculation is performed for this addition. The four histories share the starting acquisition and declared draws but subsequently diverge. This is an intervention on the entire repeated channel history, not a comparison of channel orders applied to one fixed packet. The review source record binds protocol, implementation and summaries.

The saved independent audit replays all dynamics, channels, mechanism diagnostics and assays, checks 655,360 rebuilding-body and 556,032 terminal cue observations and 2,561 summary statistics, and binds the completion receipt. These observations are nested within 32 blocks. The

manuscript verifies retained report/summary hashes and the audit's completion identity; it does not rerun this audit.

### C.32. Record-feature interventions: E054

All E051 blocks 34000–34031 remain, with four founders and four recipients per block. Acquired records and four horizon-32 histories (full rewrite or strong preservation, each under matched or independent cue noise) supply 512 records per history. Acquired records repeated across recipient slots do not create new acquisitions. Eleven intervention arms use fresh E054 reset/noise and injury draws paired across histories and arms, float64 computation, zero denominator floor, 2,048 reading steps, 1,024 settling steps and the strict output margin 0.25 over the last 32 observations. Intact, scattered-24 and square-9 injuries preserve task ports. The assay performs no writing.

Uniform scaling by one half and exact deletion/restoration are null controls. Magnitude permutation occurs within positive and negative classes; equalization preserves signs, support and total absolute mass. Anchored switching selects eight non-port cells, transforms every incident edge as $w'_{ij} = s_i w_{ij} s_j$, and transforms reset and all noise draws by the same signs; task ports have $s = +1$. Isolated sign flips match the occupied edit count within eight magnitude bins, but not exact mass or positions. Another arm deletes exactly those flipped edges. Two further arms compare deletion of returning repaired edges with other edges matched by count within the same bins, retaining zero-exposure cases. Erasure is the negative control.

The primary family comprises magnitude-permutation minus untreated, isolated-sign minus anchored-switch, and same-edge deletion minus untreated, for the matched-noise full-rewrite history under scattered injury. Ten thousand paired whole-block bootstrap draws provide 98.333333% two-sided intervals (Bonferroni family of three). Loss exceeds five points only if the upper bound is below $-5$; a lower bound above $-5$ establishes loss within that allowance; otherwise the classification is unresolved. Other history/injury comparisons use pointwise 95% intervals. Numerical scale, restoration and switching controls pass, allowing the prespecified sensitivity classifications. E76

**Post hoc history-by-permutation interaction, added in v0.20.** For full-rewrite records under scattered injury, let $p^h_{b,a}$ denote both-cue success averaged over the sixteen founder/recipient combinations in block $b$, intervention $a$ and rebuilding history $h$. Define loss $L^h_b = p^h_{b,\text{untouched}} - p^h_{b,\text{permuted}}$ and estimate

$$I = \frac{1}{32} \sum_{b=1}^{32} \left( L_b^{\text{matched}} - L_b^{\text{independent}} \right).$$

Positive $I$ means a larger permutation loss after matched-noise rebuilding. All four cell means retain the same block identities, and the contrast is formed before resampling. The analysis reuses E054's 10,000 whole-block bootstrap draws and reports the 2.5th/97.5th percentiles with linear interpolation. Its semantic random key is the compact JSON array `["E054","bootstrap","recorded_interventions"]`, SHA256-hashed to the first sixteen bytes, interpreted as a little-endian integer for NumPy PCG64. This is a reviewer-requested, outcome-informed analysis with a pointwise 95% interval, not a prespecified primary test or a multiplicity-adjusted claim. Its estimate is 72.8516 points [68.5547, 76.9531]; the original qualifications remain unchanged.

The analysis specification, source manifest, script and block counts make the calculation portable. Frozen run/inference code establishes block order and common-reader pairing; the summary is bound to completion and the recorded audit. Eight original means/intervals reproduce exactly, and resampling by integer multiplicities independently reproduces the new bootstrap distribution. This is new statistical reaggregation of saved summaries, without raw-array recounting, dynamics or another scientific audit. Permutation maps are history-specific, so the interaction compares sensitivity to the declared intervention procedure, not identical edge edits across histories. The original policy-by-intervention contrasts compare full rewriting with strong preservation within each rebuilding regime and remain separate secondary analyses (D.8.7).

The saved audit independently reconstructs source extraction, perturbations, graph/coupling diagnostics, reader trajectories, scores and all 1,632 summaries. Its 168,960 terminal cue observations do not increase the 32-block inference unit. Sign-cycle counts depend on the fixed cycle basis; comparison across changed supports is descriptive. Deletion changes both access and normalization. The interventions are not equal doses of a common quantity and do not isolate a unique mechanism for the observed writer-policy differences.

### C.33. Repair development, protected resources and serial scope

TT010–TT013 retain 64 failure-selected cases from eight exposed blocks. Equal-block averaging and descriptive whole-block bootstrap intervals preserve each source protocol; the same cases are not new confirmations at each stage. TT010 combines outgoing values on available edges with protected incoming values elsewhere. TT011 tests imported donor signs and exact compatibility, checking all 256 contradiction certificates and retained task flags but not replaying every nonlinear trajectory. TT012 implements approximate alignment and a recipient-based solver while retaining protected support/magnitudes. TT013 removes those protected resources: only the surviving outgoing record O, lattice, input pins and fixed optimizer options enter completion. Its 16-start, at-most-64-sweep sign solver uses fixed seed 2026101004, fills every exactly zero edge at the median nonzero absolute O weight and preserves all existing nonzero weights. An empty record stays empty. Shuffled controls preserve added support, magnitudes and sign counts. Native replay is complete for TT010/TT012/TT013; independent optimizer checks do not establish global optimality. Algorithm 1 in C.34 now gives the objective and complete update procedure. E77

Reported serial development instead starts from 256 exposed generation-14 records in sixteen blocks, four per history stratum. Every branch inherits its own body-written successor across generations 15–32. Untreated, one-time repair, repeated repair and matched-sign-shuffle schedules are compared under paired cues; a later study varies ongoing cue-noise correlation while preserving teaching and resets. Withdrawal adds one new branch with only the first repair, copying verified untreated/repeated controls from the noise study. The primary is uninterrupted success over eighteen subsequent bodies; generation-32 body success is not a separate fresh read of its outgoing record. Intervals are descriptive stratified block bootstraps, without a practical-superiority margin. Record size is unchanged but repeated repair adds computation and edge interventions.

These serial reports contain independent construction/scoring checks and selected trajectory checks within the same lane. TT015 was running at the v0.19 cutoff and is now complete, with full native replay and separate scoring of the serial/noise studies (C.38). The withdrawal comparison uses exposed controls; because the repeated arm already retained every first-pass lineage and

first reconstructions match, the possible direction of its uninterrupted contrast is constrained. These results are leads about continued supplied correction, not prospective fresh-cohort serial validation or autonomous maintenance.

### C.34. Prospective surviving-record repair: TT014

The frozen TT013 repair is tested on 64 new block identities 14000000–14000063 under a distinct seed namespace, excluding exposed development cohorts. Four writing histories are cyclically assigned between blocks: exact rewrite, one tolerance pulse, spaced pulses and clustered pulses, sixteen blocks each. Each block has four paired founders and four recipients. Every generation-14 outgoing record is enrolled before assessment; no success filter is applied. Consistent teaching has 512 steps and 128 captured observations; fourteen rebuilds retain the native 2,048-step reader, 1,024-step settling period, 24 missing non-port cells, dt 0.2, noise SD 0.02, birth SD 0.05 and writer beta 3.

The repair receives O and fixed options only. Incoming backup, donor, injury mask, body, history label, target-output label and future scores are unavailable. Input pins, lattice, solver and reader remain supplied. Algorithm 1 specifies the frozen TT013/TT014 procedure, also retained by TT016. A separately constructed shuffle preserves support, magnitudes and added-sign counts. All 1,024 shuffles change sign placement; 94,524 edge additions are recorded. Unmodified, repaired, shuffled and erased records each receive eight paired fresh injury/reset/noise draws, totaling 32,768 paired-cue contexts. Strict intended-task minimum margin above 0.25 in the final 32 observations defines success. No assessment-time writing or serial follow-through is included in TT014.

**Algorithm 1. Completion from surviving record structure.**
**Inputs and objective.** The float64 record $O$ contains 512 undirected edges on a periodic $16 \times 16$ lattice. Cells have row-major indices $i = 16y + x$; edge order is all rightward edges, then all downward edges, each in cell order. For each cell, set $a_i = \sum_{j \sim i} |O_{ij}|$. Define

$$c_{ij} = \tfrac{1}{2} \operatorname{sign}(O_{ij}) \left( \frac{|O_{ij}|}{a_i} + \frac{|O_{ij}|}{a_j} \right),$$
$$J(g) = \sum_{\{i,j\}: c_{ij} \neq 0} |c_{ij}| \, \mathbf{1}\{g_i g_j \neq \operatorname{sign}(c_{ij})\}.$$

A fraction with zero denominator is zero. Search $g_i \in \{-1, +1\}$ for small $J$, with the four input cells $i \in \{113, 114, 129, 130\}$ fixed at $+1$; no output signs are pinned. The objective measures disagreements with surviving edges, not task error.

1. **Initialize sixteen starts.** Use all-positive signs; a signed maximum-weight forest; and fourteen random sign arrays. For the latter, reset NumPy PCG64 to seed 2026101004 on each call and draw one $(14, 256)$ int8 binary array $b$, setting $g = 1 - 2b$. Set every input pin positive.
2. **Construct the forest start.** Join all input pins to a virtual root. On nonzero $c$, Boruvka component merging prefers larger $|c|$, then smaller edge index; process the chosen edges in that same order, rejecting cycles. Traverse selected edges from pins first, then unvisited cells in increasing index, assigning each new root $+1$ and propagating $g_j = \operatorname{sign}(c_{ij}) g_i$.
3. **Fix free-component orientation.** In every connected component of nonzero $c$ containing no input pin, multiply all signs by the sign of its smallest-index cell, making that cell positive. Apply this to all starts before updates and again to all final states. This convention supplies an otherwise undetermined orientation, including isolated cells.
4. **Update at most 64 sweeps.** Visit checkerboard colors $(x + y) \bmod 2 = 0$, then 1. Within each color, simultaneously flip every unpinned cell with $g_i \sum_j c_{ij} g_j < 0$. Sum its four terms in right, left, down, up order. A zero local product does not flip. Stop after a complete sweep changes no sign in any start, or at the sweep limit.
5. **Select a candidate.** Retain all sixteen initial and sixteen final states. Minimize $J$, summed in edge order with Python `math.fsum`; tie-break lexicographically on row-major negative-sign bits (positive precedes negative). Identical tied arrays retain their first occurrence. This bounded search does not certify global optimality.
6. **Complete only zero edges.** Let $m$ be the median of all nonzero $|O_{ij}|$, averaging the two middle values when needed. Preserve every nonzero $O_{ij}$ unchanged; replace every exactly zero edge by $m g_i g_j$, including zeros outside the previous injury. No injury mask is consulted. If no nonzero weight exists, return the unchanged empty record.

**Implementation and presentation.** The portable algorithm and figure sources preserve the registered `optimizer.py`, `repair.py`, `native.py`, `optimizer_audit.py`, `support.py` and `plan.json`, together with the original registration, completion and selected result files. Their hashes match the frozen registration or completion receipts. Figure 14 displays the original four totals, full paired cross-tabulation and all 64 block differences; the renderer checks these against integer counts and preserves the saved aggregate interval without new inference. These are documentation and presentation checks, not another optimizer audit or simulation replay.

The fixed analysis averages draws within records, records within blocks and blocks within four equally weighted strata. For paired block differences, let $v_h = s_h^2/16$ for each history stratum. Then $\mathrm{SE}^2 = \sum_h v_h/16$, with Satterthwaite degrees of freedom $(\sum_h v_h)^2 / \sum_h (v_h^2/15)$. Both one-sided 95%

lower bounds must exceed their fixed practical margins: 0.02 over unmodified O and 0.05 over shuffled repair. The reported two-sided intervals use the same approximate Student-t analysis; descriptive stratified bootstraps do not replace these gates. Both pass. E78

Registration and all 47 frozen executable inputs precede test generation; block enrollment and prepared-record commitments precede assessments. Recorded independent checks cover 64 teachings, 14,336 lineage reconstruction contexts, 32,768 assessment contexts, all optimizer/construction outputs and 65,536 dual-task flags. Maximum scalar-window discrepancy is $6.661\times10^{-16}$, with the nearest decision threshold 0.001091 away. The manuscript binds completion, selected outputs and closeout identities without repeating these checks. Procedural ordering in a shared filesystem is not external custody or independent-laboratory replication.

### C.35. Four future reconstructions of reused tissue records: D181

The same D180 four founder blocks, four lives each and two target families give 32 F0 records. Four new future IDs 2/3/4/5, kept separate from earlier 0/1, cross coupling multipliers 0.75/1/1.25, ordinary/aligned writing and four controls. Three serial read/write/reset boundaries precede common/native terminal assays. The completed panel contains 2,304 labelled lineages, 6,912 serial pairs and 4,608 assay rows; independent founder information remains four exposed blocks.

The focal group stays common-reader B with ordinary writing; aligned B remains an outcome-motivated secondary lead. Fixed disjoint pairs (2,3)/(4,5), all six pair products, donor/future omissions and four-future consistent rank reversals preserve the protocol. Signed components use native squared-score units, retain negative estimates and are not reported as variance percentages or population uncertainty intervals. Coupling changes serial reading and writing together; the common final reader isolates the utility of the resulting record under one declared decoder. All families, copying controls and shape correlations remain. E79

Completion includes full native replay; the retained saved-only formal audit checks the complete panel without new dynamics. Source selection binds the completion receipt to native record and summary outputs. The later completed fresh-founder main cohort D182 is separate in C.42/E87. L20's existing unresolved forecasting contrasts and mixed adjudication are unchanged.

### C.36. Change-at-any-task decision model and refresh provenance

CT3-anytime assigns half prior mass to no deployment change and half uniformly across 100 possible times and three supplied change kinds, giving 301 components and conditional onset hazard $1/(201-t)$. It is a one-change model, not a stationary hazard or recurrent-change process. Nine existing policies are retained unchanged. Calibration informs the old phase; weak new-phase rate priors, exact conditional Bayesian/VFE inference, free warning signals, purchased answer labels and three-decision expected-cost planning remain. Actual change indicators and unpaid labels are unavailable. Expanding support also redistributes prior mass, so these effects are not isolated.

There are 27 fixed base laws and 1,024 fresh paired blocks per base (27,648 independent blocks), with 43 physical conditions per base and 1,188,864 physical episodes evaluated by ten policies. Wrong answers cost 1 and buying the correct answer costs 0.1. Full episodes include calibration cost 0.064; post-change recovery excludes it and normalizes by each condition's remaining task count. Supported times 26/51/76 and unexpected times 13/38/63/88 remain separate. Three pri-

mary gates require upper 95% bounds below zero for recovery-minus-0.1, recovery-minus-flat and full-core-minus-flat. Two pass; the absolute recovery gate fails. E80

Native intervals are mean $\pm 1.96$ stratified Monte Carlo SE, with $\mathrm{SE}^2 = \sum_b \mathrm{Var}(\mathrm{block}_b)/(27^2 \times 1024)$ and within-block paired differences. They concern this fixed panel, not arbitrary environments. The retained independent saved-count verifier reports 735,736 assertions, maximum discrepancy $2.061 \times 10^{-13}$ and all 864 chunks checked. It is a separate aggregation within the same program, not external empirical replication; it was not rerun for the paper.

The v0.19 source manifest and portable package freeze 122 selected records at 16:08 UTC on 11 October. The extraction script verifies member hashes, available report/summary-to-completion links and retained audit identities and reproduces the native summaries without source mounts, simulation or new inference. That historical snapshot retained only P32 launch documents and no completed E055/budget-policy outcomes. The v0.20 addition now supplies their completed records in C.39/C.40/C.43, while P32's full raw archive remains unavailable locally. This package is a reviewable evidence selection, not the full raw/executable release.

## C.37. Active-form comparisons and their verification boundaries

The main section 4.2 synthesis reuses the reference-acquisition, reciprocal-repair and writer-placement methods in C.2/C.3/C.8. Two additional historical comparisons are report-based additions, not new simulations. The active-form source manifest and portable records preserve the selected logs, protocols and implementation files separately from the earlier 122-record refresh.

**CA recognition and writeback.** The Stage 6F account describes a local $4 \times 4$ visible patch after the incoming carrier is destroyed. A held-out-pair ridge diagnostic learns a cross-bit map for decoding environmental history. The live writing routes use a different mapping into the Walsh carrier representation and must produce a record that works in subsequent dynamics. Diagnostic accuracy, sham accuracy and the writer read-effect contrast retain their native endpoints. None of twelve screened writers qualifies, and the 62-pair final reserve is not opened. The later Stage 6G protocol explicitly identifies the alignment assumption separating decoder and writer. This is neither a resource-matched predictor contest nor a proof that all possible writers fail. The retained records were checked for compatible claims; original arrays and uncertainties were not reanalyzed. E81

**Wagner interpreter protection.** The Architecture model adds an inherited repair strength to one regulatory matrix, or supplies multi-copy alternatives. For a weight selected for mutation toward value $u$, the repair operation has the form $w' = w + (1 - r)(u - w)$, with inherited strength $r$. The Adaptive Maintenance extension permits global or row-specific strengths, mutation of those traits, and a cost proportional to mean repair. Its global strength is clipped to $[0, 0.9]$. These are supplied mutation-attenuation mechanisms: no activity-dependent detector or corrective reconstruction of the matrix is implemented by this operation. A topology-by-repair factorial and repair-off mutation pedigrees distinguish protection from the topology that evolved with it. Retention means that a descendant keeps at least 70% of its positive root's measured information; it is not ordinary expression accuracy. The fresh Architecture comparison and later dormant-start development are separate campaigns, both with failed complete qualifications. The primary report source is the dated discovery log; absent native run trees prevent an independent replay or reconstruction of its intervals for this manuscript. E82

The active-form extension asks whether differing body organization can write a distinction when incoming record bytes and auxiliary writing states are matched. At the v0.20 selection only technical qualification was available. Version 1.2 adds the completed D184/D186 panels and D185 diagnosis in C.45/E90–E91; technical tests and the excluded benchmark are not additional scientific cohorts.

### C.38. Prospective serial repair and completion of the exposed replay

TT015 completes full native replay of the earlier serial and cue-noise studies described in C.33: 256 exposed generation-14 records in sixteen blocks. It independently checks 36,864 distinct reconstruction contexts; a total of 46,080 replay entries includes 9,216 overlapping contexts. Scoring, optimizer and provenance checks pass. These checks upgrade verification, not prospective status or the number of independent records. The separately reported withdrawal branch remains outcome-informed development. E77

TT016 generates 64 fresh blocks 16000000–16000063, sixteen per each of the four C.34 acquisition histories, and enrolls all 1,024 generation-14 outgoing records before continuation. Of these, 180 initially fail and 844 initially pass. The same TT013/TT014 surviving-record sign solver, median-magnitude completion and strict intended-task margin greater than 0.25 are retained. Repair receives the current packet and fixed input pins/options; no protected previous record, target-output label, injury mask or body state enters it. Policies apply no repair, one repair at generation 15, repair before each generation, or shuffled added signs before each generation. They cross matched and independent ongoing cue-noise regimes; teaching and birth/reset pairing remain unchanged. All eight branches run through generations 15–32 and transmit their own actual successors after failures.

The primary is intended function at **every one of the eighteen rebuilding-body readouts**, under independent ongoing noise, comparing repeated repair with no repair. The block means are averaged equally within four history strata. The prespecified approximate Welch–Satterthwaite calculation in C.34 supplies the one-sided 95% lower bound, required to exceed 0.05; its two-sided 95% interval is reported in the main text. Stratified bootstrap intervals for secondary contrasts are descriptive and unadjusted. Terminal success means the generation-32 rebuilding body, not an additional read of its outgoing packet. Each record receives one continuation per policy/noise condition; this is not repeated sampling of entire lineages from the same frozen state. E83

All policies allocate 4,096 transmitted bytes and 36,864 reader updates per record and continuation condition: eighteen rebuilds times 2,048 integration steps. Each counted step advances both cue-specific 256-site bodies together, including the settling period. Thus the allocation equals 73,728 cue-specific whole-body steps; it is not a count of scalar cell updates or optimizer sweeps. Founder teaching and the fourteen preliminary rebuilds are excluded from this continuation budget. Repeated repair adds 18,432 optimizer calls per noise regime, compared with 1,024 for one-time repair and zero for baseline. Shuffles preserve added support, magnitudes and sign counts relative to their own input; divergent histories can accumulate different added weights. Computation and total intervention dose are not matched. Full recorded replay covers 64 teachings, 14,336 preliminary and 147,456 continuation contexts, 75,776 optimizer checks, 17,472 native-array comparisons and 1,152 forcing checks. Separate scoring verifies 147,456 endpoints and 294,912 task

flags. These are checks within this program, not independent laboratories; the manuscript verifies selected receipt bindings without executing these audits again.

### C.39. Stress, recovery and sign–magnitude substitution at capture

E055 uses all 32 exposed E051 blocks, retaining acquired packets and two generation-32 renewal histories (matched or independent cue noise). In a new injured body, incoming weights remain fixed. Each cue trial runs 2,048 updates, the first 1,024 without amplification. Ongoing noise in this body is independent across cues for all histories, while births remain paired. A rule-blind signed pulse of amplitude one acts on 64 randomly chosen living nonport cells for 32 steps with dt 0.2; the same pulse field is used in both cue trials. A matched sham retains all other forcing. Capture occurs immediately after the pulse and again after 512 unforced recovery updates. Early capture is stored without feeding back into later development.

E056 adds four packets at each capture: unstressed signs and magnitudes (CC), pulse signs and magnitudes (PP), unstressed signs with pulse magnitudes (CP), and pulse signs with unstressed magnitudes (PC). Equal support is required; a mismatch fails the run rather than excluding the case. An erased packet is the negative control. Only a packet enters a fresh recipient, read for 2,048 updates with 1,024 settling and the last 32 scored at the strict greater-than-0.25 four-margin criterion. Intact, scattered 24-cell and square nine-cell injuries retain protected task ports. The hybrids explicitly use the matched unstressed reference; they are not internally discovered corrections. E055's originally primary independent-noise criterion remains failed. E056's matched-noise choice follows the E055 result and remains development.

E057 freezes this apparatus before 32 new blocks 38000–38031: four founders and four recipient slots per block, 512 nested lineages per history. Teaching uses 512 steps with the last 128 captured. Both 32-renewal source histories are newly generated and fully retained; no acquisition or intermediate failure is excluded. Current-body and fresh-reader settings are unchanged. The primary is the matched-noise source history with scattered recipient injury. If Q is a block's mean success, subscripts E/L indicate early/late capture, and CC/PP/CP/PC identify packet combinations, the four contrasts are:

- Pulse loss: $Q_{CC,E} - Q_{PP,E}$.
- Sham-adjusted timing benefit: $(Q_{PP,L} - Q_{CC,L}) - (Q_{PP,E} - Q_{CC,E})$.
- Magnitude restoration on pulse signs: $Q_{PC,E} - Q_{PP,E}$.
- Magnitude restoration on unstressed signs: $Q_{CC,E} - Q_{CP,E}$.

The primary panel's fresh-recipient means are:

| Record capture | Sham record (CC) | Pulse record (PP) | Pulse minus sham |
|---|---|---|---|
| Early: immediately after pulse | 449/512 (87.70%) | 434/512 (84.77%) | -2.93 points |
| Late: after 512 recovery updates | 447/512 (87.30%) | 444/512 (86.72%) | -0.59 points |

These are outcomes after reading captured records in fresh recipients, not immediate measurements of the pulsed body. Their difference in pulse-minus-sham effects gives the +2.34-point timing benefit. The 512 nested lineages remain grouped in 32 independent blocks for inference.

Ten thousand paired whole-block bootstrap draws supply two-sided 98.75% intervals, a Bonferroni family of four. All lower endpoints must exceed zero and numerical controls must pass. This tests positive direction, not E055's five-point loss/recovery thresholds, equivalence or complete restoration. Other comparisons retain pointwise 95% intervals. Acquired records repeated across recipient slots are not independent acquisitions. Float64, the zero normalization floor and the exact-zero fallback rule remain; this is separate from numerical-robustness qualification. The recorded independent audit checks fresh acquisition, all source boundaries, pulse/capture/recipient dynamics and 1,868 summaries, including 92,160 terminal cue observations. E84

### C.40. Completed cellular pause boundary and onward transmission

P32 retains the native BITSEY equations and C.7/C.24 information boundary, with the registered finer numerical procedures. Its thirteen schedules are continuous RNA communication; immediate pauses of 5/10/15/20/25/100 model-time units; 100-unit pauses starting at 5/10/15/20/25; and full blockade throughout the 300-unit first challenge. Each crosses two historical origins, two fixed incoming-cell sequences, two archived source bundles and two resolutions q32/q64. The resulting sixteen variants per schedule are related deterministic panels, not sixteen independent samples from a biological population. All parents, including failed retention cases, enter onward tests.

After the fixed replacement schedule, all original carriers are absent. Actual parent trios undergo two isolated 10,000-unit holds. Their final complete states then contact three fresh M recipients for 20,000 units, with all six cells evolving. Donors are removed and actual recipients undergo two more isolated 10,000-unit holds. Retention and transmission require all three relevant cells to meet the native E endpoint and at least 95% E classification in the final quarter of both holds. Open, RNA-blocked and boundary-cut readouts are matched; no extra pause, donor clamp or recipient reset is introduced at readout. The stricter every-renewal-window E endpoint remains separate and fails for all parents. E85

| Communication schedule during challenge | Retention | Open transmission | RNA-blocked transmission | Boundary-cut transmission |
|---|---|---|---|---|
| Continuous communication | 0/16 | 0/16 | 0/16 | 0/16 |
| Immediate pause: 5, 10, 15, 20, 25 or 100 units | 16/16 | 16/16 | 0/16 | 0/16 |
| 100-unit pause starting at 5 or 10 | 16/16 | 16/16 | 0/16 | 0/16 |
| 100-unit pause starting at 15, 20 or 25 | 0/16 | 0/16 | 0/16 | 0/16 |
| Full blockade for 300 units | 0/16 | 0/16 | 0/16 | 0/16 |

Grouped rows report the identical counts for **each listed schedule separately**, not pooled denominators. Start times and durations are model-time units within the challenge. Every schedule replaces all original carriers, and none passes the stricter every-renewal-window endpoint. These counts establish the observed timing boundary across the related model panels, not a population success rate.

All 428 q32/q64 and 214 source-bundle comparisons pass the normalized primary-sample error limit $|x - y|/\max(1, |x|, |y|) \leq 0.01$, together with the registered classification and inventory requirements. This is paired numerical/source agreement, not full-grid three-resolution contraction or independent biological replication. The recorded closeout checks 856 cases, 7,352 integrations

and 525,505 assertions. Native counts, input identities and verification results are available in the retained completion-integrity JSON; the referenced external raw tree is unavailable locally. The paper did not rerun those integrations or audits. P27/P30's original failed gates and unresolved source-paper table/geometry correspondence in C.7 remain unchanged.

### C.41. Hopfield reader intervention and complete activity replacement

H004 uses a separate symmetric 64-node binary Hopfield model. A fixed shared base matrix B and acquired record W determine the effective couplings. Asynchronous updates set each unpinned activity to the sign of its local field; an exactly zero field retains its previous bit. Each sweep visits all sites once in a pseudorandom permutation. This is a taught two-orientation task, not arbitrary competence discovery.

| Component | Frozen setup |
|---|---|
| Base generation | Draw independent standard-normal entries of G per network; set $B_{ij} = (0.12/\sqrt{64})(G_{ij} + G_{ji})/2$ for $i \neq j$, with $B_{ii} = 0$. B is shared, fixed and neither rewritten nor damaged. |
| Teaching | Four independently initialized founders per base, two per orientation $s = \pm 1$. Under each cue $c = \pm 1$, clamp sites 0–7 to c, sites 8–15 to sc and sites 16–23 to -sc for 24 sweeps. |
| Initial record | Apply the body-only writer below to the two final teaching states. A further 24-sweep stability check does not replace those states as the record source. |
| Subsequent reading | Fresh independent binary activity; clamp only sites 0–3 to c. Release the other teaching sites. Read both cues for eight sweeps each. |
| Readout | Both eight-site outputs must have mean activity strictly beyond 0.5 in their assigned directions, under both cues. |
| Reader intervention | Multiply B+W couplings between selected input sites and the remaining sites by gain 1, 2 or 4. The base and acquired record remain separately accounted for. |

These details follow the exact public acquisition implementation, `acquire`, update and writer model, and H003 reader reused by H004. Their contents match H004's frozen source copies. The Gaussian base supplies random shared dynamics; the taught output orientation enters through the acquisition clamps and resulting record.

At each boundary, activity is wholly replaced and independent undirected edge deletion removes 50% of the record's connections before reading. Let $x_i^{(t)}$ denote the final body activity in cue episode t. The complete body-only writer is

$$W'_{ij} = \frac{1}{64T} \sum_{t=1}^{T} x_i^{(t)} x_j^{(t)}, \qquad T = 2, \quad W'_{ii} = 0.$$

It has no access to incoming W, teaching labels or the desired outputs; protected and cumulative-loss copying explicitly bypass it. Gain 1/2/4 multiplies the symmetric B+W connections joining the four cue sites to other sites, leaving within-group connections unchanged. Gain 4 at unrelated sites 24–27 is a control. Each gain changes the effective reader, not the body-only writing formula. The intervention covers 240 unique undirected edges and uses supplied addresses. The packet has 2,016 unique float64 entries (16,128 payload bytes), stored as a 32,768-byte dense matrix; the shared base has another 2,016 independent values. Eight sweeps under both cues entail 960 unpinned-site

updates per body. This apparatus does not implement E041's contrast/fallback writer or strength-normalized reader.

Seventy-two fresh base networks (14000–14071) each contain four founders and four descendants per founder. Sixteen replacements and all failed cases are retained. Nine arms include own renewal, opposite-rule foreign records, coordinate scrambling, erasure, reading/writing ablation, protected copying, cumulative-loss copying and scrambled writer addresses. Births, update orders and injury tapes are paired. Extra controls read the baseline lineage's actual terminal record with gain 4 only at the end, or extend its baseline read from eight to 32 sweeps. Sustained and late-only gain-4 terminal readers are matched; their record histories differ.

Twenty thousand whole-network bootstrap draws provide pointwise two-sided 95% intervals. Seventeen primary directional lower bounds use one-sided level .05/17; specified minimum effects apply to point estimates. All must pass before the secondary sustained-minus-late criterion can qualify. No sample extension follows the result. All primary and eligible secondary gates pass; protected-copy inferiority remains visible. The retained audit reports 3,334 checks and four scientific fixtures. H002–H003 development and H005–H008 follow-ups are not pooled into confirmation. Further Hopfield work remains ongoing. E86

## C.42. Fresh-founder extension of record–recipient interaction

D182 uses four new founder blocks 120000/120005/120010/120015, four lives and two target families each, producing 32 records sealed before descendants. Four futures (2/3/4/5), three recipient-coupling settings (0.75/1/1.25), ordinary/aligned writers and retained controls follow C.35's three-rewrite boundary and common/native final readers. The 2,304 labelled lineages, 6,912 read/write pairs and 4,608 final assays remain nested under four independent founder blocks. No earlier cohort is pooled.

The focal group is still common-reader B with ordinary writing; aligned B is the prospectively named secondary lead inherited from exposed development. Both fixed disjoint future pairs, all six pair products, omissions and four-future ranking reversals are reported without converting signed components to probabilities or adding a new significance test. All eight family/writer/assay groups and shape/copying controls remain. A separately frozen saved-only audit follows generation-time native replay and checks complete-panel membership, all saved outputs and statistics. Native summaries, record tables and completion/audit receipts are copied from the available external storage into the portable paper package; underlying simulations were not rerun. E87

For record r, recipient setting j and future f, let $q_{rj}^{(f)}$ be qualified preference on its native fractional scale. The interaction residual removes the record mean and recipient mean within that future:

$$u_{rj}^{(f)} = q_{rj}^{(f)} - \bar{q}_{r\cdot}^{(f)} - \bar{q}_{\cdot j}^{(f)} + \bar{q}_{\cdot\cdot}^{(f)}, \qquad I_{fg} = \frac{1}{RJ} \sum_{r,j} u_{rj}^{(f)} u_{rj}^{(g)}.$$

Here R = 16 records per family and J = 3 recipient settings. The reported cross-future component averages $I_{fg}$ over all six pairs of the four futures. It measures recurrence of the nonadditive record–recipient pattern, in squared preference units; negative estimates are retained. It is neither a success probability nor a fraction of variance explained.

On the two fixed pairs (2,3) and (4,5), focal ordinary-writer B remains mixed: +0.0006162 and −0.0000346. The named secondary aligned-writer B is positive in both: +0.0006162 and +0.0001788,

and also in all six pairings and the averaged components after each founder-block or future omission. One aligned record-pair ranking reverses consistently across all four futures. The secondary lead therefore recurs on **four new founder blocks**, while the focal result remains mixed. This panel does not establish population precision or a prospective prediction advantage. E87

### C.43. Repair scheduling, call budgets and prospective efficiency

The P08 budgeted pilot enrolls 256 generation-14 records from sixteen fresh blocks, four per acquisition-history stratum. Four policies continue every lineage through generations 15–32 using independent ongoing cue noise and paired resets/teaching. Warning, periodic and random policies each spend exactly nine repair calls; a continual reference spends eighteen. All share the first repair. The warning model uses six incoming-record measurements: missing/weak edges, weak nodes, strength variability, unsigned reach and signed output separation. A fixed ridge logistic model and rank-based quota rule are fitted on previously exposed histories with whole-block cross-validation. No current body, injury or future outcome enters scheduling. The fitted/scaled/calibration state uses 60,528 numeric bytes in shared apparatus. Calls, CPU, added edge events and weight are recorded separately.

The warning-minus-periodic primary concerns uninterrupted function. A nonpositive estimate fails the strict directional prediction; intervals from 20,000 stratified block bootstrap draws are descriptive, without a practical-effect or equivalence margin. The later five-budget development curve reuses this exposed cohort and controls where declared. It motivates a separate 132-block, 2,112-record prospective test with entirely new teaching, fourteen preliminary renewals and eighteen continuations, comparing one/six/eighteen fixed repair calls. All initial failures remain enrolled. E88

With p1, p6 and p18 denoting uninterrupted rates, both one-sided 95% lower bounds must be positive for

$$G = p_{18} - p_1, \qquad D = p_6 - \frac{10}{17}p_{18} - \frac{7}{17}p_1.$$

The second inequality requires six calls to deliver more than twice the added benefit per extra call of eighteen, relative to one initial repair: five versus seventeen additional calls. A negative verdict requires either one-sided 97.5% upper bound below zero; otherwise the result is unresolved. Native paired bootstraps retain four equally weighted strata.

| Fixed repair calls | All eighteen bodies pass | Last rebuilding body passes |
|---|---|---|
| One | 1,761/2,112 (83.38%) | 1,884/2,112 (89.20%) |
| Six | 1,833/2,112 (86.79%) | 1,911/2,112 (90.48%) |
| Eighteen | 1,879/2,112 (88.97%) | 1,925/2,112 (91.15%) |

For uninterrupted success, **G = +5.59 points**, with a two-sided 95% interval **[4.59, 6.58]** and the primary one-sided 95% lower bound **+4.78**. The efficiency contrast **D = +0.12 points** has a two-sided 95% interval **[-0.49, +0.72]** and primary lower bound **-0.40**. G passes; D remains unresolved under the registered positive/negative rules. These are the retained 20,000-draw stratified paired block-bootstrap bounds, not a new analysis. This cohort is separate from TT016, and terminal success remains distinct from uninterrupted success. E88

This is not noninferiority, an optimum or a minimum safe budget. Separate optimizer, score and resource checks cover all cases; independent flat dynamics cover teaching and selected continuation states, not exhaustive native replay. This verification scope differs from TT016's full replay.

### C.44. Matched saved-boundary rescue and revision provenance

The failure-boundary study selects every first later failure in the six-call prospective cohort: 46 records from 36 independent source blocks, all at the second consecutive unrepaired reconstruction. Selection and analysis are frozen before the counterfactual reads. Each factual/intervention pair restores the same incoming packet, injury, birth and ongoing cue noise. Current repair is the only intervention. The original strict four-margin threshold remains; no successor record is written and no later continuation is simulated. Separate flat dynamics match all 92 reads byte for byte, and separate optimizer checks agree for every repair. Two already exposed technical examples, including one immediate harm, are excluded from the 46-case scientific cohort. All 46 immediate rescues therefore estimate an effect on these selected failures, not harm rates, safety or a better policy at the original call budget. E89

The v0.20 manifest identifies 150 selected source records and a portable package. The extractor verifies all package/member hashes and available completion, summary and audit bindings, then reproduces native summaries without mounting their original locations. It does not run models, fit predictors, resample intervals or repeat original scientific audits. P32 is limited to its retained completion-integrity counts and documents, with its external raw archive still absent. At that historical cutoff, TT017, the active-form extension and the FEP horizon comparison supplied prospective status only. C.45 now adds completed D184–D186 development; no further TT017, FEP-horizon or Hopfield outcome is incorporated. The package is a reviewable manuscript selection, not the complete raw/executable release.

### C.45. Identical records, surviving body information and early capture

D183 defines the information boundary; D184 applies the ordinary writer, D185 diagnoses its saved trajectories, and D186 adds early reference capture. All substantive panel comparisons reuse four exposed D079 blocks, 30056–30059, with four half-plane messages and four records per message: 64 records. Within each block, a message is paired with the message two indices away, preserving replicate index. The two members receive the same normalized mean of their acquired theta fields, byte for byte. This packet retains pair-level information but cannot distinguish its two intended messages. A separate residual condition preserves the individually damaged packet. The surviving half-body, matched fresh half, source substitutions and noise follow the frozen contract. R and the activity average initially equal zero in the ordinary comparison. E90

D186 changes one initialization: $\mathbf{R}_0 = \mathbf{V}_0$, copying the permitted post-injury body into the existing full-field reference. It does not read the parent's old reference, original record or target label. Reference update rate 0.002, reference-to-body coupling 0.2, writing rate 0.001, record normalization 0.3 and 1000 repair sweeps remain fixed. The reference affects both subsequent body dynamics and writing; the intervention does not isolate either route. The average remains zero at birth and is not an additional causal input to the rung-A activity drift. Only the final 2048-byte theta record is read in a fresh body; the surviving body and its reference do not cross that boundary. This is a different outgoing-record construction from C.8's final-reference-derived packet. E91

Both initializations are crossed with common/residual records and own, partner, scrambled-own, scrambled-partner and history-erased bodies. Live feedback, fixed-record reading and writing-disabled conditions remain. Holding captured R unchanged is explicitly a protected working-memory comparator, not information removal. The original 36 repair paths are preserved exactly and 32 captured paths are added per block. The conventional writer receives the same fixed-reader activity stream within each initialization: V1 through V1000 for zero-start, and V0 through V1000 when capture is supplied. It uses the same final record normalization. Captured and uncaptured starts therefore differ by the available birth observation, while native and averaging writers within a start have matched observational access. E91

Every named resulting record receives four paired fresh reads of 1000 sweeps. The same exact packet bytes and reading flag may alias one physical assay; names are retained without counting aliases as independent observations. The full panel contains 272 repair paths and 372 named reads, corresponding to 288 unique conditions. All 144 original D184 repair paths and 204 original named reads reproduce their saved states and scores exactly according to the retained replay records. No performance filtering, new threshold, fitted model or serial continuation is introduced.

Recognition is determined from each terminal activity field's projection onto four supplied templates. The largest projection identifies the message; ties within 1e-12 divide recognition credit equally. Accuracy and own-minus-partner recognition are averaged over four fresh draws, then records and the four equally weighted blocks. These are message-level endpoints, distinct from Q and qualified preference in C.8/C.16. Record fidelity here is the coordinate-based centered correlation between outgoing theta and the original acquired theta, without translation optimization. Sign fidelity concerns fresh-body signs. Neither total recognition nor correlation measures the amount of historical information in bits. All panel results are descriptive; four exposed blocks supply no new confidence interval or fresh-cohort qualification.

D185 uses retained checkpoints 0, 1, 50, 250, 500 and 1000. Its contract was specified after terminal outcomes were known. Body, reference and record projections have different units; a roughly 95% fall in the pooled body projection by step 50 is not a 95% loss of all information. All 135,936 field rows and 204 terminal-read bridges are independently checked in the retained diagnostic. No intermediate fresh readers or new trajectories were generated.

The v1.2 selection binds 90 source records in a portable package. Intake checks all 420 D184 and 1,392 D186 frozen input hashes, 649 and 1,157 output hashes, and D186's four block receipts. The extractor recomputes 19,669 saved-score aggregate comparisons from packaged records without external arrays. Original field replay is attributed to retained native verification; the original Python 3.13.5/NumPy 2.5.2 environment was unavailable, and the native saved verifier stopped at its runtime guard. No guard was bypassed, and this manuscript check is not another dynamics replay or independent laboratory replication. The full 64-record D186 panel, not its disposable benchmark, supplies E91. D074's criterion and L22's mixed qualification remain unchanged.

## C.46. Fresh body-source and early-capture confirmation: TT021

**Cohort and boundary.** The prelaunch protocol enrolls all 64 new tissue blocks 21000000–21000063, with four messages and four parent replicas per message: 1,024 records per condition. Settling, teaching, repair and fresh reading retain C.45's 120/1000/1000/1000 updates, native rates and binary64 operation order. Each record has four receiving draws. Arms share repair and receiving

challenges within a block; the original coupling across messages remains. Distinct blocks and receiving draws have separate declared pseudorandom streams. No case is excluded for poor acquisition or expression. This confirms the original pairing regime, not independently perturbed message environments. E92

Common incoming records are identical within the same opposite-message pairs as C.45. A separate informative-record family retains each individually damaged record. Ordinary repair begins with zero reference; early capture sets $R_0 = V_0$ from the current injured body, including its fresh fill. The held-snapshot comparator freezes that same reference. Fixed reading holds incoming theta while writing; live reading uses evolving theta. Only the final 256 float64 theta values cross to a fresh recipient. Neither body nor reference is transmitted.

**Controls and observer.** Twenty repair paths retain own, partner, spatially scrambled and history-erased bodies, live/fixed readers, held reference and disabled writing across both record families. Corresponding matched averages, unchanged inputs, supplied positive records and reading-disabled conditions give 29 named packet conditions. Captured averaging receives the birth observation $V_0$ as well as later activity, whereas ordinary averaging begins at $V_1$; within the fixed-reader comparison, memory writing and averaging receive the same stream. The live stream can differ, so a live-versus-average contrast is not an isolated writer effect.

Recognition uses C.45's four-template projection with equal fractional credit for ties within $10^{-12}$. Average the four draws within a record, sixteen records within a tissue block, then the 64 paired block contrasts. Neither draws nor records replace blocks as independent population units. Message recognition, record correlation and site-sign fidelity remain different endpoints.

| Registered directional contrast | Gain, percentage points | One-sided 99% lower bound | Required margin | Verdict |
|---|---|---|---|---|
| Common record: live capture minus ordinary repair | 25.659 | 22.157 | 5 | Pass |
| Updated capture minus held snapshot | 0.513 | −0.039 | 2 | Not confirmed |
| Captured fixed-reader writer minus matched averaging | 5.908 | 4.331 | 2 | Pass |
| Own-body minus partner-body capture | 51.416 | 44.412 | 5 | Pass |
| Informative record: ordinary minus captured repair, predicted harm | 0.098 | −1.878 | 2 | Not confirmed |

The single primary is the first row; four further claims cannot rescue it. Each uses a paired-block Student-t lower bound at 63 degrees of freedom, alpha .01, sharing a Bonferroni .05 family budget. Coverage remains approximate. The primary's separate pointwise two-sided 95% interval is [22.727, 28.591] points. The conservative bounded sensitivity lower bound is −12.276 for the primary and +13.480 for the body-source contrast: only the latter passes that stronger sensitivity calculation. Neither unconfirmed comparison establishes equivalence.

Live capture gives 75.659% common-record recognition, ordinary repair 50.000%, and the held snapshot 75.146%. Partner capture gives 24.243% intended and 75.659% partner recognition. Fixed-stream captured writing gives 68.604% versus 62.695% for matched averaging; most of that gain occurs in message 3. Informative-record recognition is 90.015% ordinary versus 89.917% captured.

In the primary comparison, 290 record means improve, twenty worsen and 714 tie. Four readouts do not estimate each record's capacity precisely.

**Execution and provenance.** The original run stopped at its storage safeguard after 62 verified blocks. An explicit execution amendment, recorded without inspecting interim scientific summaries, completed only the two unstarted identities while retaining the cohort, endpoints and criteria. The original failure remains preserved. The final receipt reports exact physical-reference checkpoint agreement, score/route checks and reproduction of all five decisions; its verifier was written after outcome inspection. These are same-program checks, not a second laboratory. This is one repair followed by fresh reconstruction; C.51 separately covers exposed serial follow-ups.

Sources: protocol, interpretation, execution amendment, closeout, v2.1 source selection. Full raw arrays were not replayed for this manuscript intake.

### C.47. Repeated stochastic continuations from restored records: TT018

**Enrollment.** Generate 64 new acquisition blocks 18000000–18000063, sixteen in each of four previously declared exact/tolerance-pulse writing histories. All sixteen candidate records per block are generated, but a rule fixed before teaching selects exactly one generation-14 outgoing record from each block. Each history uses every founder/recipient position once. The selected cohort contains nine currently failing and 55 passing bodies; these counts do not drive enrollment. Seal all 64 packet byte strings and 192 unchanged forecasts before any continuation. E93

**Futures and policies.** Restore each selected 4,096-byte weight packet independently into 64 future environments. Each environment branches into no repair, repair at generation 15 only, and repair at every generation 15–32. Use C.34's unchanged sixteen-start, sixty-four-sweep solver with fixed seed 2026101004, no desired output label and no previous packet. It preserves existing nonzero weights, fills zeros from inferred signs and median surviving magnitude, and leaves empty records empty. Each branch continues through failure and transmits its own activity-written record.

Continuation bodies have independent positive/negative cue resets and independent ongoing cue noise. Injury removes 24 non-port cells. Native dt .2, birth SD .05, noise .02, 2,048 reader updates, 1,024 settling updates, exact-zero fallback and binary64 arithmetic remain. Acquisition retains its original pairing. Record, future, generation, source and cue identify the declared pseudorandom streams; policy is omitted from environment keys so paired policies share realized challenges. Different futures and the terminal assay use distinct streams. This restores a packet, not every variable of a complete body.

**Endpoints.** After eighteen further renewals, one separately injured fresh recipient reads each generation-32 outgoing record with no additional repair or writing. This primary terminal-record outcome differs from the last rebuilding body's outcome and from uninterrupted success of all eighteen bodies. Each requires the strict original two-cue/output margin above .25 over the final 32 observations. The 4,096 futures per policy yield terminal counts 3,585/3,688/3,757, uninterrupted counts 3,191/3,415/3,692, and last-body counts 3,587/3,688/3,757 for untreated/one-time/repeated repair. Equal aggregate last-body and terminal counts do not imply identical paired outcomes.

The primary repeated-minus-untreated terminal contrast is +4.1992 points; its approximate stratified Welch–Satterthwaite two-sided 95% interval is [2.2882, 6.1102], and one-sided lower bound +2.6104 exceeds the prespecified two-point margin. Average futures within each record/block, then

equally across four history strata. There are 64 independent acquisition blocks, not 4,096 independent population units. Paired terminal counts are 213 repeated-only successes, 41 untreated-only successes, 3,544 successes under both and 298 failures under both. Record-level differences are positive for 51, negative for three and tied for ten.

**Conditional estimates and harms.** Report success fractions and exact binomial pointwise 95% intervals per starting record/policy/endpoint, plus simultaneous intervals with alpha $.05/(64 \times 3 \times 2)$. Even 64/64 has a pointwise lower bound 94.40%, and simultaneous lower bound 86.02%, rather than establishing probability one. All nine failing-body records remain below; suffixes identify the preassigned records, and the three columns share paired futures.

| Record suffix | Untreated terminal successes / 64 | One repair / 64 | Repeated / 64 |
|---|---|---|---|
| 18000004 | 62 | 64 | 64 |
| 18000007 | 48 | 43 | 46 |
| 18000019 | 34 | 14 | 14 |
| 18000038 | 1 | 1 | 0 |
| 18000039 | 0 | 0 | 0 |
| 18000045 | 30 | 52 | 54 |
| 18000059 | 0 | 8 | 9 |
| 18000060 | 0 | 1 | 0 |
| 18000063 | 51 | 60 | 61 |

These subgroup observations are descriptive, without a new gate. One terminal read per descendant estimates the starting record's serial future capacity, not that particular descendant packet's repeated-read reliability. The unchanged TT008 candidate has secondary Brier loss .054810 versus .065100 for expression/history and .067711 with unsigned access; the respective improvement intervals are [.000060, .020521] and [−.000268, .026071]. This transport analysis is not a new qualified forecast superiority claim. Repeated repair uses 73,728 optimizer calls versus 4,096 one-time calls, without equal resource budgets.

The final run retains its original cohort after an operational two-CPU amendment; registration, frozen forecasts and preexisting checkpoints remain bound by the closeout. The reported physical replay and archive checks belong to that lane. The manuscript verifies selected receipt arithmetic and source identities without resimulating or reconstructing unavailable raw arrays.

Sources: protocol, interpretation, closeout, v2.1 source selection.

### C.48. Fresh sign–magnitude compatibility: E065

**Cohort and construction.** Generate all 32 new blocks 39000–39031, four founders and four recipient positions, with two sham/pulse histories: 1,024 pooled cases per arm, clustered in blocks. Acquisition, 32 ancestral renewals, current-body pulse/sham capture and three subsequent ordinary rewrites follow the unchanged C.23/C.39 apparatus. Earlier means the pulse/sham capture at ancestral horizon 32, and final means three rewrites later; neither term means the original teacher record. No previous scientific cases or reader scores enter this fresh cohort. E94

Let $M$ be the intersection of earlier/final nonzero supports, partitioned into changed-sign connections $C$ and unchanged-sign connections $S$. Every construction retains final support and all raw weights outside $M$. On $M$, cross final/earlier signs with final/earlier magnitudes: FF, FO, OF and OO. Two further arms retain final signs but substitute earlier magnitudes only on $C$ or only on $S$. Earlier-only edges cannot reappear. Every record, including empty subsets, remains enrolled.

| Construction | Sham successes / 512 | Pulse successes / 512 | Pooled success |
|---|---|---|---|
| FF: final signs and magnitudes | 439 | 434 | 85.25% |
| FO: final signs, earlier magnitudes | 170 | 146 | 30.86% |
| OF: earlier signs, final magnitudes | 403 | 402 | 78.61% |
| OO: earlier signs and magnitudes | 408 | 372 | 76.17% |
| Earlier magnitudes on changed-sign edges only | 434 | 427 | 84.08% |
| Earlier magnitudes on unchanged-sign edges only | 380 | 333 | 69.63% |

Fresh terminal readers receive matched 24-cell injuries and paired cue birth/noise, with 2,048 updates, 1,024 settling steps and the original strict margin threshold. Binary64, exact-zero fallback and ideal absolute-strength normalization remain; zero denominators give zero drive. An erased control scores zero.

Three frozen success contrasts use 10,000 common whole-block bootstrap draws and 98.3333% percentile intervals: OO−FO = **45.31 [39.26, 51.66]** points; (OO−FO)−(OF−FF) = **51.95 [45.12, 59.38]**; and C+S−FO−FF = **37.60 [31.74, 43.62]**. Strictly positive lower bounds satisfy the three-part criterion; nominal family coverage is approximate. Histories are equally weighted within blocks. Restoration remains incomplete: OO−FF is −9.08 points, pointwise 95% interval [−12.21, −5.86].

These are causal interactions among assigned records on the success-probability scale. Subsets differ in size and weight distribution, and magnitude changes alter normalization. The result does not isolate a unique mediator or an equal-dose per-edge effect. E064's preferential changed-edge hypothesis remains failed. External archived values, paired-cue streams and exact-zero arithmetic remain supplied resources, not autonomously restored machinery.

Sources: protocol, interpretation, closeout. The closeout reports independent source/reader replay and verification of 1,489 outputs and 43 frozen inputs; this manuscript does not rerun those checks.

### C.49. Fresh recovery of an incompatible record: E068

Generate 32 new blocks 40000–40031, with the same four-founder/four-recipient and two-history structure as C.48. Reproduce acquisition and source formation anew. Native records are final source packets; mixture records retain final signs/support but receive earlier magnitudes on the earlier/final nonzero intersection. The archive is used once, for this construction. Three subsequent ordinary rewrites receive only the current packet, new body and supplied cues, with no archive, teacher, output label or persistent body. E95

Each rewrite uses a new 24-cell non-port injury, shared cue birth/noise, 2,048 reader updates, 1,024 settling steps and original exact-zero capture. The 128-step holding state is not transmitted.

At local horizons 0, 1 and 3, all policies receive common terminal-reader challenges; these assays do not feed back into the lineage. Protected-copy reads alias the unchanged h0 bytes. Cumulative-erasure copying receives corresponding injury losses without rewriting. Erased-start rewriting may form nonzero records; it is not clamped empty.

| Condition | h0 | h1 | h3 |
|---|---|---|---|
| Mixture, ordinary rewrite | 311/1,024 | 520/1,024 | 609/1,024 |
| Native, ordinary rewrite | 899/1,024 | 894/1,024 | 887/1,024 |
| Mixture, cumulative-erasure copying | 311/1,024 | 179/1,024 | 45/1,024 |
| Erased start, ordinary rewrite | 0/512 | 0/512 | 0/512 |

All three predeclared h3 contrasts pass 98.3333% whole-block percentile intervals from 10,000 paired draws: mixture absolute recovery **+29.10 [23.44, 34.83]** points, advantage over cumulative erasure **+55.08 [50.78, 59.77]**, and over erased-start rewriting **+59.47 [54.79, 64.26]**. Histories receive equal weight within each of 32 independent blocks. These are approximate family-adjusted intervals; the 1,024 cases are not independent population units.

Recovery remains below native success by 27.15 points, with secondary pointwise 95% interval [23.44, 30.96]. From h0 to h3, mixture records show 367 gains, 69 losses, 242 successes at both endpoints and 346 failures at both. Mean rule-aligned transfer also improves, a secondary endpoint. The controls separate recovery from mere attrition and the erased baseline, without demonstrating complete historical-detail preservation, robustness to changed numerical facilities, or autonomous origin of the writer.

Sources: protocol, interpretation, closeout. The retained receipt reports independent replay and 1,617 original-output/43 frozen-input checks; raw-array availability limits the present manuscript intake.

### C.50. Pruning reverses its benefit after rewriting: E073

E073 uses all previously exposed E068 g1 records from blocks 40000–40031, with four founders, four recipients and two histories per block. Prior recovery and intervention outcomes are declared; this is a saved-cohort replication of E072, not fresh acquisition or a blind holdout. E072 used the other saved cohort and contributes no pooled observations. New matched writer and terminal-reader streams are used. E96

Select an edge if the preceding incoming weight was zero, its outgoing weight is nonzero, and saved metadata identifies exact-zero fallback writing. Set those selected weights to positive zero while preserving all outside bytes. Keep every case, including empty selections. The ordinary writer receives only the resulting packet, not the selection mask or an archived reference. Apply no further pruning: physically available selected edges may be rewritten. At h0 before and h1 after one rewrite, matched terminal draws remain separate from the writer stream. Native and erased-start controls remain.

The numerical apparatus retains binary64, paired cue birth/noise, 24-cell non-port injury, natural absolute-weight normalization without a floor, 2,048 reader updates and the strict .25 margin. Equal weighting of histories, founders and recipients occurs within blocks. Two prespecified

pruned-minus-intact contrasts use 10,000 paired whole-block draws and 97.5% percentile intervals, targeting approximate 95% family coverage.

| Boundary | Intact mixture | Once-pruned mixture | Pruned minus intact, points [97.5%] |
|---|---|---|---|
| h0 | 514/1,024 (50.20%) | 589/1,024 (57.52%) | +7.3242 [4.5898, 10.2539] |
| h1 | 588/1,024 (57.42%) | 530/1,024 (51.76%) | −5.6641 [−9.9609, −2.1472] |

The frozen reversal criterion requires the immediate lower bound above zero and the post-rewrite upper bound below zero; both hold. Immediate pruning gives 84 rescues and nine losses, while after rewriting its route gives 35 gains and 93 losses against intact. Earlier E072 has the same directional pair, +5.76 and −2.73 points; their between-cohort magnitude difference is not a tested paired effect.

All physically available selected edges return. Intact writers use cue contrast on 61,867 of 62,100 selected edge instances and fallback on 233; once-pruned writers use fallback on all 62,100. These are descriptive dependent counts, not independent samples or proof of mediation. Pruning changes development and weights elsewhere; normalization also changes. The result concerns this specified subset and boundary, not all pruning or all reconstruction mechanisms.

Sources: protocol, interpretation, closeout. The receipt reports all-block replay and verification of 322 original outputs/28 frozen inputs. The post hoc branch-count aggregation is descriptive and does not change the two prespecified tests.

### C.51. Serial body capture and injury-fill subtraction: TT022–TT024

These outcome-informed development studies use four exposed TT021 blocks 21000000–21000003, all sixteen message/replica positions, both common/informative record families and four futures per record. They do not enlarge TT021's fresh-cohort qualification. TT022 follows captured packets through eight replacements; each cycle reads the actual incoming packet into a fresh body, injures that developed body, initializes reference memory, repairs and transmits only the resulting 2,048-byte theta field. The terminal fresh read is separate from development and repaired-body observations. No extra record-corruption channel is imposed. E97

TT023 tests the contribution of nominal fresh-cell fill during capture. TT024 repeats the subtraction through eight cycles, setting the captured reference to the actual injured body minus the known resting fill at new-cell positions. Actual body values and jitter are unchanged. The initializer receives the physical injury mask and resting value: this is an added information resource, not autonomous damage detection. Zero-initial-memory arms subsequently update their memory normally.

All four TT024 policies start with the same captured packet. Full capture and protected-copy controls retain their original TT022 trajectories; matching first episodes are reused only after route and byte checks. Newly generated later paths use each arm's own evolving body and packet, with the original paired TT022 environments. Reused controls are not new independent samples.

| Informative-record policy | Final fresh recipient | Every development passes | Every repaired body passes |
|---|---|---|---|
| Full capture, writing on | 48.438% | 21.875% | 12.891% |
| Fill-subtracted capture, writing on | 75.391% | 58.984% | 43.359% |
| Zero initial memory, writing on | 74.219% | 50.000% | 44.922% |
| Zero initial memory, writing off | 89.453% | 69.141% | 47.266% |

Terminal recognition averages 256 paired chain draws per family/condition, nested in 64 records and four blocks, with fractional tie credit. The two uninterrupted endpoints instead require unique correctness at every corresponding body boundary. The corrected-minus-full terminal gain is 26.953 points, positive in all four blocks; against protected preservation, corrected capture improves four record means and worsens thirty. Its common-record terminal score is 67.188%, equal to zero-memory writing and below protected preservation's 68.750%.

These are descriptive controlled development results, with no population significance gate, new confidence intervals or precise individual capacity estimate. They show that nominal fill can impair repeated capture and that the specified correction helps, while leaving continued updating inferior to preservation in this channel. TT021's one-read success, TT024's reused serial conditions and TT018's functional-network futures are different experiments.

Sources: TT022 interpretation, TT023 interpretation, TT024 protocol, TT024 interpretation, TT024 closeout. The v2.1 selection preserves source identities and distinguishes retained replay receipts from this document-only intake.

# S. Supporting figures

## S1. Reference acquisition changes restoration, while the absolute benchmark remains unmet

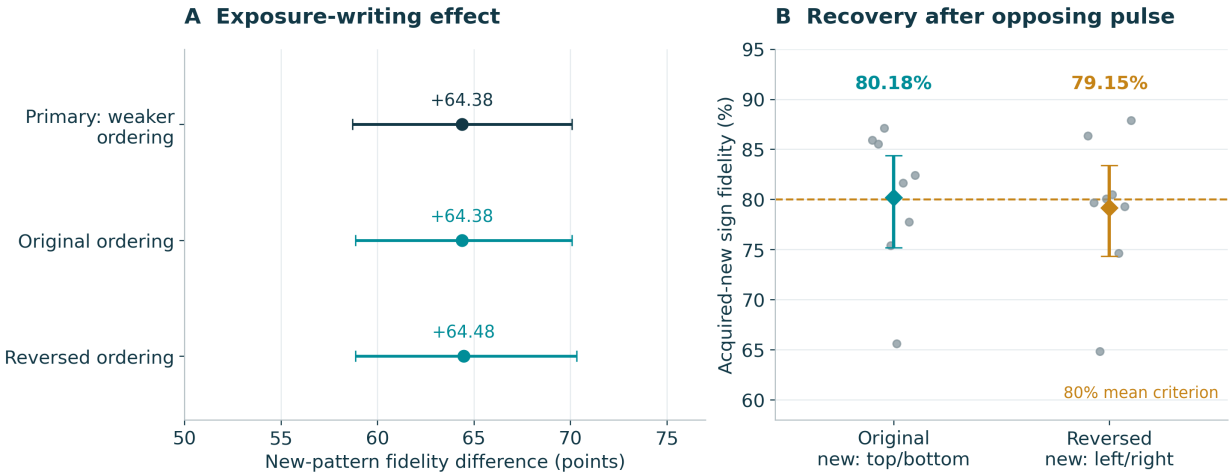



**Figure S1: Reference acquisition changes restoration, while the absolute benchmark remains unmet.** D099: eight tissue blocks, two paired lives each, and two target-role orderings. Left: the prespecified weaker-direction duration-writing effect and the two descriptive direction effects; error bars are the saved 95% tissue-bootstrap intervals. Right: acquired-new sign fidelity after an opposing pulse; dots show the eight tissue means, large markers their means, and bars pointwise 95% intervals. The dashed line is the registered 80% mean criterion, not a natural boundary. Only R crossed into readout. The two panels measure different endpoints. Values are extracted from the retained summary; no new simulations or significance tests were performed. E08

## S2. Repeated reconstruction loses readable information despite successful protected copying

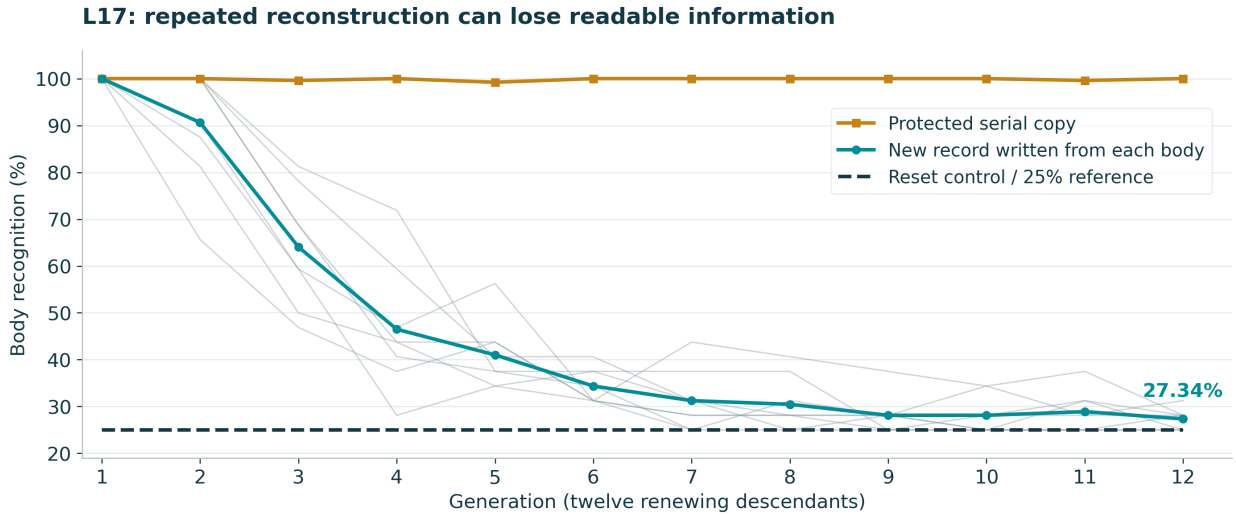



**Figure S2: Repeated reconstruction loses readable information despite successful protected copying.** L17: all twelve generations from the saved summary, eight tissue blocks with four translated stripe messages and eight paired replicates each. Faint lines show individual tissue trajectories for body-written replacement; heavy lines are equal-tissue means. The protected-copy comparator transmits the original acquired record. The reset control remains at the 25% reference level. At generation 12 the body reads Q11; the separate fresh-body assay of Q12 scored 26.95% and is not an extra renewing generation. Curves are descriptive; the three predefined terminal contrasts each failed the six-of-eight positive-tissue criterion despite positive adjusted intervals. No new uncertainty bands or statistical tests were added. E12

## S3. Three tests of pattern continuity

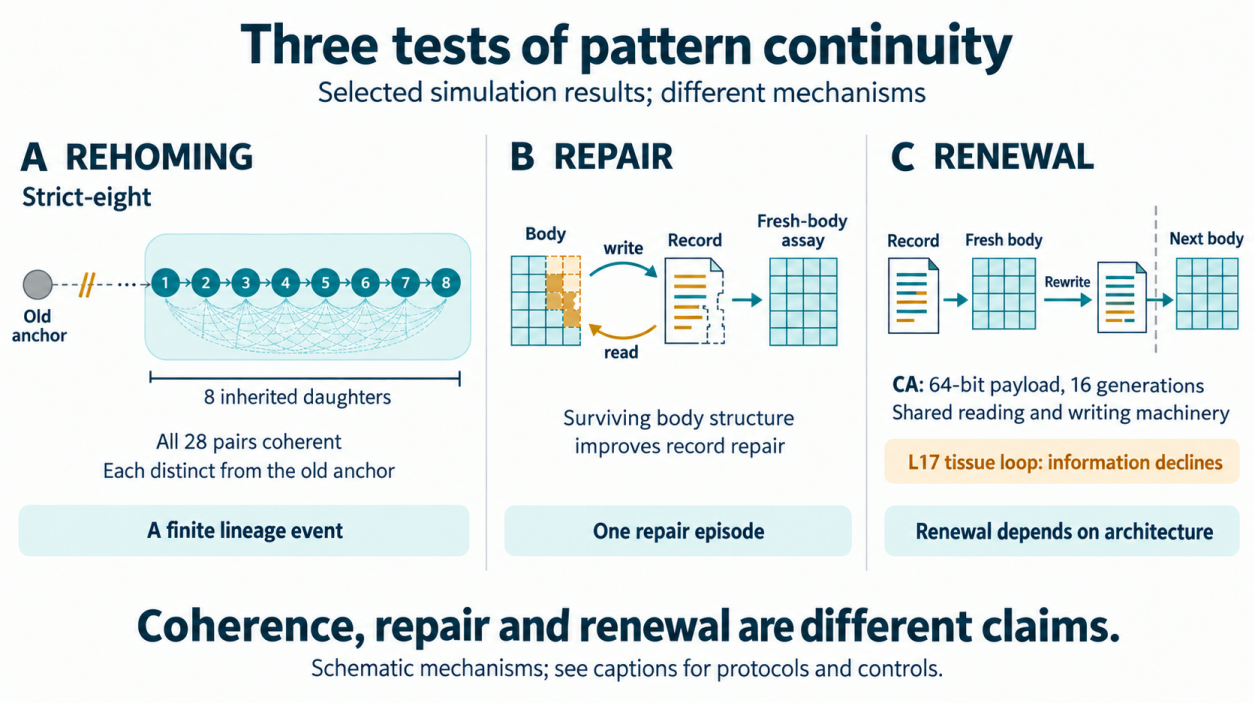



**Figure S3: Three tests of pattern continuity.** These are explanatory schematics, not reconstructed trajectories. **A:** In S8-Core, a break at similarity H <= 0.9 establishes the old anchor. A strictly later window requires eight inherited boundaries with H > 0.9, all 28 daughter-pair similarities > 0.9, and each daughter <= 0.85 relative to the anchor. One followed lineage is certified. E01 **B:** Surviving body organization contributes to record repair; the resulting record is tested in fresh bodies. This is one episode with remaining record and reader information. E09 E10 **C:** CA Stage 4 follows a 64-bit lineage payload for sixteen generations, with 656 shared codebook bits and supplied reading/writing machinery. The no-rewrite comparison imposes 50% payload decay at each boundary, rather than perfect copying. E21 The separate L17 reconstruction loop loses readable information despite writing. E12 The contrast establishes dependence on the tested architecture, not indefinite renewal.

## S4. From cell state to collective capacity

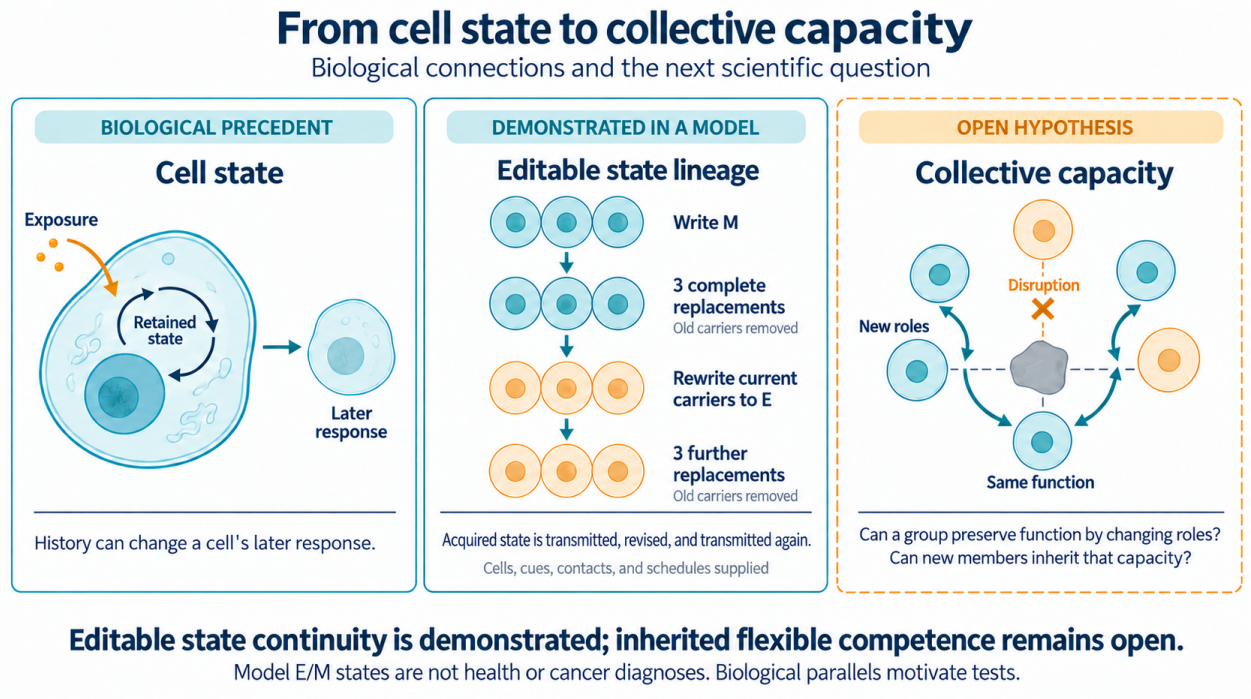



**Figure S4: From cell state to collective capacity.** Left: a generic biological comparison; acquired history can affect later responses, without specifying a particular molecular pathway. Center: P11's corrected editable lineage writes M, completes three replacements, rewrites the actual current carriers to E, and completes three further replacements. Each group contains three carriers; old teachers are removed, and fresh cells, cues, contacts, and schedules are supplied. This schematic summarizes the same P11 evidence as Figure 5, not a second experiment. E and M are model regulatory labels, not health or cancer diagnoses. Right: role reassignment that preserves function and passes that capacity to new members remains a hypothesis. The panels distinguish evidence types rather than an inevitable progression toward agency. E30

## S5. Sign agreement declines despite more frequent amplitude qualification

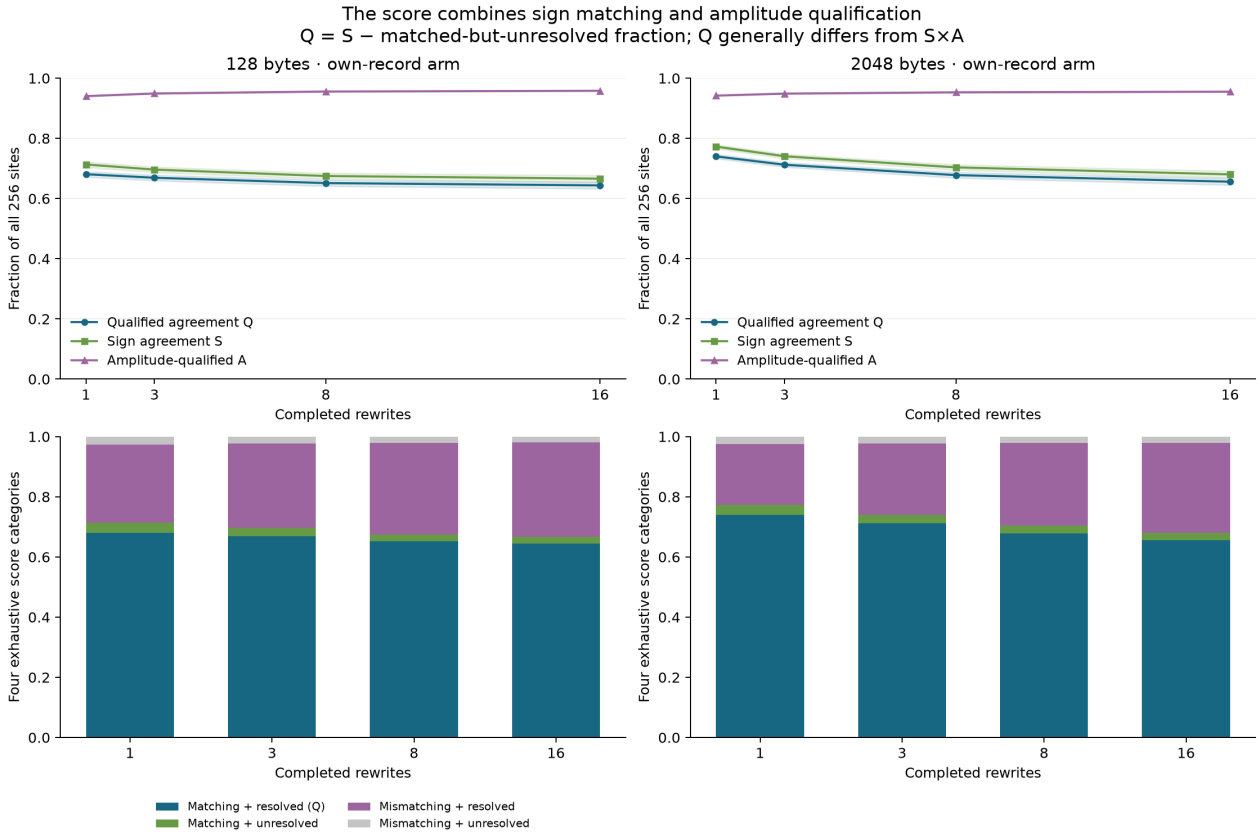



**Figure S5: Sign agreement declines despite more frequent amplitude qualification.** Post hoc analysis of E014's retained unprescribed-pattern fields, at both record budgets and all fresh checkpoints. Top: own-record qualified agreement Q, sign agreement S and fraction A above the amplitude threshold. Bottom: four exhaustive joint categories, retaining all 256 sites. Signs use $V > 0$; qualification uses strict $|V| > 0.2$. $Q = S$ minus the matching-but-subthreshold fraction, not generally $S \times A$. Between three and sixteen compact rewrites, sign agreement falls 3.005 points while amplitude qualification rises 0.892 points; a 0.476-point reduction in matching-but-subthreshold sites partly offsets the sign loss. Bands are descriptive pointwise 95% intervals from 10,000 whole-block resamples of the original 64 blocks, with all 256 founders and two recipient replicates retained. Fresh probes never enter the lineage. This is coordinate-based endpoint accounting, not a translation-tolerant shape measure or causal partition. C.16 records extraction and portable reproduction. E43

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